G E U S

Report File no

3357

TORDENSKJOLD-1

Final Well Report (Completion report)

Dansk Operatørselskab I.S.

Denmark

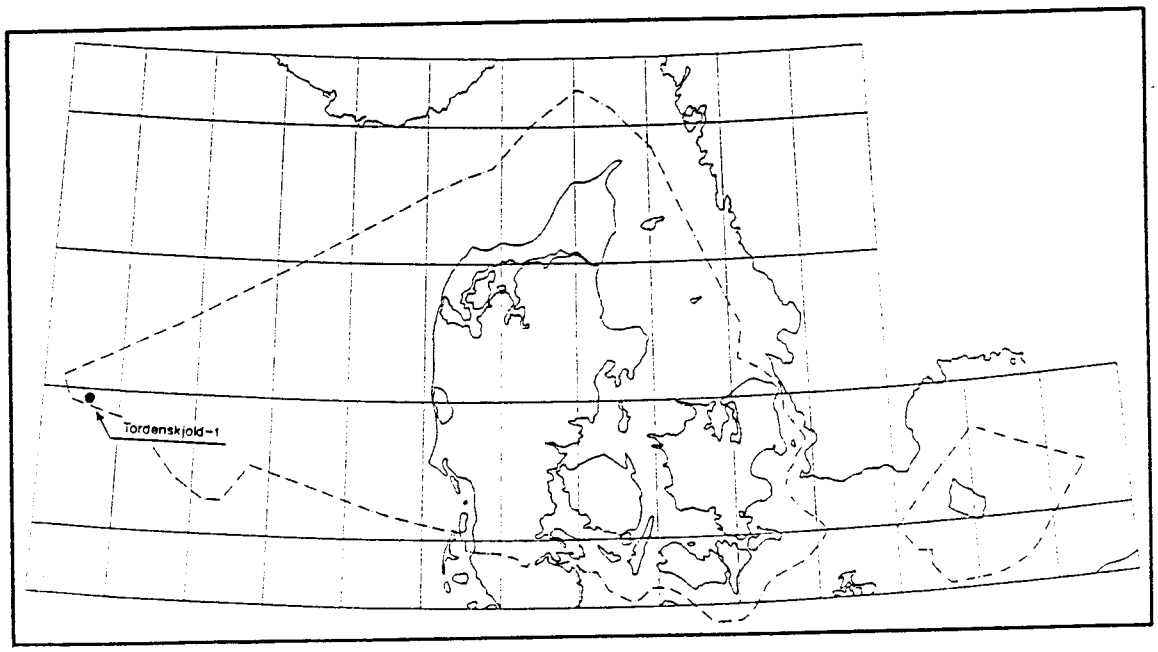
01/06-1989

MODTAGET DGU
13 JUNI 1989
UNDERGRUND
ARKIV

FINAL WELL REPORT

WELL 5503/3-2 TORDENSKJOLD-1

June 1989



212-069
83

Dansk Operatørselskab i-s

danop

TABLE OF CONTENTS:

		Page No.
1.	Pertinent Data and Summary	
	1.1 Pertinent Well Data	1
	1.2 Summary	8
2.	Operation	
	2.1 Mobilization	9
	2.2 30" Conductor Driving	10
	2.3 26" Hole Section	11
	2.4 17 1/2" Hole Section	12
	2.5 12 1/4" Hole Section	13
	2.6 8 3/8" Hole Section	14
	2.7 Plug and Abandonment	15
	2.8 Demobilization	16
3.	Geological Data	
	3.1 Stratigraphy	17
	3.1.1 Chronostratigraphical Subdivision	17
	3.1.2 Lithostratigraphical Subdivision	17
	3.1.3 Description of Lithostratigraphic Units	20
	3.1.3.1 Quaternary	20
	3.1.3.2 Tertiary	20
	3.1.3.3 Upper Cretaceous	23
	3.1.3.4 Lower Cretaceous	24
	3.1.3.5 Upper Jurassic	25
	3.1.3.6 Lower Triassic	26
	3.1.3.7 Upper Permian	26
	3.1.3.8 Lower Permian	27
	3.2 Hydrocarbon Indications	28
	3.3 Logging	31
	3.3.1 Mudlog	31
	3.3.2 Pressure Log	31
	3.3.3 MWD Log	32
	3.3.3.1 MWD Log Runs	32
	3.3.4 Wireline Logs	33
	3.3.4.1 Pertinent Log Data	33
	3.3.4.2 Wireline Log Runs	34

TABLE OF CONTENTS (CONT.):

		Page No.
	3.3.4.3 Computer Processed Logs	38
3.4	Cores and Field Core Description	39
3.5	Sidewall Cores and Description	44
3.6	Well Services	52
3.7	Formation Evaluation	54
3.8	Ditch Sample Descriptions	58
3.9	Well Data Distribution	76

APPENDICES:

Appendix 1: Location Data
- 1.1 Regional Map
- 1.2 Location Map

Appendix 2: Daily Drilling Summary

Appendix 3: Well Status

Appendix 4: Drilling Progress
- 4.1 Drilling Curve
- 4.2 Time Break Down Table
- 4.3 Time Break Down Pie

Appendix 5: Drilling Fluid Summary
- 5.1 30" - 26" Hole Section
- 5.2 17 1/2" Hole Section
- 5.3 12 1/4" Hole Section
- 5.4 8 3/8" Hole Section
- 5.5 Mud Data Summary

Appendix 6: Cementing Summary
- 6.1 General
- 6.2 Cement Product Consumption

Appendix 7: Bit and Bottom Hole Assembly Summary
- 7.1 Bit Record
- 7.2 Bottom Hole Assembly

Appendix 8: Directional Data Summary

Appendix 9: Leak Off and Pore Pressure Data Summary

Appendix 10: Casing and Wellhead Summary
- 10.1 Wellhead Summary
- 10.2 Casing Summary

Appendix 11: Cost Summary

ENCLOSURES:

Enclosure 1: Well Data Sheet

Enclosure 2: Composite Log

Enclosure 3: Log Panel

Enclosure 4: Inter Log

Enclosure 5: Graphic Well Log

1. PERTINENT DATA AND SUMMARY

1.1 PERTINENT WELL DATA

Well	:	5503/3-2, Tordenskjold-1
Location	:	Danish North Sea
Coordinates	:	55° 56' 19.08" N 3° 32' 31.28" E UTM: Zone 31 6199 519.2 N 533 859.7 E
Seismic Location	:	NH8613-212 S.P. 130
Licence	:	4/86
Group	:	Norsk Hydro Udforskning a.s Texas Eastern Denmark A/S Gas Council Exploration Ltd. Amerada Hess, Denmark A/S Dansk Olie- og Gasproduktion A/S DENERCO K/S Dansk Oliesøgning K/S A/S Korn- og Foderstof Kompagniet
Operator	:	Dansk Operatørselskab i-s
Contractor:	:	Neddrill B.V.
Drilling Rig	:	Neddrill Trigon
RKB-elevation	:	36 m
Water Depth	:	65 m

Total depth	:	Drillers depth: 3739 m RKB Loggers depth: 3737,5 m RKB
Date spudded	:	December 13, 1988
Date rig released:	:	February 8, 1989
Days on location	:	57.6 days
Objectives	:	Primary: Upper Jurassic sandstone Secondary: Lower Cretaceous sandstone
Well status	:	P & A, Oil shows
Cores	:	Core 1, cut 3331 - 3331.32 m reverse circulating (junk basket) Core 2, cut 3422 - 3429 m rec. 3422 - 3426.7 m (67%) Core 3, cut 3429 - 3436 m rec. 3429 - 3435 (87%) Core 4, cut 3562 - 3573 m rec. 3562 - 3569.7 m (70%)
Sidewall cores	:	90 shots, 36 recovered
Tests	:	None
Logs	:	MWD 152-2267 m DIL/BHC/LDL/CNL/GR/CAL/SP 2398-3339 m CBL/VDL/GR 950-2430 m (GR to surface) ISF/LSS/GR/SP 3341-3736 m LDL/CNL/NGL/CAL 3341-3737 m DLL/MSFL/GR/CAL 3341-3733 m RFT 13 pressure tests SHDT 3340-3737 m

Logs (Cont.)	CBL/VDL/GR	1400-3341 m
	VSP	200-3737 m

Casing :	30"	at 151 m RKB
	20"	at 1002 m RKB
	13 3/8"	at 2398 m RKB
	9 5/8"	at 3341 m RKB

Service Companies :	
Wellhead Equipm. & Mudline System:	Vetco Gray
Casing:	Mannesmann
30" Conductor:	Dril-Quip
Mud Chemicals and Services:	I.D.F.
Drilling Bits:	Reed Tool Norway
	Smith International
	Diamant Boart
	Hughes Tool Co.
	Eastman Christensen
Fuel Oil Supply:	Kuwait Petroleum
Drilling Rig	Neddrill B.V.
Cementing Services:	Halliburton
Coring Services:	Diamant Boart
Electric Logging	Schlumberger
Mudlogging Services	Geoservices
MWD Services:	Teleco Oilfield Services
Directional Services:	Eastman Christensen
Casing Running Services:	Salvesen
Conductor Driving:	H&S Hammers
Drilling Equipment:	C. Breinholt A/S
Test Tubing:	Dogis A/S
Underreamers and Hole Openers:	Diamant Boart
Jars and Shock Tools:	KLT-Industri
Fishing Equipment and Services:	Smith International
H_2S Services:	Maersk H_2S
Weather Forecast:	D.M.I.
Liner Hanger:	Nodeco A/S
Oil Spill Contingency Agreement:	Mærsk Olie og Gas A/S

Service Companies (continued):	
Positioning Services:	Racal
Supply Base Facilities:	Westcoast Offshore Base
Standby/Supply Vessel:	Seabrokers
Helicopter Services:	Maersk Air
Biostratigrahphic Services:	Robertson Research
Core Analysis:	DGU
Geochemical Services:	DGU
Site Survey:	Geoteam
Shallow Drilling:	Fugro-McClelland
Consultancy Services, Site Survey:	Noble Denton
R.O.V. Services:	Scan-Dive

WELL SUMMARY

Country : Denmark Area : Central Graben Licence : 4/86 Location : 55°56"19.08' N : 3°32"31.28' E UTM : Zone 31 : 6199 519.2 N : 533 859.7 E	RKB Elevation : 36 m Water Depth : 65 m Total Depth : Drlg. 3739 m : Log. 3737.5 m Spud Date : Dec. 13, 1988 Date at T.D. : Jan. 27, 1989 Rig Released : **Feb. 8, 1989** Status : P & A, Oil Shows Rig : Neddrill Trigon Contractor : Neddrill B.V.	Group: Norsk Hydro Udforskning a.s Texas Eastern Denmark A/S Gas Council (Exploration) Ltd. Amerada Hess Denmark A/S A/S Korn- & Foderstof Kompagniet DENERCO K/S Dansk Oliesøgning K/S Dansk Olie & Gasproduktion A/S

CASING

30"	151 m
20"	1002 m
13 3/8"	2398 m
9 5/8"	3341 m

CONVENTIONAL CORES

Core no.1 cut 3331 - 3331.32 m
(Junk Basket)

Core no.2 cut 3422 - 3429 m
Rec. 3422 - 3426.7 m (67%)

Core no.3 cut 3429 - 3436 m
Rec. 3429 - 3435 m (87%)

Core no.4 cut 3562 - 3573 m
Rec. 3562 - 3569.7 m (70%)

MWD

GR-Resistivity 152 - 2267 m

WIRELINE LOGGING

Suite 1 :

DIL/BHC/LDL/CNL/GR/CAL/SP
2398 - 3339 m

CBL/VDL/GR (GR to surface)
950 - 2430 m

CST/GR 1A 2580 - 3340 m
Shots: 30 Rec.: 6

CST/GR 1B 2425 - 3246 m
Shots: 30 Rec.: 8

Suite 2 :

ISF/LSS/GR/SP
3341 - 3736 m

LDL/CNL/NGL/CAL
3341 - 3737 m

DLL/MSFL/GR
3341 - 3733 m

RFT 13 Pressure Tests

SHDT
3340 - 3737 m

CBL/VDL
1400 - 3341 m

VSP
200 - 3737 m

CST 2C 3350 - 3736 m
Shots: 30 Rec.: 22

GAS RECORD

Down to:

317 m :	0	%
610 m :	0.63 - 1.16	% C_1 tr C_2C_3
1020 m :	0.87 - 0.94	% C_1
1145 m :	0.01 - 0.08	% C_1
1365 m :	0 - 0.01	% C_1
1590 m :	0.08 - 0.6	% C_1
1800 m :	0.04 - 0.09	% C_1
2100 m :	0.03 - 1.01	% C_1
2977 m :	0	%
3342 m :	0 - 0.03	% C_1
3474 m :	0	%
3645 m :	0.09 - 1.018	% $C_1 - C_3$
3713 m :	0.115 - 4.196	% $C_1 - C_3$
3739 m :	0.071 - 0.208	% $C_1 - C_3$

Gas Peak at 3535 m :

Total Gas	1.018 %
C_1	7800 ppm
C_2	800 ppm
C_3	1170 ppm

Gas Peak at 3649 m :

Total Gas	4.196 %
C_1	15910 ppm
C_2	12100 ppm
C_3	3090 ppm

Trip Gas at 3739 m :

Wiper Trip	12.6 %
C_1	8 %
C_2	7 %
C_3	2.3 %
iC_4	0.33%
nC_4	0.27%

Trip Gas at 3739 m :

After Logging	36 %
C_1	14.4 %
C_2	11.2 %
C_3	3.2 %
iC_4	0.57%
nC_4	0.44%

OPERATOR

Dansk Operatørselskab i-s (DANOP)

OBJECTIVES

Primary :
Upper Jurassic sandstone

Secondary :
Lower Cretaceous sandstone

RESULTS

The well was plugged and abandoned with traces of hydrocarbons in Zechstein dolomite

Zechstein dolomite :	3635 - 3648 m
Gross Thickness :	13 m
Porosity :	3 - 9 %
Water Saturation :	20 - 80 %

SHOWS

2880 m on chalk
very slow streaming dull white cut

3537 m on siltstone
bright yellow fluorescence, no visible cut, very slow streaming bluish white fluorescent cut, bright yellow-bluish white fluorescent residue, no visible residue.

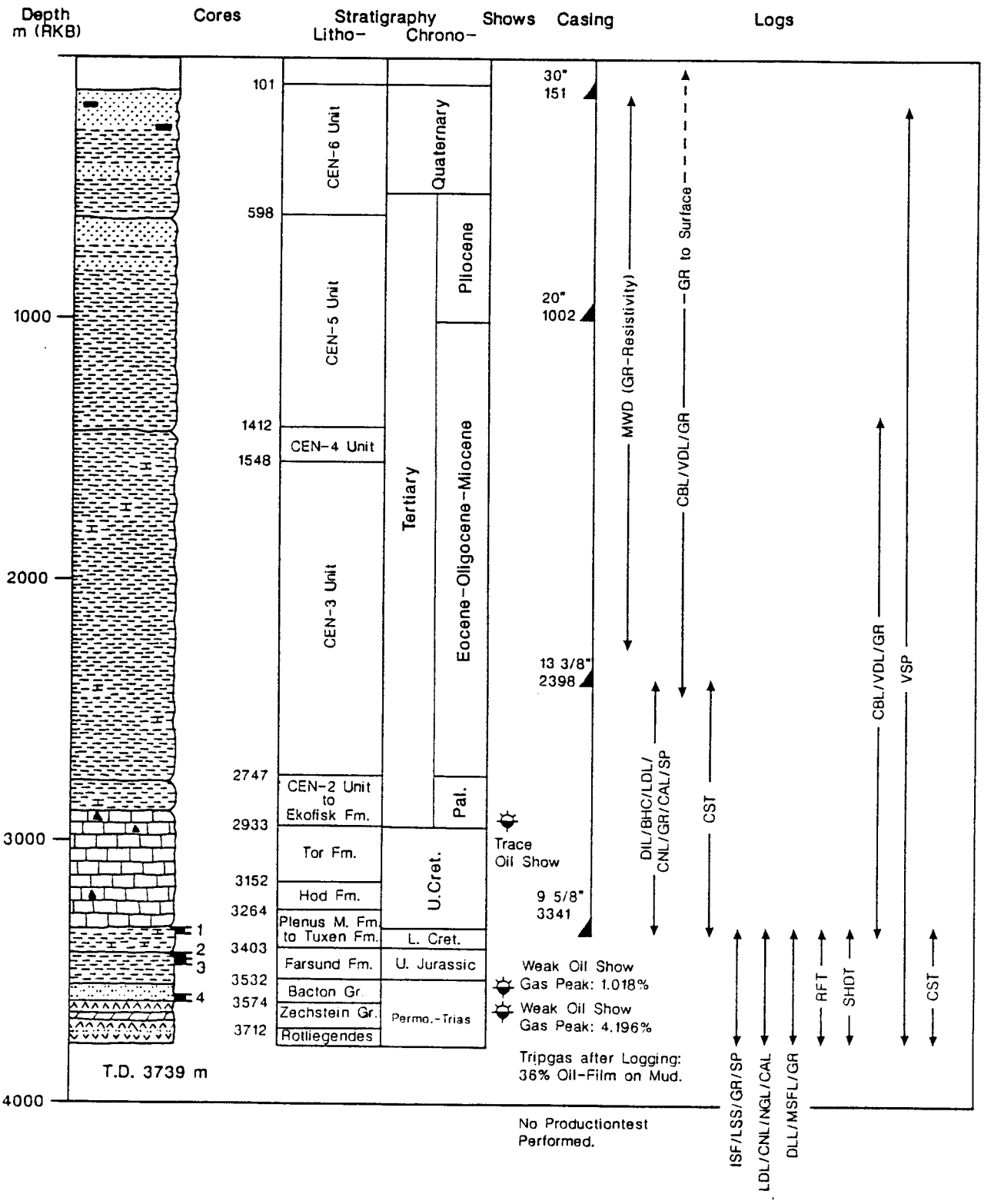
3643 - 3649 on dolomite
very dull yellow fluorescence, non to very slow streaming yellow white-bluish white fluorescent cut, no visible cut, bright yellow fluorescent residue, light straw visible residue.

Oil film on mud while circulating prior to setting first cement plug, oil evaluated to come from Zechstein dolomite at 3635 - 3648 m.

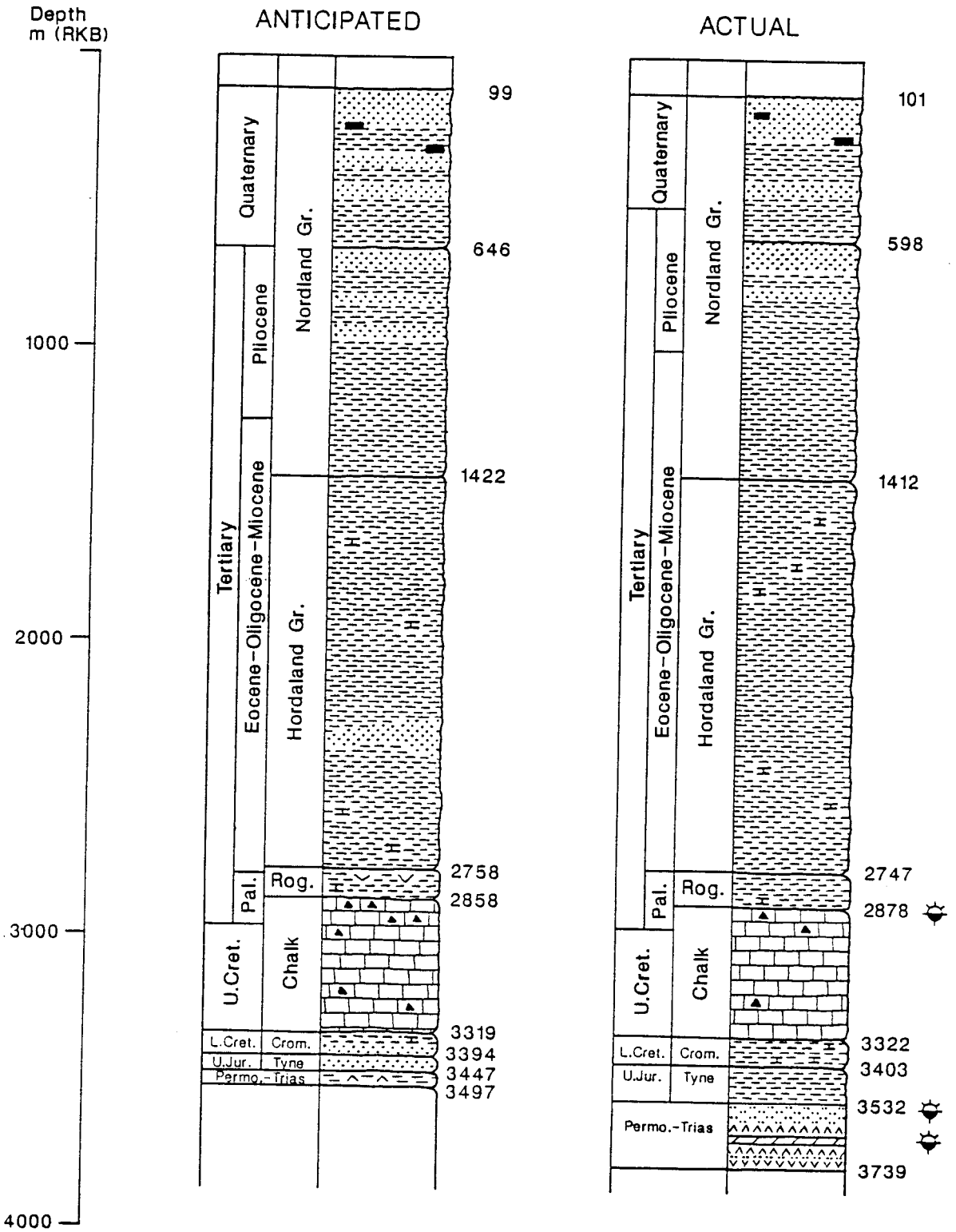
NOTE

All depths in m.b. RKB

WELL SUMMARY



WELL PROGNOSIS Vs. ACTUAL SECTION



1.2 SUMMARY

Tordenskjold-1 was drilled with Neddrill Trigon in the western most part of the Danish North Sea in 65 m of water.

The well is structurally situated in a downflank position to the Mid North Sea High in the western part of the Outer Rough Basin.

The primary objective was Upper Jurassic sandstone. A secondary objective was Lower Cretaceous sandstone.

The well was spudded on December 13, 1988, and drilled to a total depth of 3739 m terminating in Rotliegendes volcanics.

The well encountered a Quaternary/Tertiary sand/clay/-claystone sequence overlying a Tertiary/Upper Cretaceous chalk/limestone sequence. A Lower Cretaceous/Upper Jurassic claystone/limestone sequence was drilled before entering a Triassic claystone/siltstone sequence. A Permian sequence with Zechstein evaporites on top of Rotliegendes siltstone/volcanics was drilled from 3574 - 3739 m.

No reservoir was developed neither in the Upper Jurassic section nor in the Lower Cretaceous section.

A minor show was seen in a relatively tight Zechstein dolomite.

No production test was carried out.

The well was plugged and abandoned on February 6, 1989 and the rig was released on February 8, 1989.

2. OPERATION

2.1 MOBILIZATION

The jack-up rig Neddrill Trigon was mobilized from Dutch sector on December 3, 1988. The rig went on contract on December 12 at 1500 hrs after the rig was jacked up and the derrick and cantilever skidded out.

2.2 30" CONDUCTOR DRIVING

The 30" conductor was drilled and hammered to 151 m. When drilling out of the conductor at 135 m the 26" hole opener became stuck inside the 30" conductor shoe due to insufficient hole clearing.

2.3 26" HOLE SECTION

The diverter was nippled up and the 26" hole drilled to 611 m where the 26" hole opener was removed due to hole angle build up. A 17 1/2" pilot hole was drilled to 1020 m and opened to 26". The 20" casing run to 1002 m and cemented using a 5" cement stinger stung into the float collar. The 20" casing was cemented to the sea-bed. The 26" hole section was logged using MWD with gamma ray and resistivity.

After the 20" wellhead was installed, the 21 1/4" BOP was nippled up and pressure tested.

2.4 17½" SECTION

The 20" casing shoe was drilled out and the hole displaced to KCl-Polymer mud of 1.60 r.d. The formation strength test performed at 1024 m resulted in leak off at 2.25 r.d. equivalent mud weight. The 17½" hole was drilled from 1020 m to 2415 m using one PDC-bit. The MWD tool stopped pulsing at 2277 m due to turbine blockage.

A wiper trip was performed to the 20" casing shoe and the 13 3/8" casing run and cemented with the shoe at 2397 m.

The return was lost while displacing the cement resulting in top of cement at 1450 m. The 17 1/2" hole was logged using MWD with resistivity and gamma ray. After the 13 3/8" wellhead was installed the 13 5/8" BOP was nippled up and tested.

2.5 12¼" SECTION

The 13 3/8" casing shoe was drilled out and a formation strength test performed at 2418 m. resulting in leak off at 1.91. r.d. equivalent mud weight. The 12¼" hole was drilled to total depth of 3342 m.

The drilling was interrupted by loosing one bit cone at 3329 m, causing 3 extra trips to fish for junk. While fishing for junk using reverse circulating junk basket, core no. 1 was cut from 3331 m to 3331.5 m.

The following logs were run prior to running the 9 5/8" casing:

1) DIL/BHC/LDL/CNL/GR/CAL/SP
2) CBL/VDL/GR
3) CST (low recovery)
RR 3) CST

When attempting to land the 9 5/8" casing in the mud line suspension the hanger fell through. The casing was cemented and lowered so that the weight was distributed between the cement and the wellhead. The 9 5/8" casing was cemented to 1600 m.

2.6 8 3/8" SECTION

Leakages were found in the upper pipe ram and upper outer kill valve when pressure testing to BOP to 655 bar. The BOP was repaired prior to drilling out the 9 5/8" casing.

A formation integrity test was performed equivalent to 2.07 r.d. at 3345 m. The 8 3/8" hole was drilled from 3345 m to 3739 m interrupted by the following core intervals:

Core no. 2 from 3422 m to 3429 m
Core no. 3 from 3429 m to 3436 m
Core no. 4 from 3562 m to 3573 m

The following logs were run at total depth:

1) ISF/LSS/GR/SP
2) LDL/CNL/NGT/CAL (tight hole)
RR 2) LDL/CNL/NGT/CAL
3) DLL/MSFL/GR
4) RFT (tool failure)
RR 4) RFT
5) SHDT
6) CBL/VDL
7) VSP
8) CST

2.7 PLUG AND ABANDONMENT

The following cement plugs were set:

Plug no. 1 from 3739 m to 3665 m
Plug no. 2 from 3665 m to 3392 m
Plug no. 3 from 3392 m to 3192 m
Plug no. 4 from 2397 m to 2297 m

The 9 5/8" and 13 3/8" casings were cut at 251 m.

The cutting tool became stuck at 199 m when pulling out due to inserts from the 9 5/8" hanger were jammed between the cutting tool and the 9 5/8" casing.

The drill pipe was backed off and the BOP nippled down before the 9 5/8" casing was retrieved.

The 13 3/8" casing was recut at 249 m and retrieved.

Cement plug no. 5 from 351 m to 151 m was set and the 20" casing retrieved in the mud line suspension system.

The 30" casing was cut and retrieved at 106 m.

2.8 DEMOBILIZATION

65 hours were used waiting on weather before the cantilever were skidded back.

The rig was jacked down and went off contract on February 8 at 0600 hrs after having been on contract for 57.6 days.

3. GEOLOGICAL DATA

3.1 STRATIGRAPHY

3.1.1 Chronostratigraphic Subdivision

The chronostratigraphic subdivision is based on dating carried out on wet samples, sidewall cores, and core chips by the Robertson Group PLC.

160m (top not seen)	Pleistocene
500m	Pliocene
990m	Upper - Middle Miocene
1450m	Middle Miocene
1630m	Lower Miocene
----------Unconformity----------	
1690m	Upper Oligocene
1790m	Lower Oligocene
----------Unconformity----------	
1850m (log)	Middle Eocene
----------Unconformity----------	
2639m (log)	Lower Eocene
----------?Unconformity----------	
2748m (log)	Upper Paleocene
2870m	Lower Paleocene
----------Unconformity----------	
2932.3m (log)	Uppermost Maastrichtian
2980m	Upper Maastrichtian
3080m	Lower Maastrichtian - ?Campanian
----------?Unconformity----------	
3152m (log)	?Santonian
3174m	Coniacian - Turonian
3289m	Upper Cenomanian
3324m	Lower Cenomanian
3330m (log)	Upper Albian
----------Unconformity----------	
3355.5m (log)	Aptian
3387.5m	Barremian
3400m (SWC)	Upper Hauterivian
----------Unconformity----------	
3403.5m (log)	Middle Volgian
3512.5m (core)	Lower Volgian
----------Unconformity----------	
3531.5m (log)	?Upper Permian
3566m (core)	Upper Permian
3712m (log)	Lower Permian

3.1.2 Lithostratigraphic Subdivision

The following lithostratigraphic breakdown has been performed based on the formal stratigraphic units as defined in Michelsen, O. (ed.) 1982 Geology of the

Danish Central Graben – Dan. Geol. Unders. Ser. B, 8 and Jensen, T.F. et al. 1986 Jurassic - Lower Cretaceous lithostratigraphic nomenclature for the Danish Central Trough - Dan. Geol. Unders., Ser. A 12, and correlation of logs to nearby wells and use of the lithological descriptions of cuttings, SWC and cores.

The following table represents the Operator's interpretation.

Lithostratigraphic Unit	Thickness (m)	Interval (m.b.KB)	Chronostratigraphy	
CEN-6 UNIT	497	101-598	PLEISTOC.	QUATERNARY & TERTIARY
			500	
CEN-5 UNIT	814	598-1412	PLIOCENE	
			990	
CEN-4 UNIT	136	1412-1548	U.-M. MIO.	
			1450	
CEN-3 UNIT	1199	1548-2747	M.MIOCENE	
			1630	
CEN-2 UNIT	64	2747-2811	L.MIO-EOC.	
			2748	
CEN-1 UNIT	37	2811-2848	UPPER PALEOCENE	
NORTH SEA MARL	30	2848-2878		
			2870	
EKOFISK FORMATION	55	2878-2933	L.PALEOCE	
			2933	
TOR FORMATION	219	2933-3152	MAASTRICT. CAMPANIAN	UPPER CRETACEOUS
			3152	
HOD FORMATION	112	3152-3264	SANTONIAN CONIACIAN TURONIAN	
PLENUS MARL FM.	26	3264-3290		
			3289	
HIDRA FORMATION	32	3290-3322	CENOMANIAN	
			3330	
RØDBY FORMATION	33	3322-3355	U.ALBIAN	LOWER CRETACEOUS
			3355	
SOLA FORMATION	33	3355-3388	APTIAN	
			3388	
TUXEN FORMATION	15	3388-3403	BARREMIAN HAUT.	
			3403	
FARSUND FORMATION	129	3403-3532	M.-L. VOLGIAN	UPPER JURASSIC
			3532	
BACTON GROUP	42	3532-3574	SCYTHIAN	LOWER TRIASSIC
			3574	
ZECHSTEIN GROUP	138	3574-3712	U. PERMIAN	PERMIAN
ROTLIEGENDES GROUP	27	3712-TD	L. PERMIAN	

3.1.3 Description of Lithostratigraphic Units

3.1.3.1 Quaternary

Quaternary Sand

151 - 314 m

Fine to coarse grained SAND of clear, colorless, subangular to rounded, poorly sorted quartz grains. The SAND contains trace to abundant lignite, trace to moderate amount of shell fragments and trace to rare mica and glauconite. Brownish black laminae of CLAYSTONE and laminae of light grey, slightly calcareous SILTSTONE are locally observed above 230 m.

Quaternary Clay

314 - 598 m

Medium dark grey, soft and sandy to silty CLAY, varyingly calcareous with trace to rare mica and shell fragments, trace to locally rare glauconite and trace of microlignite and pyrite.

3.1.3.2 Tertiary

CEN-5 Unit

598 - 1412 m

Dark grey to medium dark grey CLAY/CLAYSTONE, soft to firm, locally sticky, silty locally sandy and non- to very calcareous. The CLAYSTONE is furthermore micaceous and microlignitic and contains none to minor glauconite and locally trace of pyrite (dispersed, as aggregates and as replaced trace fossils).

CEN-4 Unit

1412 - 1471 m

Medium dark grey, soft to firm, locally sticky CLAYSTONE, slightly silty, locally sandy, non- to very calcareous, micaceous, microlignitic and with trace of glauconite, shell fragments, microfossils and pyrite.

1471 - 1548 m

The CLAYSTONE is dusky brown to dusky yellowish brown, firm to moderately hard, slightly to non-calcareous and micaceous and contains rare to trace microlignite, minor to trace pyrite and trace to absent shell fragments. Stringers of very calcareous, light olive grey and hard CLAYSTONE occur locally.

CEN-3 Unit

1548 - 2160 m

Dusky yellowish brown to dark grey, soft to moderately hard, non- to varyingly calcareous, micaceous and slightly glauconitic and microlignitic CLAYSTONE, which contains minor to trace pyrite (dispersed, as aggregates and as replaced trace fossils), and trace to absent shell fragments. The interval contains several stringers of light olive grey, hard and very calcareous CLAYSTONE grading to MARLSTONE and light olive grey, argillaceous LIMESTONE (micrite with laminae of sparite), firm to hard and locally microlignitic.

Between 1670 and 1700 m are several stringers rich in PHOSPHORITE (as cement?), dusky green, hard to very hard, as aggregates and single grains, but PHOSPHORITE rich horizons are also encountered further down in the interval. Below 1900 m are also stringers of brownish grey to dark yellowish brown DOLOMITE (occasionally LIMESTONE), which are hard and varyingly argillaceous.

2160 - 2412 m

CLAYSTONE, olive black, firm to moderately hard, slightly to non-calcareous, micaceous and slightly microlignitic with trace of minor pyrite, trace of glauconite and locally trace of phosphorite. The CLAYSTONE contains stringers of light olive grey, moderately hard

and argillaceous LIMESTONE and brownish grey, moderately to very hard and argillaceous DOLOMITE.

2414 - 2747 m
Dark grey to medium dark grey, soft to firm CLAYSTONE with rare microlignite, varyingly interbedded with a brownish grey, soft to firm, blocky to slightly fissile CLAYSTONE and a dark greenish grey and firm to moderately hard CLAYSTONE. At base of the interval are beds of pale brown and varyingly reddish brown and grey CLAYSTONE. The interval contains furthermore scattered stringers and beds of dark to pale yellowish brown, friable and partly argillaceous LIMESTONE. Minor beds of very fine to medium grained, moderately sorted and subrounded to subangular SAND grains occur.

CEN-2 Unit
2747 - 2811 m
CLAYSTONE, light olive to medium dark grey, soft to moderately hard, slightly fissile, slightly silty and moderately glauconitic. The CLAYSTONE contains trace to rare mica and pyrite and trace of microlignite. Scattered beds of pinkish grey, grayish orange pink or yellowish brown LIMESTONE, soft to hard, argillaceous.

CEN-1 Unit
2811 - 2848 m
Interbedded CLAYSTONE of different reddish, grayish, greenish, bluegreen and brownish colours turning black and slightly calcareous towards the base. The CLAYSTONES are soft to moderately hard, slightly silty and moderately glauconitic.

North Sea Marl
2848 - 2878 m
Greenish grey to medium grey, firm to moderately hard, very calcareous CLAYSTONE, downwards grading into a very light grey MARLSTONE. The rocks contain trace of pyrite.

Ekofisk Formation
2878 - 2933 m
CHALK (micrite), white and firm with trace of pyrite and trace to minor light brown to yellowish grey, semitranslucent chert. The interval shows beds of pinkish grey, hard LIMESTONE (indurated chalk).

3.1.3.3 Upper Cretaceous

Tor Formation
2933 - 3152 m
White to pinkish grey, firm to hard CHALK (micrite) with trace of pyrite and locally trace of sand grains and sparry calcite.

Hod Formation
3152 - 3219 m
White to very light grey to medium light grey CHALK, firm to moderately hard, non- to slightly argillaceous, locally with trace of silt, sparry calcite, microfossils and glauconite.

Towards base of the interval the CHALK grades into LIMESTONE. From 3152 to 3170 m are beds of pale red to moderate orange pink, slightly argillaceous CHALK.

3219 - 3264 m
LIMESTONE, yellowish grey to pinkish grey to light olive grey, soft to hard, locally very hard, varyingly recrystallized, locally silicified grading to cherty

limestone, non- to slightly argillaceous with trace to abundant glauconite and locally trace of mica and pyrite.

Down to 3260 m trace to 20% of light olive grey to light brownish grey, locally translucent chert.

Plenus Marl Formation
3264 3290 m
LIMESTONE pale red to grayish red, soft to firm, locally dolomitic, argillaceous to very argillaceous grading to MARL.

Hidra Formation
3290 - 3322 m
The formation are dominated by laminae of dark grey to grayish black, firm to moderately hard, calcareous CLAYSTONE grading to MARL and the basal LIMESTONE is pale reddish brown, soft to firm, sticky and very argillaceous.

3.1.3.4 Lower Cretaceous

Rødby Formation
3322 - 3355 m
Pale reddish brown, soft to firm, sticky and very argillaceous LIMESTONE with trace of chert to pale to moderate to dark reddish brown, soft to moderately hard CLAYSTONE, which is slightly to moderately calcareous and show "silty texture" due to abundant calcite crystals and contains trace of shell fragments and from 3335 and downwards CLAYSTONE, medium grey to olive black, firm to hard, calcareous, microlignitic, slightly micromicaceous, locally with "silty texture" a/a. The CLAYSTONE contains laminae and beds of loose, clear, very fine to fine grained, well sorted, subangular to subrounded SAND.

Sola Formation
3355 - 3388 m
Grayish red to moderate brown, soft to firm, slightly to moderately calcareous CLAYSTONE, locally very calcareous grading to MARL interbedded/-laminated with a light to medium grey CLAYSTONE, soft to firm, calcareous and micromicaceous. Both types of CLAYSTONE locally show "silty texture" a/a. Few beds of loose, clear, very fine to fine grained, well sorted and subangular to subrounded SAND are present in the interval.

Tuxen Formation
3388 - 3403 m
Interbedded/-laminated white to very light grey, grayish red to moderate brown and light to medium grey LIMESTONES, which are soft to firm, occasionally moderately hard, slightly argillaceous, occasionally very argillaceous grading to MARL, and show trace of "silty texture" due to abundant calcite crystals.

3.1.3.5 Upper Jurassic

Farsund Formation
3403 - 3532 m
CLAYSTONE/SHALE, dark grey to grayish/brownish/olive black to black, soft to hard, subfissile to fissile, non- to moderately calcareous, slightly to very carbonaceous, micromicaceous and micropyritic.
Local thin SANDS of very fine to fine grained, well sorted, subangular to subrounded, clear quartz grains, minor with orange tint and with trace of pyrite and heavy minerals (?). In the lower part of the interval are stringers of white, soft to firm LIMESTONE locally with carbonaceous streaks, and olive grey, hard to very hard, cryptocrystalline DOLOMITE.

3.1.3.6 Lower Triassic

Bacton Group

3532 - 3574 m

Interbedded SILTSTONE and DOLOMITE.

The SILTSTONE is white to very light grey and grayish red to light olive grey, soft to very hard, non- to well cemented (silica/dolomite/calcite), argillaceous to locally very argillaceous, locally very sandy grading to very fine grained SANDSTONE.

The SANDSTONE is pale red to grayish red to light brownish grey to light olive grey, very fine to fine grained, locally grading to SILTSTONE, loose grains to hard, non- to well cemented (silica/dolomite), well sorted, subangular to subrounded, with trace of pyrite and feldspar.

At the base of the interval is a light olive grey to light brownish grey, very hard and microcrystalline to granular DOLOSTONE.

3.1.3.7 Upper Permian

Zechstein Group

3574 - 3712 m

ANHYDRITE interbedded with minor SHALE and DOLOSTONE. ANHYDRITE, white to very light grey, soft to firm, amorphous, locally with argillaceous laminae, locally clear dolomite crystals; or white to pinkish grey to light brownish grey, moderately hard to hard, brittle, locally argillaceous, microcrystalline to granular, locally dolomitic and grey to pale yellowish brown.

SHALE, dark grey to grayish black, moderately hard to hard, blocky to fissile, slightly calcareous, micromicaceous. DOLOSTONE, white to pinkish grey or light

brownish grey to very pale yellowish brown, hard, brittle and microcrystalline to granular.

The interval from 3639 m to 3648 m consists of a light brownish grey to very pale yellowish brown DOLOSTONE, hard, brittle, slightly argillaceous and microcrystalline to granular.

3.1.3.8 Lower Permian

Rotliegendes Group

3712 - 3722 m

Interbedded SANDSTONE and CLAYSTONE.

The SANDSTONE consists of colorless or reddish, very fine grained, moderately sorted and subangular loose quartz grains and the CLAYSTONE is moderate reddish brown to grayish red and firm, but can also be brownish black, hard and slightly dolomitic.

3722 - 3739 m (TD)

VOLCANICS, light grey to light greenish grey with black and reddish brown specs., firm, "silty texture", very fine grained, slightly calcareous and with black and dark reddish brown phenocrysts.

3.2 HYDROCARBON INDICATIONS

The background gas was in general very low. A minor hydrocarbon show and a gas peak were encountered at 3535 m in the upper part of a siltstone sequence interpreted to belong to the Triassic. A minor hydrocarbon show and a gas peak were also seen at 3649 m in a Zechstein dolomite. While circulating prior to set the first cement plug for abandonment trip gas and oil film were seen. This hydrocarbon most probably originated from the same source as the hydrocarbon mentioned above.

Gas record:

Down to:				
317 m:			0	%
610 m:	0.63	-	0.16	% C_1 trC_2 C_3
1020 m:	0.87	-	0.94	% C_1
1145 m:	0.01	-	0.08	% C_1
1365 m:	0	-	0.01	% C_1
1590 m:	0.08	-	0.6	% C_1
1800 m:	0.04	-	0.09	% C_1
2100 m:	0.03	-	1.01	% C_1
2977 m:			0	%
3342 m:	0	-	0.03	% C_1
3474 m:			0	%
3645 m:	0.09	-	1.018	% C_1-C_3
3713 m:	0.115	-	4.196	% C_1-C_3
3739 m:	0.071	-	0.208	% C_1-C_3

Gas Peak at 3535 m:

Total Gas:	1.018 %
C_1	7800 ppm
C_2	800 ppm
C_3	1170 ppm

Gas Peak at 3649 m:

Total Gas:		4.196 %
	C_1	15910 ppm
	C_2	12100 ppm
	C_3	3090 ppm

Trip Gas at 3739 m (T.D.):

Total Gas:		36 %
	C_1	14.4 %
	C_2	11.2 %
	C_3	3.2 %
	iC_4	0.57%
	nC_4	0.44%

Hydrocarbon shows: 2880 m on chalk

Very slow streaming dull white cut (later described by wellsite geologist as slow streaming chalk dust and thereby representing mineral fluorescence).

3537 m on siltstone

Bright yellow fluorescence, no visible cut, very slow streaming bluish white fluorescent cut, bright yellow-bluish white fluorescent residue, no visible residue.

3643 - 3649 m on dolomite

Very dull yellow fluorescence, non to very slow streaming yellow white-bluish white fluorescent cut, no visible cut, bright yellow fluorescent residue, light straw visible residue.

Oil film on mud while circulating prior to setting first cement plug, oil evaluated to come from Zechstein dolomite at 3635 - 3648 m.

3.3 LOGGING

3.3.1 Mudlog

Mudlogging service was provided by Geoservice. The mudlog included as appendix was prepared in scale 1:500 and contain information on the following parameters:

- Bit data and other pertinent engineering details.
- Rate of penetration.
- Graphic cuttings lithology and interpreted lithology.
- Gas detection by chromatography.
- Oil fluorescence.
- Core intervals and description.
- Various information on mud, casing etc.
- Deviation surveys.
- Down to 2267 additional information on Gamma Ray and Resistivity measurements from the MWD tool (see below).

In addition to chromatographic hydrocarbon analysis, H_2S detection was carried out.

Ditch samples were taken at 10 m interval from 151 to 2600 m, at 5 m interval from 2600 to 3150 m and finally at 3 m interval from 3150 to TD.

In addition canned wet samples for geochemical analysis were collected at 30 m interval from 1400 to 3150 m and at 9 m interval from 3150 m to TD.

3.3.2 Pressure Log

Geoservice also provided a pressure log including information on:

- Bit data and other pertinent engineering details.
- Rate of penetration.

- Graphic cuttings lithology and interpreted lithology.
- Core intervals and description.
- Various information on mud, casing etc.
- D exponent.
- P.f., E.D.C., P. frac. and overburden.
- Temperature in and out.

3.3.3 MWD Log

Teleco provided MWD logging services, recording Gamma Ray, Resistivity and directional information from 151 to 2267 m. The log should have been run to the bottom of the 17 1/2" hole at 2414 m, but the tool failed at 2267 m and it was decided to drill ahead. Consequently no open hole logging has been made between 2267 and 2398 m.

3.3.3.1 MWD Log Runs

Run No. 2

Type:	Gamma Ray-Resistivity-Directional
Bit run No.:	1RR
Interval:	150 - 609 m
Scale:	1:500
Date:	December 14-16, 1988

Run No. 3

Type:	Gamma Ray-Resistivity-Directional
Bit run No.:	2
Interval:	609 - 1018 m
Scale:	1:500
Date:	December 17, 1988

Run No. 4

Type:	Gamma Ray-Resistivity-Directional
Bit run No.:	3
Interval:	1018 - 2267 m

Scale: 1:500
Date: December 21-25, 1988
Comments: The tool failed at 2267 m.

3.3.4 Wireline Logs

Wireline logging was provided by Schlumberger. No open hole wireline logging was carried out in the interval from 152 to 2398 m. The interval was covered by MWD logging as mentioned above. The log runs and intervals are summarized below. A log panel of the objective zones is included as Enclosure 3.

3.3.4.1 Pertinent Log Data

Drillers total depth: 3739.0 m
Loggers total depth: 3737.5 m

		Drillers depth	Loggers depth
Casing:	30"	151 m	
	20"	1002 m	
	13 3/8"	2397 m	2398 m
	9 5/8"	3342 m	3341 m

Hole size: 26", 17 1/2", 12 1/4", 8 1/2".

Mudtype: KCL polymer.

Run 1: Rm: 0.0977 Ohm m at 29° C
Rmf: 0.0584 Ohm m at 23° C
Rmc: 0.192 Ohm m at 23° C

Run 2: Rm: 0.164 Ohm m at 20° C
Rmf: 0.0959 Ohm m at 20° C
Rmc: 0.304 Ohm m at 20° C

3.3.4.2 Wireline Log Runs

Run No. 1A

Type: ISF-BHC-CNL-LDL-CAL-SP-GR
Interval: 3339.5 - 2398.0 m
Scale: 1:200, 1:500, 1:1000
Date: January 15, 1989
Time since circ: 8 1/4 hours
Comments: The SFLU read unreliably high values and ILM is used on the file instead.

Run No. 1A

Type: CBL-VDL-GR-CCL
Interval: 2398.0 - 950.0 (GR to surface)
Scale: 1:200
Date: January 15, 1989
Time since circ: 14 1/4 hours
BHT: 76.7°C
Comments: VDL recorded 4.2 m too deep as VDL curve cannot be memorized and played back on depth. GR affected by high KCL concentration in the mud.

Run No. 1A

Type: CST-GR
Interval: 6 samples recovered between 3340.0 and 3158 m.
Date: January 15, 1989
Time since circ: 21 3/4 hours
Comments: 2 bullets were lost, 1 empty and 12 misfired, why it was decided to POOH.

Run No. 1B

Type: CST-GR

Interval: 8 samples recovered between 3246.0 and 2464 m.

Date: January 15, 1989

Time since circ: 26 hours

Comments: 22 bullets were lost. Large overpull was often observed when retrieving the bullets, but it should be noted, that there is no consistency between amount of overpull and whether the bullet is lost or recovered.

Run No. 2B

Type: ISF-LSS-GR-SP

Interval: 3736.2 - 3341.0 m

Scale: 1:200, 1:500, 1:1000

Date: January 28, 1989

Time since circ: 7 1/4 hours

BHT: 112.8°C

Comments: Hole sticky at 3440 and 3380 m (main log). Repeat log OK.

Run No. 2A/2B

Type: LDL-CNL-NGT-CAL-GR

Interval: 3736.8 - 3341.0 m

Scale: 1:200, 1:500, 1:1000

Date: January 29, 1989

Time since circ: 8 3/4 hours

BHT: 114.6°C

Comments: Run without standoffs due to sticky hole (two previous misruns). Slightly sticky hole from 3560 m to casing shoe, very sticky from 3560 to 3500 m and tool stuck at 3367 and 3341 m (had to close caliper to get free).

Run No. 2A

Type: DLL-MSFL-GR
Interval: 3733.0 - 3341.0 m
Scale: 1:200, 1:500, 1:1000
Date: January 29, 1989
Time since circ: 14 1/2 hours
BHT: 119.0° C
Comments: Tool stuck at 3370.0 m (had to close caliper).

Run No. 2A

Type: RFT-AMS-GR
Interval: 5 Stations between 3627.5 and 3464.0 m
Date: January 29, 1989
Time since circ: 23 3/4 hours
BHT: 122.0° C
Comments: Tool failure after testing at 3627.5 m. Problems with power supply. POOH to change tool. Run without sample chambers.

Run No. 2B

Type: RFT-AMS-GR
Interval: 8 stations between 3719.5 and 3637.0 m
Date: January 30, 1989
Time since circ: 30 hours
BHT: 130.0° C
Comments: Some tool setting problems in dolomite, probably due to brittle material. Run without sample chambers.

Run No. 2A

Type: SHDT-GR
Interval: 3737.0 - 3341.0 m
Scale: 1:200
Date: January 30, 1989
Time since circ: 34 hours
BHT: 125.5° C
Comments: Sticky hole. Tension overpull at several depths.

Run No. 2B

Type: CBL-VDL-GR-CCL
Interval: 3341.0 - 1400.0 m
Scale: 1:200
Date: January 30, 1989
Time since circ: 38 1/2 hours
BHT: 125.5° C (from SHDT-run)

Run No. 2A

Type: VSP
Interval: TD to surface
Date: January 30-31, 1989
Comments: Above 3450 m the transit times are uncertain due to lack of cement and hence poor coupling to formation.

Run No. 2C

Type: CST-GR
Interval: 22 samples were recovered between 3736.0 and 3350.0 m.
Date: January 31, 1989
Time since circ: 71 1/4 hours
BHT: 126.7° C
Comments: 1 bullet lost and 7 empty.

3.3.4.3 Computer Processed Logs

On the basis of wireline logs, the following computer processed logs have been produced:

Cyberdip	3335 - 3738 m
ELAN (petrophysical evaluation)	3340 - 3730 m

3.4 CORES AND FIELD CORE DESCRIPTIONS

It was planned to cut a core in the top of the primary objective i.e. Upper Jurassic sandstone provided reservoir development, irrespective of the presence of hydrocarbons. Additional cores should be taken if hydrocarbon shows were encountered. Furthermore it was planned to cut a core in the Lower Cretaceous section if a reservoir unit was encountered.

The following 4 cores were cut:

Core No.	Interval (m.B.KB)	Recovery	stratigraphy
1	3331 - 3331.32	(reverse circulating junk basket)	Sola Fm.
2	3422 - 3429	67%	Farsund Fm.
3	3429 - 3436	87%	Farsund Fm.
4	3562 - 3573	70%	Triassic/ Zechstein.

The cores (except core No. 1 cut with junk basket) were recovered in aluminum barrels, cut in 1 m pieces and forwarded to the Geological Survey of Denmark (DGU).

danop

FIELD CORE LOG

COMMENTS: Core recovered by Reverse circulating Junk Basket.

WELL: 5503/3-2 CORE NO.: 1 CORED INTERVAL: 3321-3031.32mb.KB

REC.: 100 % DATE: 14 Jan 1989 INITIALS: C.D./P.H.D

mb TOP CORE	LITHOLOGY	GRAIN SIZE (CLAY, SILT, SAND: vf, f, m, c, vc)	SED. STRUCTURE	POROSITY (POOR, FAIR, GOOD)	SHOWS (CUT, FLUOR)	DRILL. RATE M/HR	DESCRIPTION
10					NONE	27	CLAYSTONE MODERATE BROWN TO DARK REDBROWN, HARD CALCAREOUS, ABUNDANT SHELLFRAGMENTS, ABUNDANT PATCHES OF GREENISH GREY TO DARK GREENISH GREY COLOUR PROBABLY DUE TO REDUCTION, PRED. HORIZONTALLY LAMINATED (DIP < 5°), LOCALLY RIPPLE CROSS LAMINATION (CURRENT & WAVE) LOCALLY APPARENT LIQUEFACTION STRUCTURES, LOCALLY BIOTURBATED, LOCALLY SMALL CALCITE NODULES (1-3 mm).
20							
30							
40							

danop

. IELD CORE LOG

WELL: 5503/3-2 CORE NO.: 2 CORED INTERVAL: 3422-3429 mb.KB

REC.: 3422-3426.7m (67%) DATE: 22. jan. 1989 INITIALS: P.H.D / R.R

mb TOP CORE	LITHOLOGY	GRAIN SIZE (CLAY, SILT, SAND: vf, f, m, c, vc)	SED. STRUCTURE	POROSITY (POOR, FAIR, GOOD)	SHOWS (CUT, FLUOR)	DRILL. RATE M/HR	DESCRIPTION
3423					NONE	1.7	SCALE : DARK GREY - GREY BLACK - BLACK, MOD. HARD - HARD, FISSILE, NON - LOCALLY SLIGHTLY CALCAREOUS, MICROMICACEOUS
3424							← = CORE CHIPS
3425							
3426							
							3426.7m
3428							
3429							

FIELD CORE LOG

WELL: 5503/3-2 CORE NO.: 3 CORED INTERVAL: 3429-3436 mb.KB

REC.: 3429-3435 m (87%) DATE: 23 jan 1989 INITIALS: P.H.O./N.E./R.R.

mb TOP CORE	LITHOLOGY	GRAIN SIZE (CLAY, SILT, SAND: vf, f, m, c, vc)	SED. STRUCTURE	POROSITY (POOR, FAIR, GOOD)	SHOWS (CUT, FLUOR)	DRILL. RATE M/HR	DESCRIPTION
3430							SH/CLYST: GREY BLACK - BROWN BLACK, HARD - VERY HARD, SPLINTERY - FISSILE, NON - LOCALLY CALCAREOUS, MICRO-MICACEOUS, TRACE CARBONACEOUS / MICROLIGNITIC, TRACE PYRITE, LOCALLY PYRITIZED BURROWS, LOCALLY SLICKENSIDES
3431							
3432							
3433							SHALE: a/a WITH ABUNDANT PYRITE.
3434							
3435							3435 m
							Not recovered
3436							3436 m

← = CORE CHIPS

danop

FIELD CORE LOG

WELL: 5503/3-2 CORE NO.: 4 CORED INTERVAL: 3562-3573 mb. KB

REC.: 3562-3569.7 (70%) DATE: 25.1.89 INITIALS: N.F./R.R.

mb TOP CORE	LITHOLOGY	GRAIN SIZE (CLAY / SILT / SAND: VF, F, M, C, VC)	SED. STRUCTURE	POROSITY (POOR / FAIR / GOOD)	SHOWS (CUT / FLUOR)	DRILL. RATE M/HR	DESCRIPTION
35 62							← SST INTLAM SLST SST: PALE RED, HARD, VF, WELL CMT, TR MICA, NO VIS POR. SLST: GREY RED, HD-V HD, WELL CMT, V ARG.
63			// 60-70°				← SST: GREY RED, W/PALE RED LAM, HARD, VF, GRDG SLT IN LAM, WELL CMT, SIL CMT, TR CALC CMT, NO VIS POR. LAM DIPPING 60-70°.
35 64			Slumping				← SST W/CLYST CLASTS SST: LT BRN GY, VF-F, GEN F, HD, WELL CMT, DOL CMT, TR CARB, TR MICA, NO VIS POR CLYST: LT BRN SH, HD, BLKY, TR MICRONIC, DOL, LOC SLTY. SLUMPING STRUCTURES.
35 65							← SST: LT BRN GY, VF GRDG SLT, V HD, WELL CMT, DOL CMT, MICRONIC, ARG LAM, NO VIS POR.
35 66		Micro-faults	// 30-60°				← SLST W/ SST LAM SLST: LT OLV GY, HD, WELL CMT, DOL CMT, MICRONIC, SST: LT OLV GY, VF-F, WELL CMT (DOL), MICRONIC, NO VIS POR. MICROFAULTED SST LAM.
67			// ~60°				← SLST W/ SST LAM AS 3566 m LOC W/ V MICR LAM, ABN DOL CMT
35 68		Micro-faults	// ~60°				← SLST INTLAM W/ SST, RARE W/ WELL DOL CMT.
35 69							← SLST: LT OLV GY, V HD, V DOL CMT GRDG DOL IN LAM
		Very fractured					← 3569.7 m. DOL: LT OLV SH, V HARD, ARG, CRYPTOXLN AND DOL: LT BRN GY, V HARD, BLKY, MICROXLN TO GRANULAR.
35 70							← CORE CHIPS
71							NOTE! ABN FRACTURES IN CORE.
35 72							NOTE! MOST LIKELY TOP OF CORE (~3M) WASHED OUT, I.E. 3562 m IN CORE CORRESPONDS TO 3565 m IN HOLE.

3.5 SIDEWALL CORES AND DESCRIPTION

It was planned to do one run in 12 1/4" hole section, but due to several misfires in the first run an extra run was done. The cores were shot in the Rotliegendes to Lower Tertiary section and they were shot to gather further information on age of the formation and to confirm lithologies interpreted from the cuttings.

Run No. 1A

2 lost, 1 empty, 12 misfires, 9 not fired, 6 recovered:
3340 - 3337 - 3334 - 3328 - 3313 - 3158

Run No. 1B

22 lost, 8 recovered:
3246 - 3158 - 3148 - 2950 - 2882 - 2779 - 2746 - 2464

Run No. 2C

1 lost, 7 empty, 22 recovered:
3736 - 3730 - 3722 - 3720 - 3652 - 3644 - 3578 - 3576-
3531.5 - 3528 - 3512.5 - 3505 - 3492.5 - 3480 - 3458-
3445 - 3425 - 3410 - 3405 - 3400 - 3360 - 3350.

IDE WALL CORE DESCRIPTION

ELL: 5503/3-2 RUN NO: 1A

NO OF SHOTS: 30

SAMPLE INTERVAL: 3340-2580 DATE: 15.01.89 SIGN: C.DONS

NO. OF SAMPLES REC.: 6 NO. OF LOST BULLETS: 24

danop

SAMPLE NO.	DEPTH mb KB	RECOVERY cm	SHOWS CUT	SHOWS FLUOR	DESCRIPTION (LITHOLOGY, GR. SIZE, STRUCTURE, POROSITY, CONTAM.)
1	3340	1.5	—	—	CLAYSTONE: MEDIUM DARK-DARK GREY, MODERATE HARD, VERY CALCAREOUS GRADING MARL, BLOCKY, TRACE BLACK SPECKLES, RARE MICRO PYRITE, SLIGHTLY LAMINATED. TRACE RUST CONTAMINATION.
2	3337	1.5	—	—	CLAYSTONE: DARK GREY - OLIVE BLACK, MODERATELY HARD, MODERATELY CALCAREOUS, BLOCKY, RARE BLACK MICRO SPECKLES, NO LAMINATION.
3	3334	1.0	—	—	CLAYSTONE: LIGHT GREY - LIGHT OLIVE GREY, MODERATELY HARD, VERY CALCAREOUS GRADING MARL, BLOCKY, NO LAMINATION.
4	3331	LOST			
5	3328	1.0	—	—	CLAYSTONE: PALE RED BROWN - MODERATE RED BROWN (BRICK RED), HARD, BLOCKY, VERY CALCAREOUS GRADING MARL, RECRYSTALLIZED SILTY TEXTURE. SAME FORMATION AS CORE #1.
6	3318	EMPTY			
7	3313	2.0	—	—	LIMESTONE: VERY LIGHT GREY - OFF WHITE, HARD, BLOCKY, RECRYSTALLIZED, SLIGHTLY ARGILLACEOUS, TRACE GREEN GLAUCONITE SPECKLES. SIMILAR TO 3158 M.
8	3273.5	LOST			
9	3246	MISFIRE			
10	3158	1.5	—	—	LIMESTONE: LIGHT BROWNISH GREY (WET) - LIGHT GREY (DRY), HARD, BLOCKY, ARGILLACEOUS., RECRYSTALLIZED.
11	3148	MISFIRE			
12	2987	MISFIRE			
13	2981	MISFIRE			
14	2950	MISFIRE			

ECOVERY: INDICATE MISFIRE, NO RECOVERY, LOST BULLET OR LENGTH OF SAMPLE (cm)

SIDE WALL CORE DESCRIPTION

WELL: 5503/3-2 RUN NO: 1A SAMPLE INTERVAL: 3340-2580 DATE: 15.01.89 SIGN: C. Dons

NO OF SHOTS: 30 NO. OF SAMPLES REC.: 6 NO. OF LOST BULLETS: 24

danop

SAMPLE NO.	DEPTH mb KB	RECOVERY	SHOWS CUT	SHOWS FLUOR	DESCRIPTION (LITHOLOGY, GR. SIZE, STRUCTURE, POROSITY, CONTAM.)
15	2943	MISFIRE			
16	2935	MISFIRE			
17	2925	MISFIRE			
18	2882	MISFIRE			
19	2876	MISFIRE			
20	2867.5	MISFIRE			
21	2845	MISFIRE			
22	2811.5	MISFIRE			
23	2807	MISFIRE			
24	2790	MISFIRE			
25	2779	MISFIRE			
26	2751	MISFIRE			
27	2746	MISFIRE			
28	2642	MISFIRE			
29	2629	MISFIRE			
30	2580	MISFIRE			

RECOVERY: INDICATE MISFIRE, NO RECOVERY, LOST BULLET OR LENGTH OF SAMPLE (cm)

SIDE WALL CORE DESCRIPTION

WELL: 5503/3-2 RUN NO: 1B

NO OF SHOTS: 30

SAMPLE INTERVAL: 3246-2425 DATE: 15.01.89 SIGN: C. DONS

NO. OF SAMPLES REC.: 8 NO. OF LOST BULLETS: 22

danop

SAMPLE NO.	DEPTH mb KB	RECOVERY	SHOWS CUT	SHOWS FLUOR	DESCRIPTION (LITHOLOGY, GR. SIZE, STRUCTURE, POROSITY, CONTAM.)
46	2807	LOST			
47	2790	LOST			
48	2779	1.0	—	—	CLAYSTONE: OLV BLACK, MODERATE HARD, VERY FINE < 0.2 mm LAMINATIONS WITH MEDIUM DARK GREY-DARK GREENGREY CLAYSTONE, TRACE CARBONACEOUS - ARGILLACEOUS LAMINAL LININGS, NON CALCAREOUS.
49	2758	LOST			
50	2751	LOST			
51	2746	6.0	—	—	CLAYSTONE: MODERATE BROWN, FIRM - MODERATE HARD, SLIGHTLY LAMINATED, NON CALCAREOUS, SLIGHTLY MICROMICACEOUS, TRACE VERY FINE SILTY LAMINATIONS < 0.2 mm.
52	2730	LOST			
53	2675	LOST			
54	2642	LOST			
55	2629	LOST			
56	2593	LOST			
57	2580	LOST			
58	2557	LOST			
59	2464	5.0	—	—	CLAYSTONE: OLIVE BLACK, MODERATE HARD, BLOCKY, NON CALCAREOUS, MICROMICACEOUS,
60	2425	LOST			TRACE SILTY, RARE GREEN BLACK GLAUCONITE.

RECOVERY: INDICATE MISFIRE, NO RECOVERY, LOST BULLET OR LENGTH OF SAMPLE (cm)

SIDE WALL CORE DESCRIPTION

WELL: 5503/3-2 RUN NO: 1B

NO OF SHOTS: 30

SAMPLE INTERVAL: 3246-2425 DATE: 15.01.89 SIGN: C. DONS

NO. OF SAMPLES REC.: 8 NO. OF LOST BULLETS: 22

danop

SAMPLE NO.	DEPTH mb KB	RECOVERY cm	SHOWS CUT	SHOWS FLUOR	DESCRIPTION (LITHOLOGY, GR. SIZE, STRUCTURE, POROSITY, CONTAM.)
31	3246	1.0	—	—	LIMESTONE: LIGHT GREY - MEDIUM LIGHT GREY, HARD, BLOCKY, SLIGHTLY ARGILLACEOUS SLIGHTLY RECRYSTALLIZED, TRACE ARGILLACEOUS LAMINATIONS.
32	3158	2.0	—	—	LIMESTONE: LIGHT GREY - LIGHT BROWN GREY, HARD, BLOCKY, MICROCRYSTALLINE, DARK YELLOW BROWN - BROWN BLACK LAMINATIONS OR MICROSTYLOLITE LININGS, TRACE CALCAREOUS VEINS
33	3148	1.5	—	—	CHALK: OFF WHITE - VERY LIGHT GREY, HARD, CRYPTOCRYSTALLINE - CHALKY, NON ARGILLACEOUS - TRACE ARGILLACEOUS, HOMOGENEOUS
34	2987	LOST			
35	2981	LOST			
36	2950	1.5	—	—	CHALK: OFF WHITE - VERY LIGHT PINKISH GREY, HARD, CHALKY - CRYPTOCRYSTALLINE, NON - TRACE ARGILLACEOUS, HOMOGENEOUS, TRACE BLACK MICROSTYLOLITE OR MICROLAMINATIONS < 0.2 mm.
37	2943	LOST			
38	2935	LOST			
39	2925	LOST			
40	2882	2.0	—	—	CHALK: OFF WHITE - VERY LIGHT GREY, HARD, CHALKY - CRYPTOCRYSTALLINE, NON - TRACE ARGILLACEOUS, HOMOGENEOUS
41	2876	LOST			
42	2867.5	LOST			
43	2845	LOST			
44	2838.5	LOST			
45	2811.5	LOST			

RECOVERY: INDICATE MISFIRE, NO RECOVERY, LOST BULLET OR LENGTH OF SAMPLE (cm)

SIDE WALL CORE DESCRIPTION

WELL: TORDENSKJOLD-1 RUN NO: 2C
550313-2 NO OF SHOTS: 30

SAMPLE INTERVAL: 3736 - 3350m DATE: 31.1.89 SIGN: RR
NO. OF SAMPLES REC.: 22 NO. OF LOST BULLETS: 1

SAMPLE NO.	DEPTH mb KB	RECOVERY cm	SHOWS CUT	SHOWS FLUOR	DESCRIPTION (LITHOLOGY, GR. SIZE, STRUCTURE, POROSITY, CONTAM.)
61	3736	0.5	–	–	VOLCANIC (?) ROCK : LIGHT GREY - LIGHT GREEN GREY WITH BLACK AND RED BROWN SPECS, FIRM, „SILTY" TEXTURE, VERY ARGILLACEOUS, SLIGHTLY CALCAREOUS, BLACK AND DARK RED BROWN MINERALS
62	3730	1.0	–	–	VOLCANIC (?) ROCK : DARK GREEN GREY, MOD HARD - HARD, SILTY - SANDY TEXTURE, NON CALCAREOUS, ABN. CHLORITE (?), BLACK MICROSPECS, FRACTURE (?) W/ CALC & CHLORITE
63	3722	1.5	–	–	VOLCANIC (?) ROCK : SPECKLED: DARK GREEN GREY - LIGHT GREEN GREY AND LIGHT BROWN GREY - DARK RED BROWN WITH WHITE (CALC) IRREGULAR BANDS / SPECS, ARGILLACEOUS, SLIGHTLY CALCAREOUS, LOC. ABUNDANT CHLORITE (?)
64	3720	1.0	–	–	VOLCANIC (?) ROCK : SPECKLED AS NO. 63, MOD. HARD - HARD, SILTY TEXTURE, TRACE FELDSPAR ? (PALE REDDISH BROWN), BLACK MICROSPECS.
65	3652	0.5	–	–	ANHYDRITE : WHITE - VERY LIGHT GREY WITH THIN PINKISH GREY - LIGHT OLIVE GREY BDL. LAM, HARD - VERY HARD, BRITTLE, MICROCRYST. - GRANULAR, NO VIS POR.
66	3644	1.0	–	–	DOLOSTONE : YELLOWISH GREY - PINKISH GREY, HARD - VERY HARD, BRITTLE, ARGILLACEOUS, GRANULAR, NO VIS POR.
67	3578	1.0	–	–	SILTSTONE / GYPSUM (?) / SANDSTONE = MICROBRECCIA (?) : IRREGULAR PARTS OF
					SILTSTONE : DARK RED BROWN - GREY RED, FIRM, DOLOMITIC, VERY ARGILLACEOUS, ~~SLIGHTLY SANDY~~ BEARING SHALE.
					SANDSTONE: LIGHT OLIVE GREY, VERY FINE - SILT, VERY HARD, WELL CEMENTED (DOL) NO VIS POR
					GYPSUM (?) : WHITE - CLEAR, BRITTLE, COARSE CRYSTALS („FIBROUS" TEXT.)
68	3576	0.5	–	–	ANHYDRITE : WHITE, FIRM - MOD. HARD, AMORPH - GRANULAR, MICRO-FRACTURES.
69	3551	–			EMPTY
70	3542.5	–			EMPTY
71	3522.5	–			EMPTY

RECOVERY: INDICATE MISFIRE, NO RECOVERY, LOST BULLET OR LENGTH OF SAMPLE (cm)

SIDE WALL CORE DESCRIPTION

WELL: TORDENSKJOLD-1 RUN NO: 2C
5503/3-2 NO OF SHOTS: 30

SAMPLE INTERVAL: 3736 - 3350m DATE: 31.1.89 SIGN: RR
NO. OF SAMPLES REC.: 22 NO. OF LOST BULLETS: 1

SAMPLE NO.	DEPTH mb KB	RECOVERY cm	SHOWS CUT	SHOWS FLUOR	DESCRIPTION (LITHOLOGY, GR. SIZE, STRUCTURE, POROSITY, CONTAM.)
72	3531.5	1	-	-	MICROBRECCIA (?) OR REWORKED SEDIMENT (?) : LENSES / IREGULAR PARTS OF SHALE AND SANDSTONE W/ ABUNDANT DISSEMINATED PYRITE
					SHALE: GREEN GREY - OLIVE GREY, HARD, NON CALC AND
					SHALE: BROWN BLACK - BROWN GREY, HARD, MICROMIC AS LAM/STREAKS
					SANDSTONE: LIGHT GREY - LIGHT BROWN GREY, VERY FINE - SILT, HARD, WELL SORTED, MICROPYRITE, NO VIS POR.
73	3528	1	-	-	MICROBRECCIA (?): AS NO. 72 W/ ABUNDANT QUARTZ GRAINS, CHLORITE (?) AND PYRITE (SILTY - VERY FINE SANDY TEXTURE)
74	3523	-			EMPTY
75	3512.5	0.5	-	-	SHALE : BROWN BLACK - OLIVE BLACK, HARD - VERY HARD, CALCAREOUS, MICROPYRITIC, TRACE SANDY, TRACE CHLORITE.
76	3505	1	-	-	SHALE / CLAYSTONE : OLIVE GREY, FIRM - MOD. HARD, SLIGHTLY CALCAREOUS, MICROMICACEOUS, SILTY - VERY FINE SANDY, TR MICACEOUS, MICROPYR.
77	3492.5	0.5	-	-	MICROBRECCIA (?) : ABUNDANT OLIVE GREY - OLIVE BLACK ARGILLACEOUS WITH ABN. WHITE SPECS AND COARSE CALCITE CRYSTALS, ABN MICROPYRITE, LOC SILTY - VERY FINE SANDY
78	3480	2	-	-	SHALE / CLAYSTONE : OLIVE GREY, FIRM - MOD. HARD, BLOCKY - FISSILE, SLIGHTLY CALCAREOUS, SILTY, MICROMICACEOUS.
79	3475	-			LOST
80	3458	3	-	-	SHALE / CLAYSTONE : OLIVE GREY - OLIVE BLACK, MOD. HARD, BLOCKY - FISSILE, CALCAREOUS, MICROMICACEOUS.
81	3445	1	-	-	SHALE / CLAYSTONE : OLIVE GREY - OLIVE BLACK, MOD. HARD - HARD, BLOCKY - FISSILE, SLIGHTLY CALCAREOUS, MICROMICACEOUS SILTY - LOC V FINE SANDY
82	3425	1.5	-	-	SHALE / CLAYSTONE : OLIVE GREY - OLIVE BLACK, A/A BUT VERY CALCAREOUS, SILTY
83	3410	2.5	-	-	SHALE / CLAYSTONE : OLIVE GREY - OLIVE BLACK, FIRM - MOD. HARD, BLOCKY, MICROMICACEOUS, CALCAREOUS, SILTY.

RECOVERY: INDICATE MISFIRE, NO RECOVERY, LOST BULLET OR LENGTH OF SAMPLE (cm)

SIDE WALL CORE DESCRIPTION

WELL: TORDENSKJOLD-1 RUN NO: 2C

550313-2 NO OF SHOTS: 30

SAMPLE INTERVAL: 3736-3350m DATE: 31.1.89 SIGN: RR

NO. OF SAMPLES REC.: 22 NO. OF LOST BULLETS: 1

SAMPLE NO.	DEPTH mb KB	RECOVERY cm	SHOWS CUT	SHOWS FLUOR	DESCRIPTION (LITHOLOGY, GR. SIZE, STRUCTURE, POROSITY, CONTAM.)
84	3405	2.5	-	-	CLAYSTONE/SHALE: OLIVE GREY - OLIVE BLACK, MOD. HARD, BLOCKY, SLIGHTLY CALCAREOUS, MICROMICACEOUS, SILTY.
85	3404	-			EMPTY
86	3400	1.0	-	-	CLAYSTONE/SHALE: OLIVE GREY, FIRM - MOD. HARD, BLOCKY, VERY CALCAREOUS GRADING MARL.
87	3382.5	-			EMPTY
88	3370	tr			CLYSTONE: DARK RED BROWN - GREY RED, HARD, CALCAREOUS, MICROMICACEOUS.
89	3360	0.5	-	-	CLAYSTONE: DARK RED BROWN - GREY RED, HARD, BLOCKY, SLIGHTLY CALCAREOUS, TRACE LIGHT GREEN MIN (CHLORITE), TRACE ROUNDED QTZ GRAINS.
90	3350	1.5	-	-	CLAYSTONE: LIGHT OLIVE GREY - LIGHT GREEN GREY, MOD. HARD, CALCAREOUS, TRACE SANDY

COVERY: INDICATE MISFIRE, NO RECOVERY, LOST BULLET OR LENGTH OF SAMPLE (cm)

3.6 WELL SERVICES

Biostratigraphy

The Robertson Research Group performed the biostratigraphic work on the Tordenskjold-1 well. The following analyses were performed:

Lithology:	227 ditch cuttings, 37 sidewall cores and 6 core samples from the entire section.
Micropalaeontoloty:	200 ditch cuttings, 8 sidewall cores and 1 core sample from the interval 160m-3711m.
Palynology:	34 ditch cuttings, 12 sidewall cores and 3 core samples from the interval 3267m-3793mTD.
Nannofossils:	11 ditch cuttings and 4 sidewall core samples from the interval 3258m-3410m.
'Hot shots':	2 batches of samples from this well were analysed as 'hot shots'.

A separate report has been distributed to partners.

Core Analysis

The Core Analysis Laboratory of the Geological Survey of Denmark performed the conventional core analysis programme on the Tordenskjold-1 cores. The programme includes preparation of a spectral core gamma log, conventional core analysis, slabbing, core photography, and mounting of a thin cut. Beside this the Geological Survey performed a sedimentological description and environmental interpretation of the cores.

A separate report has been distributed to partners.

Geochemistry

The Geological Survey of Denmark evaluated the source rock potential of the interval 3300 - 3739 m by means of vitrinite reflectance measurements, TC/TOC determinations, Rock-Eval pyrolysis. Analysis of light gases (headspace and adsorbed gas) was performed and carbon-13 isotopic analysis was performed on the methane.

A separate report has been distributed to partners.

3.7 FORMATION EVALUATION

From suite 1: DIL/BHC/LDL/CNL/GR/CAL/SP.
from 2398m to 3339m (GR to surface)

From suite 2: ISF/LSS/GR/SP + LDL/CNL/NGL/CAL
+ DLL/MSFL/GR.
from 3341m to 3737m.

Refer to tabel 1.1: "Pertinent Well Data" for further details.

The log quality was good. Some sticking occurred during suite 2. Sidewall cores in the 12 1/4 " section were largely unsuccessfull and a motor burnout occurred during the RFT run in the 8 1/2 " hole.

No hydrocarbon indications were found during drilling of the 12 1/4" section except for a weak flouresence in the top of the chalk most probably related to mineral flouresence.

Since logs only gave weak apparent hydrocarbon indication in the uppermost chalk no CPI was performed over this section.

Logging suite 2 in the 8 1/2" section was analysed throughout using the Schlumberger "ELAN" interpretation routine. It confirmed the presence of hydrocarbons in Zechstein Carbonates.

Zonation of the logs in the 12 1/4" hole section, suite 1, can be made as follows:

- Casing shoe (2398m)to 2878m: Mainly shale with no reservoirs.

- 2878m Top chalk.

- 2878m to 2933m Danian chalk (Ekofisk formation) is a uniform chalk which can be subdivided into two zones where the upper zone (2878m to 2900m) has an average porosity of 20% whereas the lower zone (2900m to 2933m) has an average porosity of 12%.

- 2933m to 3264m Upper Cretaceous chalk (Tor and Hod formations). Tor is less uniform with porosity varying between 6% and 20% whereas Hod exibits a constant porosity of 8%.

For determination of the formation water resistivity it is assumed that the Upper Cretaceous chalk is water-bearing. This assumption results in a range of Rw of 0.035 Ohm m to 0.045 Ohm m at formation temperature 88°C corresponding to a salinity of 50,000 - 60,000 ppm NaCl. The magnitude of Rw is supported by a positive deflection of the SP curve in the chalk, relative to the shale base line above. Regional values of Rw from this interval correlate well with this range.

For determination of water saturations in the Danian chalk the same Rw is assumed as well as values for a=1 , m=2 & n=2. This assumption results in an aparent water saturation in the uppermost chalk zone of some 80%.
Since only a highly questionable show was detected in this section, and as no closure is seen at top chalk level, this apparent saturation must be referred to an unknown parameter variation of a, m & n between the chalk members. Such variations are known to exist in the Danish North Sea chalk.

The 8 1/2" hole section logs can be zoned as follows:

- Down to 3355m is the reminder of the Rødby formation. It is a claystone/marl with no reservoir.

- 3355m to 3388m is a shaly interval: The Sola formation exhibiting no reservoir.

- 3388m to 3403m is a Lower Cretaceous limestone interval, the Tuxen formation. The porosity is as low as 1% to 4% in this interval and is fully water saturated.

- 3403m top of Jurassic.

- 3403m to 3532m is a claystone interval with no signs of reservoir. Sand recovered at the shale-shaker while drilling was most probably related to thin (less than 15cm) sand stringers. Near the bottom of this interval (3525m to 3528m) considerable amounts of carbonaceous material is identified by the logs together with some limited hydrocarbon saturation Sw ave = 80%.

- 3532m to 3574m is a very shaly siltstone interval exibiting no reservoir and the logs did not show any sign of the gas detected during drilling of this section.

- 3574m top of Zechstein Evaporites (Permian)

- 3574m to 3712m is an interbedded section of dolomite and anhydrite. The dolomite sections are of very low porosity: 3% to 9%. The interval 3635m to 3648m exibits considerable oil saturation Sw = 20%

corresponding to oil and gas shows during drilling of this section.

- 3712m to TD 3739m is a shaly sandstone (Rotliegendes) with very low porosity: 1% to 5% , no reservoir present. The formation changes into heavy volcanics around 3720m after which the well was terminated at 3739m.

3.8 DITCH SAMPLE DESCRIPTIONS

The cuttings were sampled at 10 m intervals from 151 to 2600 m, at 5 m intervals from 2600 to 3150 m and at 3 m intervals from 3150 m to T.D. The presence of sand in the interval 3413-3422 m observed while drilling, cannot be confirmed on the wireline logs.

Depth (m b. KB)	Lithology
151-160 m	SAND, quartz, loose grains, clear, colorless, fine grained, well sorted, rounded; bivalve shells, trace.
160-190 m	SAND, quartz, loose grains, clear, colorless, fine to coarse grained, poorly sorted, subrounded to rounded; shells, moderate; lignite, argillaceous, rare; rare finely interlaminated CLAYSTONE, brownish black, and SILTSTONE, light grey, slightly calcareous.
190-260 m	SAND, quartz, loose grains, clear, colorless, fine to coarse grained, subangular to rounded; wood fragments, light brown, moderate to abundant; lignite, rare; at 200 m minor rock fragments (?), black, hard, fine to very coarse grains; CLAYSTONE/SILTSTONE a/a, rare, disappears at 230 m; shell fragments, rare to moderate; mica, rare; pyrite, glauconite and foraminifera, trace.
260-317 m	SAND, quartz, loose grains, clear, colorless, fine to coarse grained

becoming finer downwards, poorly sorted, subangular to rounded; wood fragments a/a, rare to minor; lignite, rare to minor; shell fragments, mica and glauconite, trace to rare; pyrite, trace.

317-611 m — CLAY, medium dark grey, soft, sandy (very fine to fine grains) grading to silty downwards, varyingly calcareous; mica and shell fragments, trace to rare; microlignite, trace from 440 m; glauconite, trace to locally rare; pyrite, trace.

611-1020 m — CLAY, dark grey, soft, silty getting slightly sandy below 860 m, varyingly calcareous; mica, trace to rare; shell fragments, locally trace; microlignite, trace; foraminifera, trace below 920 m; glauconite, locally trace; pyrite, trace.

1020-1110 m — CLAYSTONE, medium dark grey, soft, sticky, silty, locally trace of sand, slightly to very calcareous, micaceous, microlignitic; glauconite, minor to absent.

1110-1471 m — CLAYSTONE, medium dark grey, soft to firm, locally sticky, slightly fissile, slightly silty and locally sandy and silty, very- to non-calcareous, micaceous, microlignitic; glauconite, shell fragments and microfossils, trace; pyrite, dis-

persed, as aggregates and replaced tracefossils, trace; locally laminated.

At 1408-1410 m and 1439 m stringers of CLAYSTONE, light olive grey, hard, slightly silty, very calcareous, micaceous; pyrite, trace.

1471-1801 m — CLAYSTONE w/ stringers of MARLSTONE and LIMESTONE and stringers rich in PHOSPHORITE.

CLAYSTONE, dusky brown to dusky yellowish brown, firm to moderately hard, slightly to non-calcareous, micaceous; microlignite, rare to trace; pyrite, minor to trace; shell fragments, trace to absent.

Stringers of CLAYSTONE, light olive grey, hard, very calcareous, downwards grading to MARLSTONE and argillaceous LIMESTONE.

1670-1700 m several stringers rich in PHOSPHORITE (?), dusky green, hard to very hard, aggregates and grains.

1801-2160 m — CLAYSTONE w/ stringers of LIMESTONE and DOLOMITE.

CLAYSTONE, dusky yellowish brown to dark grey, soft to moderately hard, non- to varyingly calcareous, mica-

ceous, slightly microlignitic, slightly glauconitic; pyrite, dispersed, as laminae and as aggregates, minor to trace; shell fragments, trace.

Abundant stringers of LIMESTONE (micrite with laminae of sparite), light olive grey, firm to hard, varyingly argillaceous, locally microlignitic.

Below 1900 m also stringers of DOLOMITE (occasionally LIMESTONE), brownish grey to dark yellowish brown, hard, varyingly argillaceous.

Locally stringers rich in PHOSPHORITE, dusky green, hard, aggregates and single grains.

2160-2414 m

CLAYSTONE w/ stringers of LIMESTONE and DOLOMITE.

CLAYSTONE, olive black, firm to moderately hard, slightly to non-calcareous, micaceous, slightly microlignitic; pyrite, dispersed and as aggregates, trace to minor; glauconite, trace; phosporite, locally trace.

LIMESTONE (micrite), light olive grey, moderately hard, argillaceous, locally microlignitic.

DOLOMITE, brownish grey, moderately to very hard, argillaceous.

2414-2460 m — CLAYSTONE, dark grey, soft to firm, rare microlignite; mica and pyrite, trace; sand, loose grains, clear, very fine to medium grained, moderately sorted, rounded, trace; trace of SILTSTONE, moderate yellowish brown, moderately hard. Strong cement contamination down to 2450 m.

2460-2700 m — CLAYSTONE, medium dark grey, firm; microlignite, rare. Interbedded in varying proportions with CLAYSTONE, brownish grey, soft to firm, blocky to slightly fissile; microlignite, trace; and CLAYSTONE, dark greenish grey, firm to moderately hard; microlignite and glauconite, trace. Scattered stringers and a few beds of LIMESTONE, dark to pale yellowish brown, friable, partly argillaceous, occasionally microlignitic. Below 2630 m varying amount of SAND, loose grains, clear, very fine to medium grained, moderately sorted, subrounded getting more angular downwards.

2700-2757 m — CLAYSTONE, brownish grey a/a interbedded with CLAYSTONE, pale brown, soft.

From 2715 to 2740 m beds with SAND a/a, but more angular. Below 2735 m beds of CLAYSTONE, varyingly reddish brown and grey, and a bed of pinkish grey LIMESTONE.

2757-2860 m — CLAYSTONE, light olive to medium dark grey, soft to moderately hard, slightly fissile, slightly silty, moderately glauconitic; mica, trace to rare; microlignite, trace; pyrite, trace to rare; weakly laminated.

Beds of LIMESTONE, grayish orange pink or yellowish brown, soft to hard, argillaceous.

Below 2815 m the CLAYSTONE variate between reddish, grayish, greenish, bluegreen and brownish colours turning black downwards, slightly calcareous towards base.

2860-2880 m — CLAYSTONE, greenish grey to medium grey, firm to moderately hard, very calcareous; pyrite, trace; downwards grading into a very light grey MARLSTONE.

2880-2940 m — CHALK (micrite), white, firm; pyrite, trace; CLAYSTONE grayish green and reddish brown, minor decreasing to trace towards 2925 m; chert, light brown, yellowish grey, semitranslucent, minor decreasing to

trace towards 2910 m, slight increase towards 2935 m.

Beds of pinkish grey, hard LIMESTONE (indurated chalk).

2940-3136 m — CHALK (micrite), white to pinkish grey, firm to hard; pyrite, trace; locally trace of SAND grains and trace of sparry calcite.

3136-3198 m — CHALK (micrite), white to very light grey, firm to moderately hard; locally trace of pyrite; silt grains, trace to rare; sparry calcite, locally trace.

From 3150 to 3170 m beds of pale red to moderate orange pink CHALK, slightly argillaceous. Also minor light grey to medium light grey CHALK, slightly argillaceous. Below 3170 m interbedding of whitish CHALK a/a and grey CHALK a/a.

3198-3213 m — CHALK (micrite), white to very light grey, minor light to medium light grey, getting pale red to pinkish grey downwards, firm to moderately hard, slightly argillaceous; rare silt to very fine sand grains; microfossils, trace.

3213-3228 m — CHALK downwards (below 3219 m) grading to LIMESTONE.

CHALK (micrite), very light grey to medium light grey, moderately hard, occasionally hard, brittle, locally dolomitized, slightly argillaceous; glauconite and microfossils, trace.

LIMESTONE, very light grey to medium light grey, hard, locally very hard, locally silicified, locally recrystallized, slightly argillaceous; glauconite and microfossils, trace; locally trace to 10% chert, light olive grey, very hard, locally translucent.

3228-3260 m

LIMESTONE with CHERT.

LIMESTONE, yellowish grey to pinkish grey to light olive grey, hard, locally very hard, generally very recrystallized, locally silicified grading to cherty limestone, slightly to non-argillaceous; glauconite, trace; below 3252 m pyrite, trace.

Trace to 20% CHERT, light olive grey to light brownish grey, very hard, brittle, locally translucent.

3260-3305 m

LIMESTONE locally with MARL/MARLY LIMESTONE.

LIMESTONE, yellowish grey to light olive grey, soft to hard, locally very hard, varyingly recrystallized, slightly argillaceous; glauconite, trace to abundant; mica and pyrite, locally trace; calcite, clear crystals/grains, locally trace.

Locally grading to LIMESTONE, light to dark greenish grey, hard, occasionally very hard, fully recrystallized (sparite), micaceous, very glauconitic.

MARL grading to LIMESTONE, pale red to grayish red, soft to firm, locally hard, locally dolomitic, argillaceous to very argillaceous; glauconite, trace.

At 3266 m abundant SAND, quartz, loose, clear, occasionally haematite stained, very fine grained, moderately to well sorted, subangular to subrounded, minor heavy minerals; downwards becoming trace to absent.

3305-3312 m — LIMESTONE with CLAYSTONE (trace to 20%).

LIMESTONE, very light grey to yellowish grey to greenish grey, soft to hard, chalky to recrystallized, slightly argillaceous; glauconite, trace to abundant; mica, locally trace to rare; sand, locally trace; chert, trace.

CLAYSTONE, dark grey to grayish black, firm to moderately hard, calcareous grading to MARL, locally with chalky laminae.

3312-3329 m — LIMESTONE, pale reddish brown, soft to firm, sticky, very argillaceous; chert, trace.

3329-3331 m — Cuttings recovered were 100% carvings, but the interval probably consists of a transition between the LIMESTONE described above the CLAYSTONE described below.

3331-3335 m — CLAYSTONE, dark reddish brown to moderate brown, soft to moderately hard, slightly to moderately calcareous, "silty texture" due to abundant calcite crystals; shell fragments, trace.

3335-3345 m — CLAYSTONE, dark grey to black, firm to hard, locally very hard, calcareous, slightly micromicaceous,

varyingly microlignitic, "silty texture" due to abundant calcite crystals; sand, loose, trace; slightly bioturbated.

3345-3354 m — CLAYSTONE with beds of SAND.

CLAYSTONE, medium grey to olive black, firm to moderately hard, calcareous, slightly micromicaceous, microlaminated due to varying content of microlignite, locally "silty texture" due to abundant calcite crystals.

SAND, loose, clear quartz, minor with orange tint, very fine to fine grained, well sorted, subangular to subrounded, slightly micaceous.

3354-3390 m — CLAYSTONES with few SANDSTONE beds.

CLAYSTONE, grayish red to moderate brown, soft to firm, occasionally moderately hard, slightly to moderately calcareous, trace of "silty texture" due to calcite crystals, locally very calcareous grading to MARL.

Interbedded/-laminated with CLAYSTONE, light grey to medium grey, occasionally very dark grey, soft to firm, occasionally moderately hard, calcareous, micromicaceous, locally "silty texture" due to

calcite crystals, locally micro-laminated.

At 3356, 3360 and 3379 m beds of SAND, loose, clear quartz, very fine to fine grained, well sorted, subangular to subrounded; mica, trace.

3390-3405 m — Interbedded/-laminated LIMESTONES.

LIMESTONE, white to very light grey, firm, occasionally moderately hard, slightly argillaceous, cryptocrystalline, slightly dolomitic.

LIMESTONE, grayish red to moderate brown, soft to firm, occasionally moderately hard, trace of "silty texture" due to calcite crystals, argillaceous, occasionally very argillaceous grading to MARL.

LIMESTONE, light grey to medium grey, soft to firm, occasionally moderately hard, micromicaceous, locally "silty texture" due to calcite crystals, argillaceous, occasionally argillaceous grading to MARL.

3405-3413 m — CLAYSTONE, olive black, soft to firm, sticky, moderately calcareous, slightly dolomitic, very carbonaceous, "silty texture" due to

abundant calcite crystals, micropyritic.

3413-3422 m — SAND with interbedded CLAYSTONE.

SAND, loose, predominantly clear quartz, minor with orange tint, very fine to fine grained, well sorted, subangular to subrounded, occasionally rounded; pyrite and heavy minerals (?), trace.

CLAYSTONE, olive grey, soft to firm, calcareous, carbonaceous, micropyritic; calcite crystals, trace.

3422-3429 m — SHALE, dark grey to grayish black to black, moderately hard to hard, fissile, non- to locally slightly calcareous, micromicaceous (core 2; description based on core chips).

3429-3436 m — SHALE, grayish black to brownish black, moderately hard to hard, locally very hard, fissile, non- to locally calcareous, micromicaceous, slightly carbonaceous, locally pyritic, locally slickensides (core 3; description based on core chips).

3436-3468 m

CLAYSTONE/SHALE interbedded with SANDSTONE.

CLAYSTONE/SHALE, dark grey to grayish black, firm to moderately hard, locally hard, subfissile, non- to slightly calcareous, micromicaceous, slightly carbonaceous; pyrite, trace; locally slickensides.

SANDSTONE, clear quartz, loose, fine grained, well sorted, subangular to subrounded; pyrite, trace.

3468-3470 m

CLAYSTONE, brownish black to brownish grey, firm, subfissile, "earthy", moderately calcareous (white specs of calcite), micromicaceous, slightly carbonaceous.

3470-3533 m

CLAYSTONE/SHALE with thin stringers of LIMESTONE and DOLOMITE.

CLAYSTONE/SHALE, dark grey to grayish black, firm to moderately hard, locally hard, non- to locally slightly calcareous, micromicaceous, slightly carbonaceous, locally silty; pyrite, minor to trace; locally slickensides.

LIMESTONE, white, soft to firm, locally black carbonaceous (?) streaks.

DOLOMITE, olive grey, hard to very hard, blocky, cryptocrystalline. Locally CALCITE, clear to white, crystalline, locally coarse crystals with pyrite on crystal faces (in fractures?).

3533-3535 m — CLAYSTONE, brownish grey, soft to firm, non calcareous, "earthy".

3535-3551 m — SILTSTONE, at top interbedded with CLAYSTONE.

SILTSTONE, white to light grey, soft to very hard, non- to well cemented (silica/dolomite/calcite), argillaceous, slightly carbonaceous and micropyritic; locally very sandy grading to very fine SANDSTONE.

CLAYSTONE, dark grey to grayish black a/a.

3551-3562 m — SANDSTONE, clear quartz, loose grains, very fine to fine grained, well sorted, subangular to subrounded; pyrite and feldspar, trace.

3562-3573 m — SANDSTONE with SILTSTONE and DOLOSTONE.

SANDSTONE, pale red to grayish red to light brownish grey to light olive grey, very fine to fine

grained locally grading to silt, hard, well cemented (silica/dolomite), well sorted, slightly carbonaceous; mica, trace. Downwards increasing number of laminae/beds of SILTSTONE, grayish red to light olive grey, hard to very hard, well cemented (dolomite), locally very argillaceous, locally laminae with abundant mica or sand. At base of core DOLOSTONE, light olive grey to light brownish grey, very hard, blocky, microcrystalline to granular, fractured (core 4; description based on core chips).

3573-3643 m

ANHYDRITE interbedded with SHALE and minor DOLOSTONE.

ANHYDRITE, white to very light grey, soft to firm, amorphous, locally with argillaceous laminae, locally clear dolomite crystals (coarse). Also beds of ANHYDRITE, white to pinkish grey, moderately hard to hard, brittle, argillaceous, microcrystalline to granular, locally dolomitic and grey to pale yellowish brown.

SHALE, dark grey to grayish black, moderately hard to hard, blocky to fissile, slightly calcareous, micromicaceous, locally with slickensides.

DOLOSTONE, white to pinkish grey, hard, brittle, microcrystalline to granular.

3643-3648 m — DOLOSTONE, light brownish grey to very pale yellowish brown, hard, brittle, slightly argillaceous, microcrystalline to granular.

3648-3713 m — ANHYDRITE with minor beds of DOLOSTONE and SHALE.

ANHYDRITE, white to very light grey, firm, amorphous, locally argillaceous, locally as clear crystals, fine to coarse grained. Also beds of ANHYDRITE, pinkish grey to light brownish grey, moderately hard to hard, brittle, locally argillaceous, microcrystalline to granular, locally dolomitic (?).

DOLOSTONE, light brownish grey to very pale yellowish brown and pinkish grey to very light grey, hard, brittle.

SHALE, grayish black to brownish black, moderately hard to hard, blocky to fissile, slightly dolomitic, micromicaceous.

3713-3739 m

SANDSTONE, CLAYSTONE and VOLCANICS(?).

SANDSTONE, loose grains, colorless or reddish, very fine to medium grained, moderately sorted, subangular.

CLAYSTONE, moderate reddish brown to grayish red, firm, blocky.

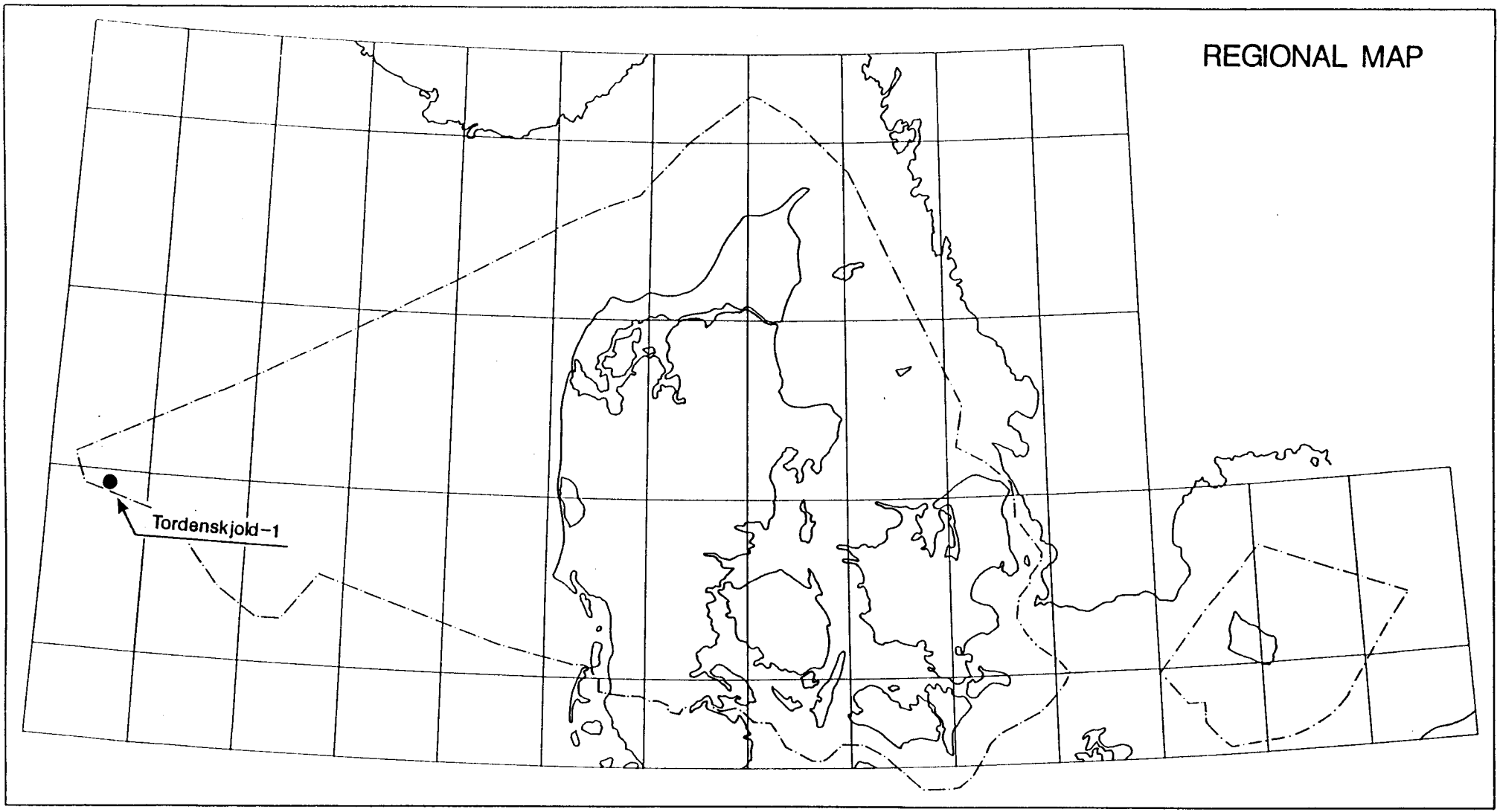
VOLCANICS(?), very light grey with very abundant and irregular black dots and laminae, locally with green and pink to purple grains. And beds of CLAYSTONE, brownish black, hard, blocky, slightly dolomitic.

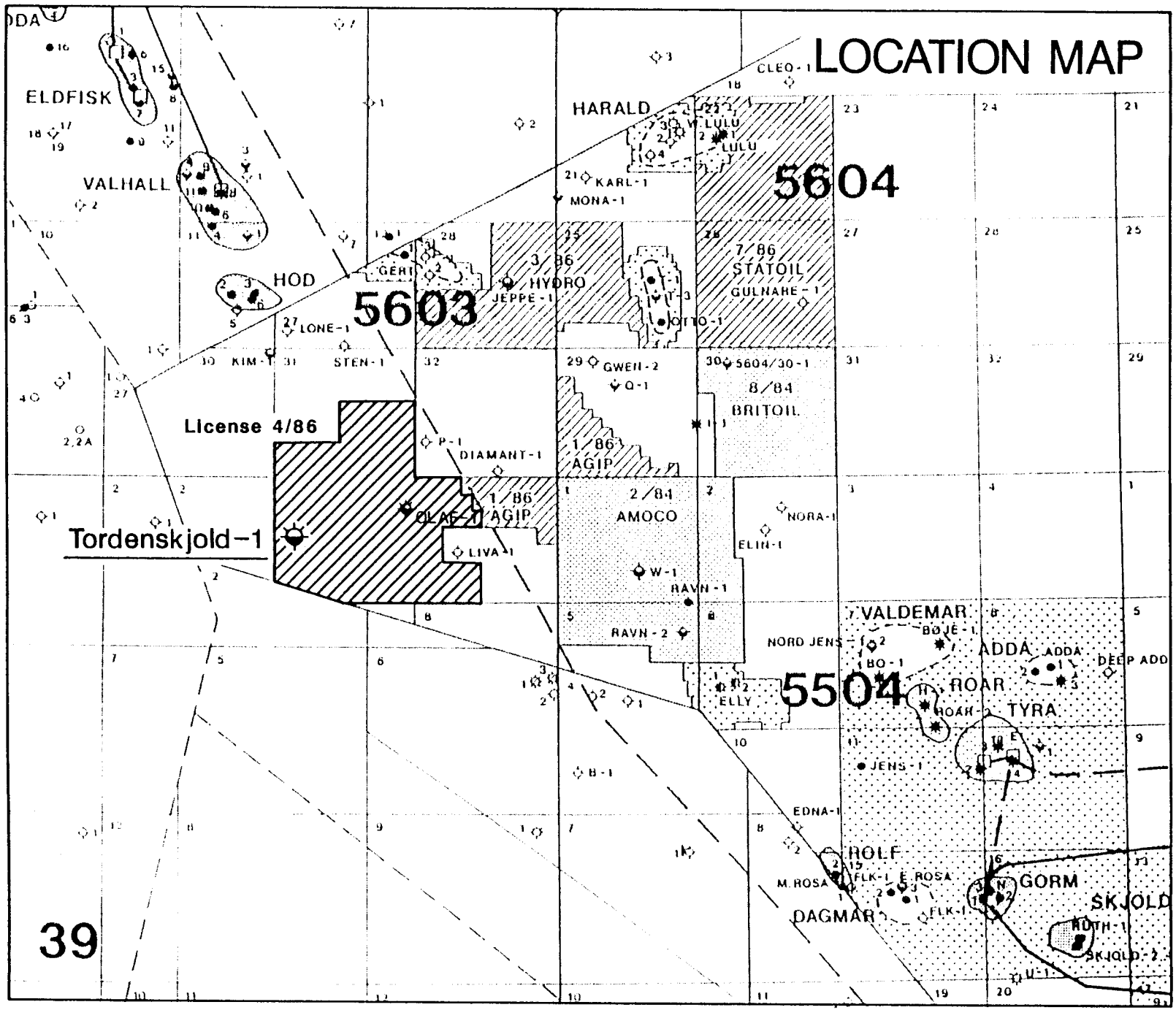
WELL DATA DISTRIBUTION 5503/3-2

		AMERADA HESS		GAS COUNSIL		TEXAS EASTERN		NORSK HYDRO		DENERCO		DOPAS		DANISH GEOL. SURVEY		DANISH ENERGY AGENCY		DANOP		MÆRSK OIL AND GAS		PALEODATING		GEOCHEMISTRY		TOTAL	REMARKS
SAMPLES	WASHED AND DRIED	1		1				7		1		1		1				1								13	DENERCO FROM TOP OF CHALK
	WET							1						1				1				1					
	WET, CANNED							1																1		2	
	MUD SAMPLE							1						1												2	1 L. EVERY 300m
	CORE													1												1	
	SWC													1								1				2	
FLUID FROM TESTS	OIL/CONDENSATE																										
	GAS																										
	WATER																										
LOGS	WIRELINE FIELD PRINTS	1		1		1		1		1		1		1		1		1		1						10	1:200 1:500 (PRINTS AND
	WIRELINE EDITED	1		1		1		1		1		1		1		1		1		1						10	SEPIAS) 1:1000 DEA+DGU
	WIRELINE LIS TAPES	1		1		1		1				1		1				1		1						8	1600 BPI
	MUD LOG/MWD LOG	1		1		1		1		1		1		1		1		1		1						10	PRINTS (SEPIA AT COM-
	PRESSURE LOGS	1		1		1		1		1		1		1		1		1		1						10	PRINTS PLETION OF WELL)
OTHERS	GRAPHIC WELL LOG	1		1		1		1		1		1		1		1		1		1						10	

APPENDIX 1.

LOCATION DATA


REGIONAL MAP
Tordenskjold-1


LOCATION MAP
5604
5603
5504
License 4/86
Tordenskjold–1
ELDFISK
VALHALL
HOD
HARALD
LULU
W LULU
KARL-1
MONA-1
CLEO-1
7/86
STATOIL
GULNARE-1
3/86
HYDRO
JEPPE-1
GERI
OTTO-1
LONE-1
KIM-1
STEN-1
GWEN-2
Q-1
5604/30-1
8/84
BRITOIL
P-1
DIAMANT-1
1/86
AGIP
CLAE-1
LIVA-1
2/84
AMOCO
W-1
RAVN-1
RAVN-2
NORA-1
ELIN-1
VALDEMAR
BOJE-1
NORD JENS-1
BO-1
ADDA
ADDA
DEEP ADDA
ELLY
ROAR
ROAR-2
TYRA
JENS-1
B-1
EDNA-1
ROLF
FLK-1
E.ROSA
M.ROSA
GORM
SKJOLD
DAGMAR
RUTH-1
SKJOLD-2
39

APPENDIX 2.

DAILY DRILLING SUMMARY

Date	Hours	Description
12/12/88		RIG ON CONTRACT AT 15:00 HRS.
	1500-1900	MADE UP THE 26" BOTTOM HOLE ASSEMBLY.
	1900-2400	MIXED SPUD MUD AND RIGGED UP THE BOP DECK.

Date	Hours	Description
13/12/88	0000-0130	MIXED SPUD MUD WHILE RIGGING UP THE BOP DECK.
	0130-0400	RIGGED UP TO RUN THE CONDUCTOR.
	0400-1030	RAN THE 30" CONDUCTOR TO THE SEABED.
	1030-1530	PICKED UP THE DRIVE HAMMER AND RIGGED UP TO DRIVE THE CONDUCTOR.
	1530-2400	DROVE THE CONDUCTOR TO 135 M.

Date	Hours	Description
14/12/88	0000-0030	INSTALLED THE CONDUCTOR TENSION SYSTEM.
	0030-0230	LAID DOWN ONE JOINT OF CONDUCTOR AND THE HAMMER.
	0230-0330	RAN IN THE HOLE TO 101 M.
	0330-0700	DRILLED TO 147 M.
	0700-1130	WORKED STUCK PIPE. HOLE OPENER INSIDE THE CONDUCTOR SHOE AT 134 M. MAX OVERPULL: 137 MT. PIPE CAME FREE WHEN WORKED 3 M. UP.
	1130-1200	CIRCULATED THE HOLE CLEAN.
	1200-1400	PULLED OUT OF THE HOLE. LAID DOWN THE MWD.
	1400-1530	RAN IN THE HOLE TO 147 M.
	1530-1630	DRILLED 26" HOLE TO 151 M.
	1630-1900	PULLED OUT. TIGHT HOLE FROM 147 M TO 139 M.
	1900-2230	RIGGED UP THE HAMMER.
	2230-2400	DROVE THE CONDUCTOR TO 140 M.

Date	Hours	Description
15/12/88	0000-0300	DROVE THE CONDUCTOR TO 151 M.
	0300-0430	CUT THE CONDUCTOR AND LAID DOWN THE LANDING JOINT.
	0430-0630	RIGGED UP THE DIVERTER. LAID DOWN THE CONDUCTOR RUNNING EQUIPMENT.
	0630-0800	REINSTALLED THE TOP TENSION WIRES.
	0800-1200	NIPPLED UP THE DIVERTER.
	1200-1400	MADE UP THE MWD AND RAN IN THE HOLE TO 142 M.
	1400-1500	WASHED DOWN INSIDE THE CONDUCTOR TO 151 M.
	1500-2400	DRILLED 26" HOLE TO 321 M.

Date	Hours	Description

Date	Hours	Description
16/12/88	0800-0900	CHANGED OUT 3 LINERS ON MUDPUMP NO. 1.
	0900-2030	DRILLED 26" HOLE FROM 449 M TO 611 M.
	2030-2100	CIRCULATED THE HOLE CLEAN.
	2100-2400	PULLED OUT OF THE HOLE. LAID DOWN THE 26"HOLE OPENER AND THE BIT.

Date	Hours	Description
17/12/88	0000-0030	CHECKED THE MWD AND LAID DOWN THE BIT.
	0030-0330	MADE UP THE 17.5" BOTTOM HOLE ASSEMBLY.
	0330-0500	RAN IN THE HOLE.
	0500-1830	DRILLED 17.5" HOLE FROM 611 M TO 1020 M.
	1830-1930	CIRCULATED THE HOLE CLEAN.
	1930-2230	PULLED OUT OF THE HOLE AND LAID DOWN THE STABILIZER AND THE MWD.
	2230-2400	DRESSED THE BIT AND HOLE OPENER WITH NEW NOZZLES AND MADE UP NEW BOTTOM HOLE ASSEMBLY.

Date	Hours	Description
18/12/88	0000-0130	MADE UP THE 26" BOTTOM HOLE ASSEMBLY AND PICKED UP 6 EA. 8" DRILL COLLARS.
	0130-0230	INSTALLED THE DIVERTER BAG AND RAN IN THE HOLE TO 590 M.
	0230-1530	OPENED THE HOLE FROM 17.5" TO 26" FROM 590 M TO 1020 M.
	1530-1730	CIRCULATED THE HOLE CLEAN, PUMPED SLUG AND FLOW CHECKED.
	1730-2130	PULLED OUT OF THE HOLE. LAID DOWN BIT, STABILIZER AND HOLE OPENER.
	2130-2300	CLEANED THE RIG FLOOR AND MADE UP THE CEMENT STINGER.
	2300-2400	PULLED AND LAID DOWN THE DIVERTER ASSEMBLY.

Date	Hours	Description
19/12/88	0000-0200	CUT THE 30" CONDUCTOR AND RIGGED UP TO RUN THE 20" CASING.
	0200-0500	PICKED UP THE CASING HANGER AND MADE UP THE LANDING AND HANGER JOINT.
	0500-1630	MADE UP 20" FLOAT COLLAR AND SHOE JOINT, INSTALLED CENTRALIZERS AND RAN 20" CASING TO 1002 M.
	1630-1830	LANDED THE CASING. RIGGED DOWN CASING RUNNING EQUIPMENT, AND MADE UP 5" CEMENT STINGER.
	1830-2130	RAN IN THE HOLE WITH THE STINGER.
	2130-2230	STUNG INTO THE FLOAT COLLAR AT 990 M AND CIRCULATED THE CASING VOLUME WITH SEA-

Date	Hours	Description
19/12/88		WATER. RIGGED UP AND RAN 2 STRINGS OF 1.66" WASHPIPE TO 114 M IN 30" BY 20" ANNULUS. CIRCULATED SEAWATER.
	2230-2400	RIGGED UP THE CEMENT LINES, CIRCULATED AND PRESSURE TESTED THE LINES TO 207 BAR STARTED TO CEMENT THE 20" CASING.

Date	Hours	Description
20/12/88	0000-0200	CEMENTED THE 20" CASING.
	0200-0300	PUMPED AND DISPLACED THE TAIL SLURRY.
	0300-0630	CHECKED FOR BACK FLOW, PULLED OUT OF THE COLLAR, CIRCULATED THE PIPE CLEAN AND PULLED OUT.
	0630-0700	CIRCULATED THE 20" BY 30" ANNULUS WITH 2 EA 1.66" WASH PIPE.
	0700-0830	PULLED AND LAID OUT 1.66" WASH PIPE. ROUGH CUT THE 20" CASING.
	0830-1200	REMOVED EXCESS 20" CASING. PREPAIRED TO INSTALL THE WELLHEAD AND THE BOP.
	1200-1400	INSTALLED THE WELLHEAD AND THE ADAPTOR ON THE 20" CASING.
	1400-2200	NIPPLED UP AND FUNCTION TESTED THE 20 3/4" BOP, KILL AND CHOKE LINES AND CONTROL HOSES.
	2200-2400	NIPPLED UP THE DIVERTED AND THE FLOW-LINE.

Date	Hours	Description
21/12/88	0000-0030	LANDED THE DIVERTER AND HOOKED UP THE CONTROL LINES.
	0030-0100	MADE UP AND RAN THE BOP TESTPLUG.
	0100-0200	RIGGED UP AND TESTED THE SURFACE LINES TO 207 BAR.
	0200-0400	PRESSURE TESTED THE RAMS, KILL AND CHOKE LINES, CASING AND WELLHEAD TO 100 BAR. PRESSURE TESTED THE ANNULAR TO 95 BAR.
	0400-0430	RAN THE WEARBUSHING.
	0430-0930	MADE UP 17.5" BOTTOM HOLE ASSEMBLY.
	0930-1100	PRESSURE TESTED THE MUD CIRCULATION SYSTEM TO 345 BAR. PRESSURE TESTED THE KILL AND CHOKE MANIFOLD TO 138 BAR.
	1100-1200	RAN IN THE HOLE TO 989 M.
	1200-1430	BROKE CIRCULATION AND DRILLED THE FLOAT, CEMENT AND SHOE TO 1002 M. DRILLED THE 17.5" HOLE TO 1024M.
	1430-1600	DISPLACED THE HOLE TO KCL-POLYMER MUD.
	1600-1630	CIRCULATED AND

Date	Hours	Description
21/12/88		CONDITIONED THE MUD FOR LEAK OFF TEST.
	1630-1700	RACKED THE KELLY AND RIGGED UP FOR LEAK OFF TEST.
	1700-1800	PERFORMED THE LEAK OFF TEST EQUIVALENT TO 2.25 R.D. MUD WEIGHT.
	1800-2400	DRILLED 17.5" HOLE FROM 1024M TO 1110M.

Date	Hours	Description
22/12/88	0000-0400	DRILLED 17.5" HOLE 1169 M.
	0400-0500	CIRCULATED THE ANNULUS CLEAN FOR GUMBO AND CHUNKS OF CEMENT.
	0500-0630	DRILLED 17.5" HOLE TO 1180 M.
	0630-0800	CIRCULATED AND CLEANED THE FLOWLINE FOR CHUNKS OF CEMENT.
	0800-2400	DRILLED 17.5" HOLE TO 1399 M.

Date	Hours	Description
23/12/88	0000-2345	DRILLED 17.5" HOLE 1801 M.
	2345-2400	TOOK SURVEY AND SLOW CIRCULATION RATES.

Date	Hours	Description
24/12/88	0000-0200	CIRCULATED THE HOLE CLEAN.
	0200-0500	PULLED OUT TO 1002 M. MAXIMUM 36 MT OVERPULL AT 1090 M.
	0500-0515	PERFORMED A KICK DRILL.
	0515-0645	SLIPPED AND CUT THE DRILL LINE.
	0645-0815	SUSPENDED THE OPERATION DUE TO BAD WEATHER.
	0815-0900	REPAIRED BROKEN AIRHOSE TO THE ELEVATOR.
	0900-0930	LAID DOWN 9 JOINTS OF DRILL PIPE.
	0930-1000	SECURED HYDRAULIC HOSES IN THE DERRICK.
	1000-1200	RAN IN THE HOLE TO 1801 M.
	1200-2400	DRILLED 17.5" HOLE TO 2003 M.

Date	Hours	Description
25/12/88	0000-0530	DRILLED 17.5" HOLE TO 2118 M.
	0530-0930	CIRCULATED AND CONDITIONED THE MUD.
	0930-2400	DRILLED 17.5" HOLE TO 2308 M.

Date	Hours	Description
26/12/88	0000-0530	DRILLED 17.5" HOLE TO 2414 M.
	0530-0900	CIRCULATED BOTTOM UP AND CONDITIONED THE MUD.
	0900-0930	DISPLACED WEIGHTED PREMIX ON BOTTOM.
	0930-1000	DROPPED THE SINGLE SHOT SURVEY.
	1000-1130	PULLED OUT TO 2160 M. MAX OVERPULL 70 MT WORKED THE PIPE TO 2065 M.
	1130-1630	PICKED UP THE KELLY AND WASHED AND

Date	Hours	Description
26/12/88		REAMED SINGLES TO 1913 M.
	1630-1930	PULLED OUT TO 992 M.
	1930-2030	RETRIEVED THE SINGLE SHOT SURVEY.
	2030-2300	RAN IN THE HOLE TO 2414 M.
	2300-2400	CIRCULATED THE HOLE CLEAN.

Date	Hours	Description
27/12/88	0000-0230	CIRCULATED THE HOLE CLEAN.
	0230-0900	PULLED OUT OF THE HOLE. MAXIMUM OVERPULL: 60 MT AT 2070 M.
	0900-1000	REARRANGED THE BOTTOM HOLE ASSEMBLY. CHANGED BOP RAMS TO 13 3/8".
	1000-1100	PULLED THE WEAR BUSHING. TESTED THE BOP BONNETS.
	1100-1230	RIGGED UP TO RUN CASING.
	1230-2400	RAN 13 3/8" CASING TO 1233 M.

Date	Hours	Description
28/12/88	0000-0830	RAN 13.3/8" CASING TO 2319 M.
	0830-1030	WASHED TO 2346 M DUE TO TIGHT HOLE.
	1030-1130	CONTINUED TO RUN CASING, LANDED CASING WITH SHOE AT 2397 M.
	1130-1230	CIRCULATED.
	1230-1300	RIGGED UP TO CEMENT. TESTED CEMENT LINES TO 345 BAR.
	1300-2030	MIXED AND PUMPED LEAD AND TAIL SLURRIES. DISPLACED THE CEMENT. THE PLUG DID NOT BUMP. LOST RETURNS WHEN MIXED LEAD SLURRY.
	2030-2230	PREPARED TO NIPPLE DOWN THE BOP.
	2230-2400	FLUSHED ANNULUS WITH WATER THROUGH PORTS IN THE MLS.

Date	Hours	Description
29/12/88	0000-0030	RETORQUED THE MLS. FLOW CHECKED.
	0030-1030	LIFTED THE BOP AND INSTALLED CASING SLIP ASSEMBLY. ROUGH CUT CASING, RIGGED DOWN DIVERTER AND MOVED THE BOP TO TEST STUMP.
	1030-1500	DRESSED 13 3/8" CASING AND INSTALLED THE WELLHEAD.
	1500-2100	NIPPLED UP THE 13 5/8" BOP.
	2100-2230	NIPPLED UP THE DIVERTER.
	2230-2400	COMMENCED TO TEST THE BOP STACK.

Date	Hours	Description
30/12/88	0000-0430	TESTED THE BOP STACK TO 440 BAR.
	0430-0500	TESTED THE CASING TO 240 BAR.

Date	Hours	Description
30/12/88	0500-0830	TESTED THE SURFACE EQUIPMENT TO 345 BAR.
	0830-0900	RAN THE WEAR BUSHING.
	0900-1500	MADE UP BOTTOM HOLE ASSEMBLY AND RAN IN THE HOLE. TAGGED TOP OF CEMENT AT 2309 M
	1500-1530	REPAIRED MUDLOGGING SENSOR ON ELMAGCO.
	1530-2000	DRILLED CEMENT, FLOAT AND SHOE TO 2414 M
	2000-2030	DRILLED 12 1/4" HOLE FROM 2414 TO 2418 M
	2030-2130	CIRCULATED BOTTOMS UP.
	2130-2200	RAN LEAK OFF TEST TO 1.91 RD MUDWEIGHT.
	2200-2400	DRILLED 12 1/4" HOLE TO 2460 M.

Date	Hours	Description
31/12/88	0000-0200	DRILLED 12 1/4" HOLE TO 2518 M.
	0200-0400	CIRCULATED AND RAN SURVEY.
	0400-0830	DRILLED FROM 2518 TO 2623 M.
	0830-1030	CIRCULATED AND RAN SURVEY.
	1030-1500	DRILLED FROM 2623 TO 2719 M.
	1500-1730	CIRCULATED AND RAN SURVEY.
	1730-2400	DRILLED FROM 2719 TO 2784 M.

Date	Hours	Description
01/01/89	0000-0700	DRILLED 12 1/4" HOLE TO 2805 M.
	0700-0830	CIRCULATED, CHECKED FOR FLOW AND DROPPED THE SURVEY.
	0830-1400	PULLED OUT OF HOLE LAYING DOWN EXCESS DRILL PIPE.
	1400-1930	MADE UP NEW BIT AND RAN IN THE HOLE. WASHED 4 M FILL.
	1930-2400	DRILLED FROM 2805 TO 2830 M.

Date	Hours	Description
02/01/89	0000-1700	DRILLED 12/4" HOLE TO 2904 M.
	1700-1730	CIRCULATED.
	1730-1800	DROPPED SURVEY AND FLOW CHECKED.
	1800-2030	PULLED OUT OF THE HOLE TO 318 M.
	2030-2100	RETRIEVED THE SURVEY.
	2100-2230	CONTINUED TO PULL OUT OF THE HOLE.
	2230-2400	MADE UP NEW BOTTOM HOLE ASSEMBLY.

Date	Hours	Description
03/01/89	0000-0300	RAN IN THE HOLE.
	0300-2400	DRILLED 12 1/4" HOLE TO 2940 M.

Date	Hours	Description
04/01/89	0000-0700	DRILLED 12 1/4" HOLE TO 2967 M.
	0700-0730	REPAIRED POP OFF VALVE ON PUMP NO. 2.
	0730-2400	DRILLED FROM 2967 TO 3034 M.

Date	Hours	Description
05/01/89	0000-1900	DRILLED 12 1/4" HOLE TO 3095 M.

Date	Hours	Description
05/01/89	1900-1930	CIRCULATED.
	1930-2000	DROPPED DIRECTIONAL SURVEY.
	2000-2400	PULLED OUT OF THE HOLE.

Date	Hours	Description
06/01/89	0000-0200	PULLED OUT OF THE HOLE.
	0200-0630	PULLED THE WEAR BUSHING AND TESTED THE BOP TO 435 BAR. RERAN THE WEAR BUSHING.
	0630-1300	REARRANGED THE BOTTOM HOLE ASSEMBLY AND RAN IN. LOST 4 CU.M. MUD WHILE RIH.
	1300-1330	REAMED FROM 3080 TO 3095 M.
	1330-2400	DRILLED 12 1/4" HOLE TO 3136 M.

Date	Hours	Description
07/01/89	0000-2400	DRILLED 12 1/4" HOLE TO 3198 M

Date	Hours	Description
08/01/89	0000-0700	DRILLED 12 1/4" HOLE TO 3213 M.
	0700-0830	CIRCULATED BOTTOMS UP.
	0830-0900	DROPPED SURVEY, CHECKED FOR FLOW.
	0900-1300	PULLED OUT OF THE HOLE.
	1300-2100	MADE UP NEW BIT, BHA AND RAN IN HOLE T0 SHOE AT REDUCED RATE DUE TO MUD LOSSES TO THE HOLE, TOTAL LOSS 6 CU.M.
	2100-2200	SLIPPED AND CUT THE DRILL LINE.
	2200-2400	CONTINUED TO RUN IN THE HOLE.

Date	Hours	Description
09/01/89	0000-0030	RAN IN THE HOLE TO 3110 M.
	0030-0100	REAMED FROM 3110 M TO 3147 M. TIGHT SPOT AT 3118 M AND 3135 M.
	0100-0130	RAN IN THE HOLE TO 3213 M.
	0130-2400	DRILLED 12 1/4" HOLE 3260 M.

Date	Hours	Description
10/01/89	0000-2400	DRILLED 12 1/4" HOLE TO 3305 M.

Date	Hours	Description
11/01/89	0000-0400	DRILLED 12 1/4" HOLE TO 3312 M.
	0400-0430	STOPPED OPERATIONS DUE TO FALSE H2S ALARM IN THE SHAKER ROOM.
	0430-1430	DRILLED 12 1/4" HOLE TO 3327 M.
	1430-1500	CIRCULATED PRIOR TO DROP THE SURVEY.
	1500-1530	DROPPED THE SINGLE SHOT SURVEY.
	1530-1730	PULLED OUT TO 2397 M.
	1730-1830	RETRIEVED THE SURVEY.
	1830-2000	RAN IN THE HOLE TO 3327 M.
	2000-2300	DRILLED TO 3329 M.
	2300-2400	WORKED BIT ON BOTTOM AFTER CONNECTION.

Date	Hours	Description
11/01/89		WAS NOT ABLE TO ROTATE.

Date	Hours	Description
12/01/89	0000-0030	CIRCULATED AND WORKED PIPE AT 3300 M. MAX OVERPULL:55 TON.
	0030-0100	DROPPED THE SURVEY AT 3306 M
	0100-0600	PULLED OUT OF THE HOLE. LAID DOWN THE BIT AND SHOCK SUB AND RETRIEVED THE SINGLE SHOT SURVEY. 2/3 OF ONE BIT CONE WAS MISSING.
	0600-1400	MADE UP JUNK BIT AND JUNK SUB AND RAN IN THE HOLE. LOST 3 CU.M. MUD.
	1400-1500	WORKED BIT ON JUNK WHILE CIRCULATED.
	1500-1700	DRILLED 12 1/4" HOLE FROM 3329 TO 3330 M
	1700-1800	CIRCULATED AND WORKED THE JUNK BASKET.
	1800-2400	PULLED OUT OF THE HOLE.

Date	Hours	Description
13/01/89	0000-0600	RAN IN THE HOLE 3330 M.
	0600-0630	PUMPED 10 M3 HIGH VISCOUS PILL AND WORKED THE JUNK BASKET.
	0630-1130	DRILLED 12 1/4" HOLE TO 3331 M.
	1130-1630	PULLED OUT OF THE HOLE.
	1630-1700	MADE UP 10 1/8" REVERSE CIRCULATING JUNK BASKET.
	1700-2130	RAN IN THE HOLE TO 3331 M.
	2130-2200	BROKE THE CIRCULATION AND DROPPED THE BALL.
	2200-2300	CUT 0.5 M CORE.
	2300-2400	FLOW CHECKED AND PULLED OUT OF THE HOLE.

Date	Hours	Description
14/01/89	0000-0400	PULLED OUT OF THE HOLE AND RECOVERED 2 EA JUNK FROM THE BIT AND 40 CM CORE.
	0400-0830	MADE UP NEW BIT AND RAN IN THE HOLE TO 3331 M.
	0830-0900	CIRCULATED AND WORKED THE JUNK SUB.
	0900-1600	DRILLED 12 1/4" HOLE TO 3342 M.
	1600-1730	CIRCULATED BOTTOM UP.
	1730-1800	WORKED THE JUNK BASKET AND FLOW CHECKED.
	1800-2230	PULLED OUT OF THE HOLE. LAID DOWN JUNK-BASKET AND BIT.
	2230-2300	RIGGED UP FOR LOGGING.
	2300-2400	RAN IN THE HOLE WITH LOG NO.1: CNL/ISF/LDL/BHC/GR/CAL/SP.

Date	Hours	Description
15/01/89	0000-0500	COMPLETED LOG NO 1:

Date	Hours	Description
15/01/89		ISF/BHC/LDL/CNL/GR/CAL/SP.
	0500-0600	RIGGED UP FOR LOG NO 2: CBL/VDL/GR.
	0600-1230	RAN LOG NO 2.
	1230-1300	RIGGED UP FOR LOG NO 3: CST/GR.
	1300-1730	RAN LOG NO 3. WAS NOT ABLE TO FIRE LAST 21 SHOTS. RECOVERED 6, LOST 2, 1 EMPTY.
	1730-1800	RIGGED UP NEW CST GUN.
	1800-2200	RAN CST/GR LOG. RECOVERED 8, LOST 22.
	2200-2230	RIGGED DOWN AFTER LOGGING.
	2230-2300	PICKED UP 9 5/8" CASING HANGER AND MADE UP ON THE RUNNING TOOL.
	2300-2400	RAN IN THE HOLE FOR WIPER TRIP.

Date	Hours	Description
16/01/89	0000-0230	RAN IN THE HOLE TO 2397 M.
	0230-0330	SLIPPED AND CUT 96 FEET DRILL LINE.
	0330-0500	RAN IN THE HOLE TO 3342 M.
	0500-0530	WORKED THE JUNKBASKET.
	0530-0630	CIRCULATED BOTTOM UP.
	0630-1200	PUMPED THE SLUG AND PULLED OUT OF THE HOLE. LAID DOWN THE JUNK BASKETS.
	1200-1600	CHANGED OUT UPPER PIPE RAMS TO 9 5/8" CASING RAMS. PRESSURE TESTED BONNETS TO 173 BAR, AND RIGGED UP TO RUN 9 5/8" CASING.
	1600-2300	RAN 9 5/8" CASING TO 1690 M.
	2300-2400	CHANGED OUT THE ELEVATOR AND SLIPS FROM 150 MT TO 500 MT.

Date	Hours	Description
17/01/89	0000-0800	RAN THE 9 5/8" CASING TO 3336 M.
	0800-0930	ATTEMPTED TO LAND THE CASING IN MUDLINE SUSPENSION. WAS NOT ABLE TO HANG OFF THE CASING. LOWERED THE SHOE TO 3340 M.
	0930-1030	MADE UP THE CEMENT HEAD.
	1030-1200	CIRCULATED THE CASING VOLUME. BATCH MIXED THE THE TAIL SLURRY.
	1200-1230	PUMPED 9 M3 SPACER.
	1230-1300	PRESSURE TESTED THE CEMENT LINES TO 345 BAR.
	1300-1430	RELEASED THE BOTTOM PLUG. MIXED AND PUMPED THE LEAD SLURRY.
	1430-1500	PUMPED THE TAIL SLURRY.
	1500-1530	RELEASED THE TOP PLUG AND PUMPED 3.2 M3 WATER.
	1530-1630	DISPLACED THE CEMENT WITH MUD. THE PLUG DID NOT BUMP.
	1630-1730	CHECKED FOR BACK FLOW. LOWERED THE CASING TO 3342 M.

Date	Hours	Description
17/01/89	1730-2100	LAID DOWN CASING HANDLING EQUIPMENT. CHANGED UPPER PIPE RAMS TO 5" RAMS.
	2100-2130	LIFTED THE BOP AND INSTALLED THE SLIPS AND THE PACK OFF.
	2130-2400	ROUGH CUT THE 9 5/8" CASING. REMOVED THE DIVERTER AND BOP ADAPTOR.

Date	Hours	Description
18/01/89	0000-0230	NIPPLED DOWN THE BOP AND LAID DOWN THE DOUBLE STUDDED ADAPTOR.
	0230-0330	FINAL CUT THE 9 5/8" CASING 0.6 M ABOVE THE B-SECTION.
	0330-0500	INSTALLED THE 9 5/8" WELLHEAD SECTION.
	0500-1030	INSTALLED AND NIPPLED UP THE BOP. INSTALLED THE DIVERTER AND FLOWLINE.
	1030-1130	INSTALLED CONTROL LINES AND LAID DOWN RUNNING TOOLS.
	1130-1330	RIGGED UP AND TESTED THE 9 5/8" CASING TO 495 BAR.
	1330-1400	RAN IN THE HOLE WITH BOP TEST PLUG.
	1400-1730	PRESSURE TESTED THE BOP TO 655 BARS.
	1730-1800	RAN IN THE HOLE WITH CUP TESTER.
	1800-2000	PRESSURE TESTED THE WELLHEAD TO 640 BAR AGAINST LOWER PIPE RAM.
	2000-2030	LAID DOWN TEST ASSEMBLY AND RAN THE WEAR BUSHING.
	2030-2400	LAID DOWN 8" DRILL COLLARS. PRESSURE TESTED THE KILL AND CHOKE MAN. TO 655 BAR AND THE STAND PIPE MANIFOLD TO 345 BAR.

Date	Hours	Description
19/01/89	0000-0200	LAID DOWN 8" DRILL COLLAR.
	0200-0330	PRESSURE TESTED THE KELLY AND STAB-IN VALVES TO 345 BAR.
	0330-0500	DUE TO LEAKAGE DISCOVERED IN THE UPPER PIPE RAM WHEN TESTING THE WELLHEAD THE BOP TEST PLUG WAS RUN AND AN ATTEMPT TO TEST THE UPPER PIPE RAM MADE.
	0500-0700	LAID DOWN 8" DRILL COLLARS AND THE JAR WHILE CHANGING THE RUBBER ON THE UPPER PIPE RAM.
	0700-0900	RAN THE BOP TEST PLUG. ATTEMPTED TO TEST THE UPPER PIPE RAM TO 655 BAR. PULLED THE TEST PLUG AND RAN THE WEAR-BUSHING.
	0900-1700	MADE UP 8 3/8" BIT AND BOTTOM HOLE ASSEMBLY AND RAN IN THE HOLE TO 2600 M.
	1700-1730	CHANGED THE SEALS ON THE UPPER PIPE RAM.

Date	Hours	Description
19/01/89	1730-1800	ATTEMPTED TO RUN THE CUP TESTER. WAS NOT ABLE TO PASS THROUGH THE WEAR BUSHING. THE BUSHING WAS PULLED WITH THE TESTER.
	1800-1830	MADE UP AND RAN THE BOP TEST PLUG.
	1830-1900	ATTEMPTED TO TEST THE UPPER PIPE RAM.
	1900-2400	REPAIRED THE UPPER PIPE RAM.

Date	Hours	Description
20/01/89	0000-0230	REPAIRED THE BOP.
	0230-0330	PRESSURE TESTED THE UPPER PIPE RAM. FOUND LEAK IN THE UPPER OUTER KILL VALVE. TESTED THE UPPER PIPE RAM AGAINST THE UPPER INNER KILL VALVE.
	0330-0400	PULLED THE TEST PLUG AND INSTALLED THE WEAR BUSHING.
	0400-0500	RAN IN THE HOLE AND TAGGED TOP OF CEMENT AT 3298 M.
	0500-0900	DRILLED FLOAT AND CEMENT TO 3340 M. WHILE REPAIRING THE KILL VALVE.
	0900-1000	PULLED THE WEAR BUSHING AND RAN THE TEST PLUG.
	1000-1100	ATTEMPTED TO TEST THE KILL VALVE AGAINST UPPER PIPE RAM. LEAKAGE AT 445 BAR.
	1100-1400	OPENED THE BONNET BOLTS ON UPPER PIPE-RAM AND CHECKED RAM SEALS. CLOSED THE BONNETS.
	1400-1530	ATTEMPTED TO RETEST. FOUND LEAK TO BE IN THE TEST STRING. TESTED THE KILL VALVE AGAINST THE UPPER PIPE RAM TO 655 BAR.
	1530-1600	LAID DOWN THE BOP TEST PLUG,RAN THE WEAR BUSHING AND INSTALLED THE DIVERTER.
	1600-1730	DRILLED CEMENT AND THE SHOE FROM 3340M TO 3342 M.
	1730-1830	DRILLED 8 3/8" HOLE TO 3345 M.
	1830-1900	CIRCULATED PRIOR TO DO FORMATION INTEGRITY TEST.
	1900-1930	ATTEMPTED TO PULL OUT TO 3342 M. BACK REAMED DUE TO JUNK ABOVE THE BIT.
	1930-2000	PERFORMED A FORMATION INTEGTITY TEST, EQUIVALENT TO 2.07 R.D. MUD WEIGHT.
	2000-2130	DRILLED 8 3/8" HOLE TO 3351 M.
	2130-2230	CIRCULATED BOTTOM UP FOR SAMPLE.
	2230-2400	DRILLED 8 3/8" HOLE TO 3355 M.

Date	Hours	Description
21/01/89	0000-1500	DRILLED 8 3/8" HOLE FROM 3355 TO 3409 M.
	1500-1530	CIRCULATED AND DROPPED THE SINGLE SHOT SURVEY.

Date	Hours	Description
21/01/89	1530-1600	PULLED OUT TO 3320 M.
	1600-1830	ATTEMPTED TO RETRIEVE THE SURVEY.
	1830-1900	RAN IN THE HOLE TO 3409 M.
	1900-2230	DRILLED 8 3/8" HOLE TO 3422 M.
	2230-2330	CIRCULATED BOTTOM UP FOR SAMPLES.
	2330-2400	FLOWCHECKED,DROPPED THE SURVEY AND PUMPED THE SLUG PRIOR TO PULL OUT OF THE HOLE FOR CORING.

Date	Hours	Description
22/01/89	0000-0400	PULLED OUT OF THE HOLE.
	0400-0630	PICKED UP 30 M COREBARREL.
	0630-1000	RAN IN THE HOLE TO 3342 M.
	1000-1030	REPROGRAMMED THE BLOCK EMERGENCY STOP SYSTEM.
	1030-1100	RAN IN THE HOLE TO 3413 M. REAMED TO 3422 M.
	1100-1130	CIRCULATED AND CONDITIONED THE MUD.
	1130-1200	DROPPED THE BALL AND TOOK SLOW CIRCULATING RATE.
	1200-1630	CUT CORE FROM 3422 M TO 3429 M.
	1630-1700	RECORDED 14 BAR PRESSURE DROP. CHECKED THE PUMPS AND PERFORMED SLOW CIRCULATING RATE.
	1700-1730	ATTEMPTED TO RESTART THE CORE.
	1730-1800	FLOWCHECKED AND PUMPED THE SLUG.
	1800-2200	PULLED OUT OF THE HOLE.
	2200-2230	RECOVERED THE CORE. 67 % RECOVERY.
	2230-2330	CHANGED OUT 10 M INNER BARREL AND MADE UP NEW COREHEAD.
	2330-2400	RAN IN THE HOLE TO CUT CORE.

Date	Hours	Description
23/01/89	0000-0300	RAN IN HOLE WITH CORE BARREL.
	0300-0400	CIRCULATED AND DROPPED BALL.
	0400-0900	CUT CORE NO.3 FROM 3429 M TO 3436 M CORE JAMMED.
	0900-1300	PULLED OUT OF THE HOLE.
	1300-1400	RETRIEVED THE CORE. 87% RECOVERY.
	1400-1430	SERVICED THE RIG.
	1430-1930	RAN IN HOLE, REAMED FROM 3418 TO 3436 M.
	1930-2300	DRILLED 8 3/8" HOLE TO 3449 M.
	2300-2400	CIRCULATED BOTTOMS UP FOR SAMPLE.

Date	Hours	Description
24/01/89	0000-0030	CONTINUED TO CIRCULATE BOTTOMS UP FOR SAMPLE.
	0030-2400	DRILLED 8 3/8" HOLE TO 3540 M.

Date	Hours	Description
25/01/89	0000-0430	DRILLED 8 3/8" HOLE TO 3562 M.

Date	Hours	Description
25/01/89	0430-0530	CIRCULATED BOTTOMS UP.
	0530-0600	DROPPED DIRECTIONAL SURVEY.
	0600-1100	FLOW CHECKED AND PULLED OUT AF THE HOLE. FLOW CHECKED AT SHOE.
	1100-1130	RETRIEVED THE SURVEY.
	1130-1500	MADE UP CORE BARREL AND RAN IN THE HOLE.
	1500-1600	SLIPPED AND CUT DRILL LINE AT THE SHOE.
	1600-1630	CONTINUED TO RUN IN HOLE.
	1630-1700	BROKE CIRCULATION AND DROPPED THE BALL.
	1700-2200	CUT CORE NO.4 FROM 3562 TO 3573 M. CORE JAMMED.
	2200-2400	FLOW CHECKED AND PULLED OUT OF THE HOLE. FLOW CHECKED AT THE SHOE.

Date	Hours	Description
26/01/89	0000-0230	CONTINUED TO PULL OUT OF THE HOLE.
	0230-0300	RETRIEVED CORE NO.4, 70% RECOVERY.
	0300-0800	RAN IN HOLE WITH BIT TO 3560 M.
	0800-0830	REAMED FROM 3560 TO 3573 M.
	0830-1030	DRILLED 8 3/8" HOLE TO 3580 M.
	1030-1130	FLOWCHECKED AND OBSERVED THE WELL.
	1130-2400	DRILLED TO 3648 M.

Date	Hours	Description
27/01/89	0000-1830	DRILLED FROM 3648 TO 3739 M.
	1830-2000	INCREASED THE MUDWEIGHT TO 1.67 RD.
	2000-2130	MADE WIPER TRIP TO THE SHOE. NO FILL.
	2130-2300	CIRCULATED BOTTOMS UP.
	2300-2330	DROPPED DIRECTIONAL SURVEY.
	2330-2400	PULLED OUT OF THE HOLE TO LOG.

Date	Hours	Description
28/01/89	0000-0400	CONTINUED TO PULL OUT OF THE HOLE.
	0400-0430	RETRIEVED THE DIRECTIONAL SURVEY.
	0430-0900	RIGGED UP AND RAN LOG NO.1: ISF/LSS/SP
	0900-1300	RAN LOG NO.2: LDL/CNL/NGT/CAL. WAS NOT ABLE TO PASS 3368 M. PULLED OUT.
	1300-1630	RAN IN THE HOLE FOR WIPER TRIP.
	1630-2100	CIRCULATED BOTTOMS UP. INCREASED THE MUD WEIGHT TO 1.70 RD.
	2100-2400	PULLED OUT OF THE HOLE.

Date	Hours	Description
29/01/89	0000-0030	CONTINUED TO PULL OUT OF THE HOLE.
	0030-0400	RIGGED UP AND RERAN LOG NO.2: LDL/CNL/ NGT/CAL. WAS NOT ABLE TO PASS 3420 M. PULLED OUT AND REMOVED STAND OFF FROM LOGGING TOOL.
	0400-0800	RERAN LOG NO.2.

Date	Hours	Description
29/01/89	0800-1300	RAN LOG NO.3: DLL/MSFL.
	1300-2130	RAN LOG NO.4: RFT. TOOL FAILED AFTER 5 PRESSURE POINTS, PULLED OUT.
	2130-2300	REPAIRED THE RFT.
	2300-2400	RERAN LOG NO.4.

Date	Hours	Description
30/01/89	0000-0400	CONTINUED LOG NO.4: RFT.
	0400-0900	RAN LOG NO.5: SHDT.
	0900-1400	RAN LOG NO.6: CBL/VDL.
	1400-2400	RAN LOG NO.7: VSP.

Date	Hours	Description
31/01/89	0000-0330	CONTINUED LOG NO.7: VSP.
	0330-0900	RAN LOG NO.8: CST. RIGGED DOWN AFTER LOGGING.
	0900-1330	PICKED UP 30 JOINTS 3.5" DP AND RAN IN.
	1330-1530	CIRCULATED BOTTOMS UP.
	1530-1700	SAT CEMENT PLUG FROM 3739 TO 3665 M.
	1700-1730	PULLED OUT OF HOLE TO 3665 M.
	1730-1900	SAT CEMENT PLUG FROM 3665 TO 3392 M.
	1900-1930	PULLED OUT OF HOLE TO 3392 M.
	1930-2130	REVERSE CIRCULATED, PREMIXED CEMENT.
	2130-2230	SAT CEMENT PLUG FROM 3392 TO 3190 M.
	2230-2300	PULLED OUT OF THE HOLE TO 3190 M.
	2300-2400	REVERSE CIRCULATED.

Date	Hours	Description
01/02/89	0000-0030	PULLED OUT OF THE HOLE TO 2757 M.
	0030-0130	CIRCULATED.
	0130-0900	LAID DOWN DRILL PIPE AND DRILL COLLARS WHILE WAITING ON CEMENT.
	0900-0930	RAN IN HOLE, TAGGED CEMENT AT 3192 M.
	0930-1200	PULLED OUT OF HOLE TO 2397 M, LAYING DOWN DRILL PIPE.
	1200-1230	PRESSURE TESTED THE CEMENT PLUG TO 245 BAR.
	1230-1330	SAT CEMENT PLUG FROM 2397 TO 2297 M.
	1330-1400	PULLED OUT OF THE HOLE TO 2222 M.
	1400-1430	CIRCULATED.
	1430-2000	PULLED OUT OF THE HOLE LAYING DOWN DRILL PIPE.
	2000-2400	MADE UP CASING CUTTER AND CUT THE 9 5/8" CASING AT 251 M. REDRESSED CUTTER AND CUT THE 13.3/8" CASING AT 251 M. NOTE: THE 13 3/8" X 9 5/8" CASING ANNULUS IS PRESSURE TESTED TO 140 BAR. THE 20" X 13 3/8" CASING ANNULUS IS PRESSURE

Date	Hours	Description
01/02/89		TESTED TO 143 BAR.

Date	Hours	Description
02/02/89	0000-0200	CUT 13 3/8" CASING AT 251 M.
	0200-0300	PULLED OUT OF HOLE TO 199 M. CUTTING TOOL STUCK.
	0300-0330	PRESSURE TESTED THE 20" X 13 3/8" CASING ANNULUS AND THE 13 3/8" X 9 5/8" CASING ANNULUS TO 140 BAR THROUGH THE CUT.
	0330-0700	WORKED THE STRING TO 164 M.
	0700-0900	LIFTED BOP, OBSERVED 9 5/8" CASING NOT RETAINED BY SLIPS.
	0900-1100	RIGGED UP AND RAN EXPLOSIVE CHARGE AND BACKED OFF THE DRILL PIPE AT 151 M.
	1100-1200	PULLED OUT AND LAID DOWN THE DRILL PIPE.
	1200-2100	NIPPLED DOWN THE DIVERTER, THE BOP AND THE WELL HEAD.
	2100-2400	MADE UP 9 5/8" CASING SPEAR. PULLED AND LAID DOWN THE 9 5/8"CASING. NOTE: BLED OFF PRESSURE ON THE WELLHEAD IN THE 20" X 13 3/8" CASING ANNULUS TO CONFIRM COMMUNICATION TO THE ANNULUS.

Date	Hours	Description
03/02/89	0000-0300	LAID DOWN 9 5/8" CASING.
	0300-0400	RAN IN HOLE WITH 13 3/8" CASING SPEAR, ATTEMPTED TO RETRIEVE THE CASING.
	0400-0530	MADE UP 13 3/8" CASING CUTTER AND RAN IN.
	0530-0600	CUT THE 13 3/8" CASING AT 249 M.
	0600-0630	PULLED OUT OF THE HOLE.
	0630-0300	MADE UP CASING SPEAR, ENGAGED SPEAR AND PULLED THE CASING. HAD 159 TON OVERPULL.
	0800-0830	MADE UP THE CIRCULATING HEAD, CIRCULATED THROUGH CASING.
	0830-1130	CONTINUED TO PULL AND LAY OUT CASING.
	1130-1230	RAN IN HOLE WITH OPEN ENDED DRILL PIPE.
	1230-1430	SAT CEMENT PLUG FROM 351 TO 151 M.
	1430-1500	PULLED OUT OF HOLE TO 140 M, CIRCULATED.
	1500-1530	PULLED AND LAID OUT DRILL PIPE.
	1530-1600	CUT AND REMOVED 20" CASING SPOOL.
	1600-2000	MADE UP 20" CASING SPEAR, ENGAGED AND RETRIEVED THE 20" CASING.
	2000-2400	LAID OUT CORE BARREL, DRILL PIPE AND COLLARS FROM THE DERRICK.

Date	Hours	Description
04/02/89	0000-0330	MADE UP THE 30" CASING CUTTING TOOL

Date	Hours	Description
04/02/89		AND CUT THE 30"CASING AT 106 M.
	0330-1000	RETRIEVED AND LAID OUT THE 30" CASING.
	1000-1100	LAID OUT DRILL PIPE FROM THE DERRICK.
	1100-2000	PREPARED TO SKID DERRICK.
	2000-2130	WAIT ON WEATHER, MAX WIND 65 KNOTS.
	2130-2400	CONTINUED PREPARATION AND COMMENCED TO SKID THE DERRICK.

Date	Hours	Description
05/02/89	0000-0800	SKIDDED THE DERRICK.
	0800-2400	WAITED ON WEATHER.

Date	Hours	Description
06/02/89	0000-2400	WAITED ON WEATHER.

Date	Hours	Description
07/02/89	0000-2100	WAITED ON WEATHER.
	2100-2400	SKIDDED THE CANTILEVER.

Date	Hours	Description
08/02/89	0000-0300	SKIDDED THE CANTILEVER.
	0300-0600	JACKED DOWN.
		RIG OFF CONTRACT AT 06:00 HRS.

APPENDIX 3.

WELL STATUS

WELL STATUS

RKB - MSL: 36 m

RKB - Seabed: 101 m

30" Casing at 106 m
20" Casing retrieved at 114 m
Cementplug from 351 m to 151 m
30" Conductor at 151 m

13 3/8" Casing cut at 249 m

9 5/8" Casing cut at 251 m

20" Casing shoe at 1002 m

Top of Cement outside
13 3/8" Casing at 1450 m

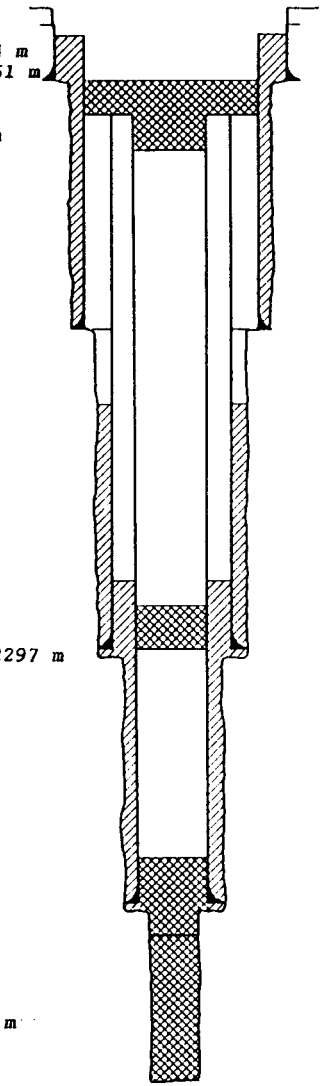
Top of Cement outside
9 5/8" Casing at 1600 m

13 3/8" Casing at 2397 m
Cementplug from 2397 m to 2297 m

9 5/8" Casing at 3342 m

Cementplug: 3739 m to 3192 m

P and A schematic figure

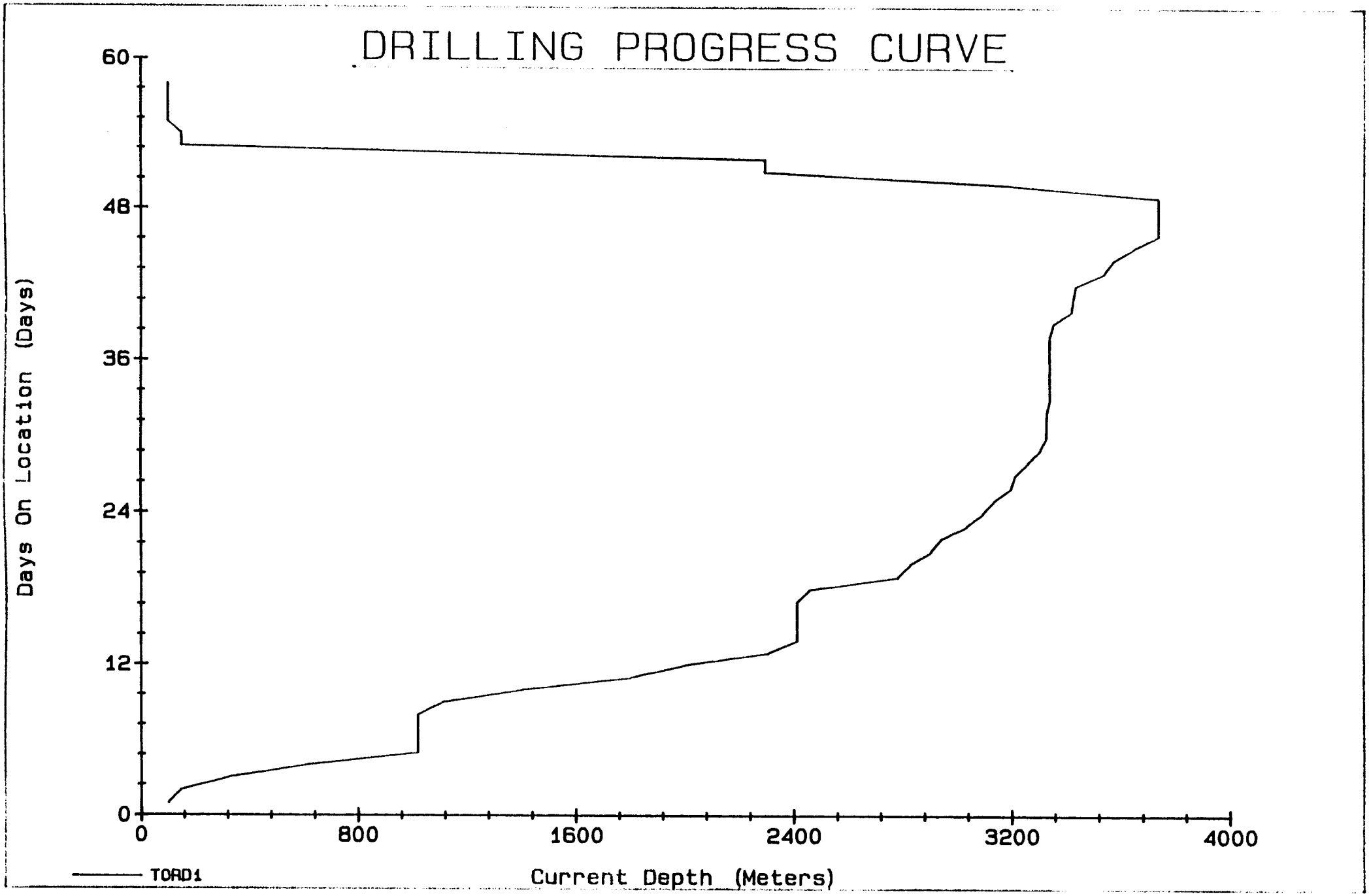


Total measured drillers depth 3739.0 m
Total measured loggers depth 3737.5 m

RIG	: NEDDRILL TRIGON		
	: Rig on contract	:	December 12, 1988 at 15.00 Hrs.
	Spud In	:	December 13, 1988 at 10.30 Hrs.
30" DRIVING	:	:	Drove the 30" casing to 151 m
	Problems	:	Stuck with 26" hole opner at 134 m, free after 5 Hrs
26" SECTION	: Drilling	:	Drilled 26" hole to 611 m , Drilled 17 1/2" pilot hole to 1020 m prior to open up to 26" hole
	Mud	:	Seawater/Bentonite
	Casing	:	20" casing shoe at 1002 m cemented to surface
	Logging	:	MWD
	Problems	:	Angle building using 26" hole opener, drilled 17 1/2" pilot hole from 611 m to 1020 m
17 1/2" SECTION	: Drilling	:	Drilled 17 1/2" hole to 2414 m
	Casing	:	13 3/8" casing shoe at 2397 m cemented to 1450 m
	Mud	:	KCL/Polymer
	Logging	:	MWD tool failed at 2267 m
	Leak Off	:	2.25 r.d. at 1002 m
12 1/4" SECTION	: Drilling	:	Drilled 12 1/4" hole to 3342 m
	Mud	:	KCL/Polymer
	Logging	:	1) CNL-ISF-LDL-BHC-GR-CAL-SP 2) CBL-VDL-GR 3-4) CST-GR
	Casing	:	9 5/8" casing shoe at 3342 m, cemented to 2100 m
	Leak Off	:	1.91 r.d. at 2397 m
	Problems	:	Lost one cone on bit no. 8, Lost 29.5 Hrs. on junk drilling/fishing, Mudline suspension did not function, set casing on bottom, Leakage in upper pipe ram and fail safe valve on 13 5/8" BOP
8 3/8" SECTION	: Drilling	:	Drilled 8 3/8" hole to 3739 m
	Mud	:	Seawater/high temp. polymers
	Coring	:	2) 3422 - 3429 m, 67% rec. 3) 3429 - 3436 m, 87% rec. 4) 3562 - 3573 m, rec. 70%
	Logging	:	1) ISF-LSS-SP 2) LDL-CNL-NGT-CAL 3) DLL-MSFL 4) RFT 5) SHDT 6) CBL-VDL 7) VSP 8) CST
	Leak Off	:	Formation integrity test to 2.07 r.d. at 3342 m
	Problems	:	BOP test string leakage, rerun of log no. 2) and 4) lost time: 35 Hrs.
PLUG & ABANDON	: Cementing	:	Set cement plug as per P & A schematic
	Problems	:	Not able to cut 9 5/8" and 13 5/8" casing in one go, Cutting tool stuck inside 9 5/8"casing at 199 m lost time: 7.5 Hrs.
DEMOBILIZATION	:	:	Waited on weather to jack down from February 4. 20.00 Hrs. until February 7. 21.00 Hrs. (62.5 Hrs.) Rig off contract February 8. 06.00 Hrs.

APPENDIX 4.

DRILLING PROGRESS


DRILLING PROGRESS CURVE
60
48
36
24
12
0
Days On Location (Days)
0
800
1600
2400
3200
4000
Current Depth (Meters)
TORD1

GROUP	CODE	SUB DESCRIPTION
Mob. - Demob.	1	Under transport
	2	Jacking and rigging up/down
Drilling Operations	5	Drilling
	6	Enlarging hole
	7	Tripping
	8	Circulation
	9	Leak-off test
	10	Directional survey
	11	BOP eqp. handling and testing
	12	Running casing
	13	Primary cementing
	14	Slipping and cutting drilling line
Formation Evaluation	20	Other formation evaluation
	21	Logging
	22	Coring
Plugging and Abandoning	35	Other P&A work
	36	Tripping
	37	Cementing
	38	Cutting casing
Downtime	40	Rig repair
	41	Other eqp. repair
	42	Fishing
	43	Wait on weather
	44	Wait on cement
	45	Repairing cement jobs
	46	Hole problems
	47	Other downtime

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
12/12/88		5.0					4.0														9.0
13/12/88		1.5										22.5									24.0
14/12/88					4.5		3.5					7.5									15.5
15/12/88					10.0		2.0				6.0	6.0									24.0
16/12/88					19.5		3.0	0.5													23.0

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
12/12/88																					0.0
13/12/88																					0.0
14/12/88																					0.0
15/12/88																					0.0
16/12/88																				1.0	1.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
12/12/88																					0.0
13/12/88																					0.0
14/12/88		8.0				0.5															8.5
15/12/88																					0.0
16/12/88																					0.0

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
17/12/88					13.5		9.5	1.0													24.0
18/12/88						13.0	6.5	2.0			1.0	1.5									24.0
19/12/88												18.5	5.5								24.0
20/12/88											10.0	5.5	8.5								24.0
21/12/88					6.0		6.0	1.5	2.0		8.0	0.5									24.0

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
17/12/88																					0.0
18/12/88																					0.0
19/12/88																					0.0
20/12/88																					0.0
21/12/88																					0.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
17/12/88																					0.0
18/12/88																					0.0
19/12/88																					0.0
20/12/88																					0.0
21/12/88																					0.0

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
22/12/88					21.5																21.5
23/12/88					23.8					0.2											24.0
24/12/88					12.0		5.5	2.2						1.5							21.2
25/12/88					20.0			4.0													24.0
26/12/88					5.5		7.0	4.0		1.5											18.0

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
22/12/88																					0.0
23/12/88																					0.0
24/12/88																				1.2	1.2
25/12/88																					0.0
26/12/88																					0.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
22/12/88						2.5															2.5
23/12/88																					0.0
24/12/88			1.5																		1.5
25/12/88																					0.0
26/12/88						6.0															6.0

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
27/12/88							7.5				1.0	13.0									21.5
28/12/88								1.0				13.0	8.0								22.0
29/12/88											9.0	15.0									24.0
30/12/88					2.5		6.0	1.0	0.5		8.5	0.5	4.5								23.5
31/12/88					17.5					6.5											24.0

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
27/12/88																					0.0
28/12/88																					0.0
29/12/88																					0.0
30/12/88																					0.0
31/12/88																					0.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
27/12/88						2.5															2.5
28/12/88						2.0															2.0
29/12/88																					0.0
30/12/88	0.5																				0.5
31/12/88																					0.0

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
01/01/89					11.5		11.0			1.5											24.0
02/01/89					17.0		6.0			1.0											24.0
03/01/89					21.0		3.0														24.0
04/01/89					23.5																23.5
05/01/89					19.0		4.0	0.5		0.5											24.0

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
01/01/89																					0.0
02/01/89																					0.0
03/01/89																					0.0
04/01/89																				0.5	0.5
05/01/89																					0.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
01/01/89																					0.0
02/01/89																					0.0
03/01/89																					0.0
04/01/89																					0.0
05/01/89																					0.0

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
06/01/89					10.5		8.5	0.5			4.5										24.0
07/01/89					24.0																24.0
08/01/89					7.0		14.0	1.5		0.5				1.0							24.0
09/01/89					22.5		1.0														23.5
10/01/89					24.0																24.0

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
06/01/89																					0.0
07/01/89																					0.0
08/01/89																					0.0
09/01/89																					0.0
10/01/89																					0.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
06/01/89																					0.0
07/01/89																					0.0
08/01/89																					0.0
09/01/89						0.5															0.5
10/01/89																					0.0

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
11/01/89					17.0		3.5			2.0											22.5
12/01/89					2.0		5.0			0.5											7.5
13/01/89					5.0		11.0														16.0
14/01/89					7.0		9.5	1.5													18.0
15/01/89												0.5									0.5

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
11/01/89																					0.0
12/01/89																					0.0
13/01/89																					0.0
14/01/89	1.5																				1.5
15/01/89	19.0																				19.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
11/01/89						1.0	0.5														1.5
12/01/89		16.5																			16.5
13/01/89		8.0																			8.0
14/01/89		4.5																			4.5
15/01/89							4.5														4.5

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
16/01/89							9.5	1.0			4.0	8.0		1.0							23.5
17/01/89								1.5			0.5	14.5	6.0								22.5
18/01/89							3.5				16.0	4.5									24.0
19/01/89							12.0				1.5										13.5
20/01/89					9.5		1.0		1.0											1.0	12.5

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
16/01/89																					0.0
17/01/89																					0.0
18/01/89																					0.0
19/01/89																				10.5	10.5
20/01/89																				11.0	11.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
16/01/89		0.5																			0.5
17/01/89							1.5														1.5
18/01/89																					0.0
19/01/89																					0.0
20/01/89						0.5															0.5

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
21/01/89					18.5		1.0			0.5										1.0	21.0
22/01/89																					0.0
23/01/89					3.5															1.0	4.5
24/01/89					23.5			0.5													24.0
25/01/89					4.5					1.0				1.0						1.0	7.5

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
21/01/89		0.5																			0.5
22/01/89		23.5																		0.5	24.0
23/01/89		19.0																		0.5	19.5
24/01/89																					0.0
25/01/89		16.5																			16.5

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
21/01/89							2.5														2.5
22/01/89																					0.0
23/01/89																					0.0
24/01/89																					0.0
25/01/89																					0.0

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
26/01/89					14.5			1.0													15.5
27/01/89					18.5		2.0	3.0		0.5											24.0
28/01/89							4.0			0.5											4.5
29/01/89																					0.0
30/01/89													5.0								5.0

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
26/01/89		8.5																			8.5
27/01/89																					0.0
28/01/89	8.5																				8.5
29/01/89	17.5																				17.5
30/01/89	15.0																				15.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
26/01/89																					0.0
27/01/89																					0.0
28/01/89						11.0															11.0
29/01/89	2.5					4.0															6.5
30/01/89	4.0																				4.0

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
31/01/89																					0.0
01/02/89																					0.0
02/02/89																					0.0
03/02/89																					0.0
04/02/89		11.5																			11.5

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
31/01/89	9.0														5.0	6.0	4.0				24.0
01/02/89															2.0	17.0	1.0	4.0			24.0
02/02/89															11.5	1.0		6.0			18.5
03/02/89															0.5	5.5	2.5	11.5			20.0
04/02/89																1.0		10.0			11.0

date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
31/01/89																					0.0
01/02/89																					0.0
02/02/89							5.5														5.5
03/02/89							4.0														4.0
04/02/89			1.5																		1.5

date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	tot.
05/02/89		8.0																			8.0
06/02/89																					0.0
07/02/89		3.0																			3.0
08/02/89		6.0																			6.0

date	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	tot.
05/02/89																					0.0
06/02/89																					0.0
07/02/89																					0.0
08/02/89																					0.0

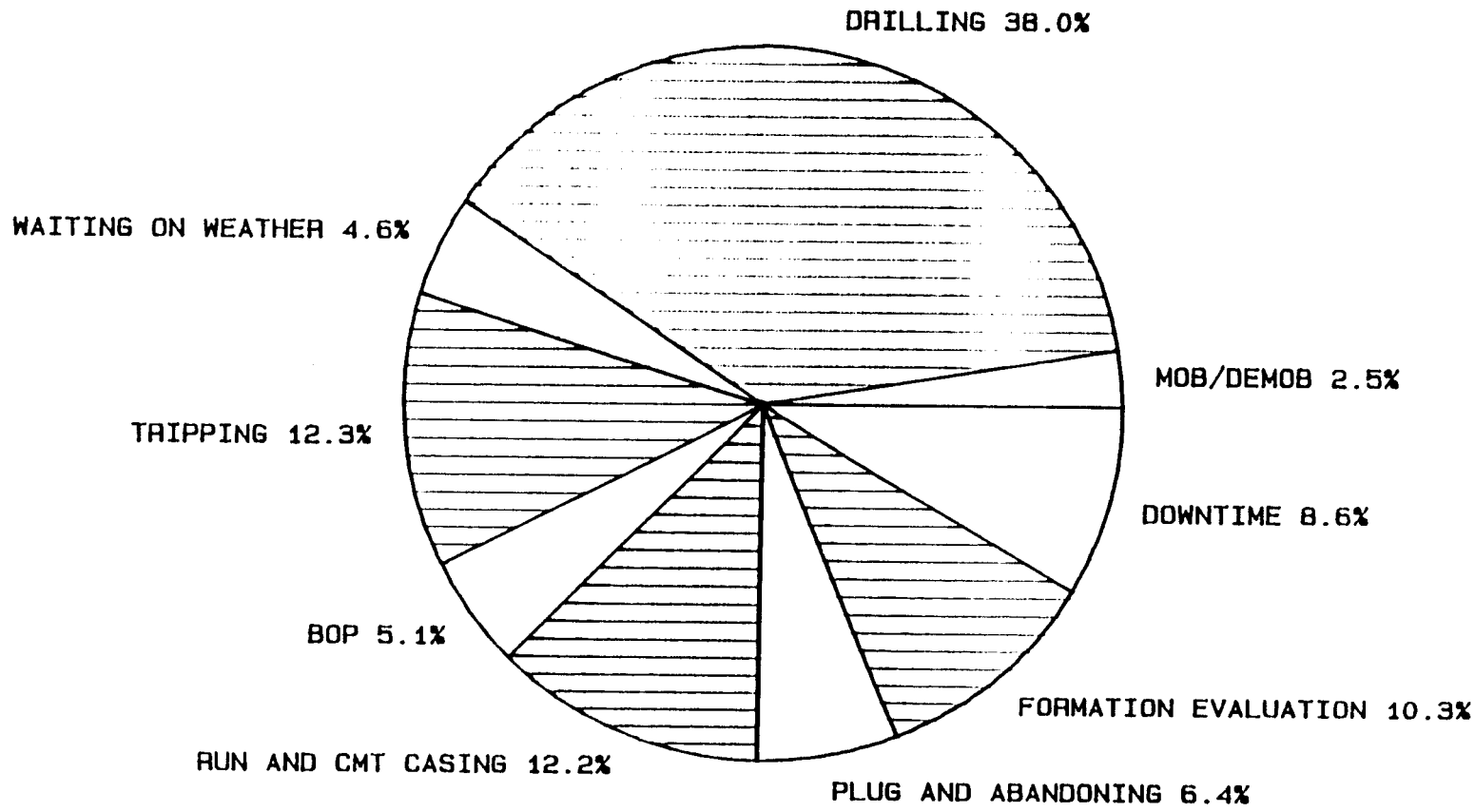
date	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	tot.
05/02/89			16.0																		16.0
06/02/89			24.0																		24.0
07/02/89			21.0																		21.0
08/02/89																					0.0

total	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
	0.00	0.00	459.75	170.00	3.50	70.00	37.50	0.00	0.00	0.00										
	35.00	0.00	13.00	28.25	16.75	131.00	4.50	0.00	0.00	4.00										

total	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
	70.50	0.00	0.00	0.00	0.00	0.00	0.00	19.00	7.50	0.00										
	68.00	0.00	0.00	0.00	0.00	0.00	0.00	30.50	31.50	25.25										

total	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
	7.00	64.00	0.00	13.50	0.00	0.00	0.00	0.00	0.00	0.00										

TIME DISTRIBUTION CHART



APPENDIX 5.

DRILLING FLUID SUMMARY

5. DRILLING FLUID SUMMARY

5.1 30" - 26" Hole Section

Depth

101 - 1020 m

Mud Type

Seawater/Bentonite

Résumé

The 26" hole for the 30" conductor and the 26" hole from 151 to 1020 were drilled using seawater and prehydrated bentonite.

The contained 25 kg/m^3 prehydrated bentonite in seawater which gave an average yield point of 10-20 Pa and mud weight varying between 1.10 and 1.16 r.d.

The properties were maintained using lime as viscosifier.

Product Consumption

Product	Quantity
Bentonite	33 MT
Caustic Soda	0.4 MT
Soda Ash	0.2 MT
Lime	1.75 MT
Barite	6.0 MT

5.2 17 1/2" Hole Section

Depth

1020 - 2414 m

Mud Type

KCl/Polymer

Résumé

The KCl brine wad deluded back to a concentration of 80 kg/m^3 and mixed with IDVIS and IDF FLR as primary viscosifier and fluid loss agent. Inhibition was provided by addition of the encapsulating polyacrylamid polymer; IDBOND throughout the section.

The mud weight was maintained at 1.60 r.d., yield point at 10-13 Pa and fluid loss at 10 ml. These rheologies together with sufficient inhibition from IDBOND and KCl gave limited hole problems while drilling continuously from 1020 m to 2415 m.

The encapsulation of the reactive clay limited the solids to stick together and the solids passed the shaker screens with no indication of binding and blanking of the screens.

Product Consumption –

Product	Quantity
Barite	690 MT
Caustic	2.65 MT
IDVIS	1.38 MT
IDF FLR	4.05 MT
IDF FLR XL	4.53 MT
IDFLO	2.30 MT
KCl brine	531 m^3
KCl (big bags)	10 MT
Defoamer	0.08 m^3
IDBOND	1.83 m^3
Bicarbonate	0.25 MT

5.3 12 1/4" Hole Section

Depth

2397 - 3342 m

Mud Type

KCl/Polymer

Résumé

The mud from the 17 1/2" section was deluded to reduce the solids contents, but otherwise the rheologies were maintained at the same level. At the end of the section the KCl contents were reduced from 80 to 60 kg/m^3. No hole problems due to the formation or the mud were encountered.

Due to barite slugs added on trips at 3329 m unweighted premix was added to control the mud weight at 1.60 r.d.

Total 18 m^3 of mud were lost to the formation while tripping.

Product Consumption

Product	Quantity
Barite	256 MT
Caustic	1.13 MT
IDVIS	0.50 MT
IDF FLR	2.13 MT
IDF FLR XL	2.53 MT
IDFLO	1.93 MT
KCl brine	27.0 m^3
KCl (big bags)	24 MT
IDBOND	1.78 m^3
Bicarbonate	0.30 MT
Defoamer	0.03 m^3

5.4 8 3/8" Hole Section

Depth

3342 - 3739 m

Mud Type

High temperature polymer

Résumé

The KCl concentration was depleted to 35 kg/m³ by addition of seawater. The mud weight was increased from 1.63 r.d. to 1.65 r.d. at 3730 m and further increased on to 1.70 r.d. at 3739 m on wipertrip while logging due to pore pressure increase in Permian.

The active system was treated with 10 kg/m³ of PTS-200 and 3.75 kg/m³ of IDSPERSE-HT prior to logging due to high temperature.

With the MBT of 20 ppb bentonite the viscosity increased rapidly so constant dilution was made to maintain the MBT at 15 ppb of bentonite.

Product Consumption

Product	Quantity
Barite	168 MT
Caustic	1.75 MT
IDVIS	0.20 MT
IDF FLR	0.30 MT
IDF FLR XL	1.58 MT
Bicarbonate	0.15 MT
IDCIDE L	0.20 MT
IDSPERSE-HT	1.10 MT
PTS 200	0.20 MT
Defoamer	0.04 MT

Note: The material consumption in this chapter is material added to the mud. No transport losses are taken into consideration.

date	Measured Depth (Meters)	Mud Weight (g/cc)	Plastic Viscosity (cp)	Yield Point (Pa)	pH	Fluid Loss (cc/30 min)	Chlorides (ppm)	KCL Concentration (mg/l)	MBT (ppb)	% Sand	% Solids
13/12/88		1.03									
14/12/88		1.03									
15/12/88	321	1.13	4	12	9.50		11000			7.0	
16/12/88	611	1.13	4	11	9.50		12000			0.5	
17/12/88	1020	1.13	8	20	10.00		19000			0.5	
18/12/88	1020	1.14	8	7	10.00		19000			0.5	
19/12/88	1020										
20/12/88	1020	1.10	26	13	9.50	6.0	38000	85000.0			
21/12/88	1110	1.10	28	14	9.00	7.0	61000	80000.0	4.0	0.5	18.0
22/12/88	1399	1.60	45	12	9.00	9.8	59000	77000.0	20.0	1.0	12.0
23/12/88	1801	1.60	68	8	9.00	10.0	52000	80000.0	35.0	3.0	15.0
24/12/88	2003	1.61	44	10	9.00	11.0	63000	80000.0	34.0	3.0	21.0
25/12/88	2308	1.60	44	12	10.00	17.0	55000	76000.0	37.0	3.0	20.0
26/12/88	2414	1.60	63	9	10.00	18.0	65000	82000.0	38.0	2.0	21.0
27/12/88	2414	1.60	40	12	10.00	18.0	69000	82000.0	35.0	2.0	21.0
28/12/88	2414	1.60	24	15	10.00	14.0	71000	83000.0	22.0	2.0	20.0
29/12/88	2414	1.60	30	13	10.00	6.2	71000	83000.0	82.0		20.0
30/12/88	2460	1.60	34	6	11.50	19.0	67000	82000.0	8.0		20.0
31/12/88	2784	1.60	36	8	11.00	9.0	67000	81000.0	14.0		19.0
01/01/89	2830	1.60	40	10	10.50	7.5	64000	79000.0	17.5	0.2	19.0
02/01/89	2904	1.60	36	8	10.00	8.0	63000	79000.0	17.5	0.3	19.0
03/01/89	2940	1.60	38	9	10.00	7.0	64000	80000.0	19.5	0.2	19.0
04/01/89	3034	1.60	33	9	10.00	7.0	64000	80000.0	20.5	0.2	19.0
05/01/89	3095	1.60	34	9	9.50	6.5	65000	80000.0	20.5	0.2	19.0
06/01/89	3136	1.60	34	9	9.50	6.5	63000	78000.0	21.0	0.2	19.0
07/01/89	3138	1.60	31	9	9.50	6.7	64000	78000.0	21.0	0.2	19.0

date	Measured Depth (Meters)	Mud Weight (g/cc)	Plastic Viscosity (cp)	Yield Point (Pa)	pH	Fluid Loss (cc/30 min)	Chlorides (ppm)	KCL Concentration (mg/l)	MBT (ppb)	% Sand	% Solids
08/01/89	3213	1.60	31	8	9.50	7.0	64000	78000.0	21.0	0.2	20.0
09/01/89	3260	1.60	30	7	9.50	7.0	63000	77000.0	21.0	0.2	20.0
10/01/89	3305	1.60	30	7	9.00	6.5	65000	79000.0	20.0	0.2	20.0
11/01/89	3329	1.60	30	7	9.00	6.0	66000	80000.0	19.0	0.2	20.0
12/01/89	3330	1.60	38	8	9.00	6.2	65000	79000.0	19.0	0.2	20.0
13/01/89	3331	1.60	31	8	9.00	6.5	65000	78000.0	19.0	0.2	20.0
14/01/89	3342	1.60	32	7	9.00	6.5	63000	77000.0	19.0	0.2	20.0
15/01/89	3342	1.60	40	9	9.00	6.0	63000	77000.0	19.0	0.2	20.0
16/01/89	3342	1.60	40	8	9.00	6.0	63000	77000.0	19.0	0.2	20.0
17/01/89	3342	1.60	29	7	9.00	6.0	62000	76000.0	19.0	0.2	20.0
18/01/89	3342	1.60	39	9	9.00	6.0	62000	74000.0	19.0	0.2	20.0
19/01/89	3342	1.62	35	10	9.00	5.6	63000	74000.0	20.0	0.2	20.0
20/01/89	3355	1.61	34	10	10.00	5.8	63000		20.0	0.2	20.0
21/01/89	3422	1.63	38	12	9.50	4.3	48000		16.0	0.2	21.0
22/01/89	3429	1.63	40	12	9.50	4.4	48000		16.0	0.2	20.0
23/01/89	3438	1.63	38	10	9.50	3.4	46000		13.0	0.5	19.0
24/01/89	3540	1.64	44	16	10.00	4.0	45000		16.0	0.2	20.0
25/01/89	3573	1.61	35	10	9.50	4.6	38000		12.5	0.1	21.0
26/01/89	3648	1.63	33	10	10.00	5.6	38000		14.0	0.5	21.0
27/01/89	3739	1.67	44	9	10.50	4.0	40000		16.0	0.5	21.0
28/01/89	3739	1.70	38	10	10.00	5.6	40000		15.0	0.5	22.0
29/01/89	3739	1.70	38	10	10.00	5.6	40000		15.0	0.5	22.0
30/01/89	3739	1.70	38	10	10.00	5.6	40000		15.0	0.5	22.0

APPENDIX 6.

CEMENTING SUMMARY

6. CEMENTING SUMMARY

6.1 General

The 20" casing was cemented until return cement was seen on the rig using a 5" drill pipe stinger stung into a stab in float collar. The annulus from the seabed up was thereafter displaced to water.

The 13 3/8" casing was planned to be cemented 200 m into the 20" casing. However, returns were lost during cementing which resulted in top of cement at 1450 m.

The 9 5/8" casing was cemented back to 1600 m with full returns during the cement job.

The abandonment cement plugs were set as shown in Appendix 3.

6.2 Cement Product Consumptions

The volumes used in this chapter are theoretical volumes and do not include transport losses and measuring errors.

20" casing

Lead slurry

Slurry volume:	211.8 m^3
Cement class G:	126.5 MT
Seawater:	154.0 m^3
Econolite:	4160 l
Density:	1.40 r.d.

Tail in slurry

Slurry volume:	12.2 m^3
Cement class G:	9.5 MT
Seawater:	4.5 m^3
Density:	1.90 r.d.

13 3/8" casing

Lead Slurry

Slurry volume:	94.8 m^3
Cement class G:	89.2 MT
Freshwater:	60.7 m^3
Econolite:	1824 l
HR-4 L:	1824 l
Density:	1.68 r.d.

Tail Slurry

Slurry volume:	10.5 m^3
Cement class G:	13.4 MT
Seawater:	6.2 m^3
HR-4 L:	154 l
Density:	1.89 r.d.

9 5/8" casing

Lead slurry

Slurry volume:	37.7 m^3
Cement class G:	40.0 MT
Freshwater:	24.5 m^3
Halad 361-B:	3250 l
FDP 365:	132 l
Density:	1.68 r.d.

Tail Slurry

Slurry Volume	11.3 m^3
Cement class G:	16.3 MT
Freshwater:	5.9 m^3
SSA-1:	5700 kg
HDP-100 A:	80 kg
HR-4 L:	70 l
CFR-3L:	200 l
Density:	1.91 r.d.

Spacer

Dual purpose spacer:	1200 kg
Barite:	6500 kg
Density:	1.60 r.d.

Plug and abandonment

Open hole plug No. 1

Volume:	5.9 m^3
Cement class G:	5.9 MT
Freshwater:	3.2 m^3
SSA-1:	2080 kg
HDP-100 A:	30 kg
HR-4 L:	150 l
CFR-3L:	1100 l
Density:	1.92 r.d.

Open hole plug No. 2

Volume:	6.6 m^3
Cement class G:	6.6 MT
Freshwater:	3.6 m^3
SSA-1:	2310 kg
HDP-100 A:	45 kg
HR-4 L:	170 l
CFR-3L:	180 l
Density:	1.92 r.d.

9 5/8" casing shoe plug

Volume:	7.4 m^3
Cement class G:	7.5 MT
Freshwater:	3.9 m^3
SSA-1:	2590 kg
HDP-100 A:	37 kg
HR-4 L:	50 l
CFR-3L:	50 l
Density:	1.93 r.d.

Inside 9 5/8" casing plug

Volume:	3.7 m^3
Cement class G:	4.7 MT
Seawater:	2.2 m^3
HR-4 L:	54 l
Density:	1.90 r.d.

Top cement plug

Volume:	22 m^3
Cement class G:	28.5 MT
Seawater:	13 m^3
Density:	1.90 r.d.

APPENDIX 7.

BIT AND BOTTOM HOLE ASSEMBLY SUMMARY

****

* Bit Record — Page 1 *
* From: 12/12/88 — Well Code: TORD1 *
* To: 08/02/89 *

****

Bit No.	Bit Size	Bit Type	Serial Number	Jet Sizes	Depth Out	Mtrs Drld	Hrs Run	Mtrs/ Hour	Cum Mtrs	Cum Hrs	Bit Wt.	RPM	Pump Pres	Pump LPM	Mud Wt	Mud T	Mud B	Mud G	Man.
1	17.50	S11C	5195	28/28/28/16	147	46	3.5	13.1	46	4	2/3	50/50	55	3000	1.0	1	1	0	REED
1RR1	17.50	S11C	5195	28/28/28/16	152	5	1.0	5.0	5	1	1/2	50/50	55	3000	1.1	1	1	0	REED
1RR2	17.50	S11C	5195	28/28/28/16	611	459	28.5	16.1	459	28	10/23	70/120	110	3900	1.1	1	1	0	REED
2	17.50	S11C	5192	28/28/28/16	1020	409	7.2	56.8	409	7	10/25	100/120	121	3900	1.1	1	1	0	REED
2RR1	17.50	S11C	5192	28/28/28/16	1020	409	9.5	43.1	409	10	10/25	100/120	121	3900	1.1	1	1	0	REED
3	17.50	PS950	JQ1029	28/28/28/22	2414	1394	71.5	19.5	1394	72	20/30	220/250	260	3100	1.6	8	0	8	SMITH
4	12.25	PS951	PS1100	20/20/20/11	2805	391	20.5	19.5	391	20	10/17	120/250	225	2910	1.6	3	0	0	SMITH
5	12.25	MXDG	722PL	14/14/16/16	2904	99	19.7	5.2	99	19	14/16	100/130	235	2500	1.6	5	4	0	HTC
6	12.25	F27DL	LZ1034	14/14/20/ 0	3095	191	57.0	3.4	191	57	25/28	70/110	240	2500	1.6	1	4	0	SMITH
7	12.25	PD1	IH286	14/14/15/15	3213	118	36.6	3.3	118	36	12/18	150/240	240	2500	1.6	8	0	0	DBS
8	12.25	F27DL	L21033	14/14/20/ 0	3329	116	58.0	2.0	116	58	30/33	100/125	220	2660	1.6	8	8	3	SMITH
9	12.25	R7	637	16/16/14/ 0	3330	1	2.0	0.5	1	2	1/20	100/125	206	2100	1.6	8	8	5	HTC
10	12.25	HP21G	BT5379	14/14/18/ 0	3331	1	4.0	0.2	1	4	1/20	100/125	235	2500	1.6	3	6	5	REED
11	10.13	REVCIR		0/ 0/ 0/ 0	3331	0	1.0	0.5	0	1	1/2	100/100	241	2975	1.6	8	0	0	BOWEN
12	12.25	HP21G	BT5378	14/14/18/ 0	3342	11	5.0	2.2	11	5	14/18	100/130	236	2344	1.6	4	2	0	REED
12RR1	12.25	HP21G	BT5378	14/24/18/ 0	3342	0	0.0	0.0	0	1	1/4	100/130	240	2619	1.6	4	2	0	REED
13	8.38	S225	123408	15/15/15/15	3422	80	20.5	3.9	80	20	12/18	150/185	245	1940	1.6	6	0	0	CHR
14C1	8.38	CT303	7890028	0/ 0/ 0/ 0	3429	7	4.5	1.6	7	4	4/11	50/500	90	725	1.6	4	0	0	DBS
15C2	8.38	CT403	7880838	0/ 0/ 0/ 0	3436	7	5.0	1.4	7	5	5/5	130/130	69	725	1.6	1	0	0	DBS
16	8.38	S225	123126	15/15/15/15	3562	126	27.2	4.6	126	27	12/14	170/180	210	1700	1.6	2	0	0	CHR
15RC2	8.38	CT403	7880838	0/ 0/ 0/ 0	3573	11	3.6	3.1	11	4	10/10	120/120	65	618	1.6	0	0	0	DBS
16RR1	8.38	S225	123126	15/15/15/15	3739	166	28.0	5.9	166	28	11/14	150/200	169	1670	1.7	4	0	0	CHR
17	8.38	SVH	XH8744	0/ 0/ 0/ 0	3739	0	0.0	0.0	0	1	0/0	0/0	140	1770	0.0	0	0	0	SMITH

B.H.A Number	Length (meters)	Hours	Description	Date Run
1	142.6	4.0	17.5"BIT-9.5" MWD-BITSUB-26" HOLEOPENER-BITSUB-8" NMDC-7X8" DC.-JAR-3*8" DC-XOVER DEPTH OUT: 147 M.	13/12/88
2	139.3	5.0	17.5"BIT-26"HO-BITSUB-8"NMDC-9X8"DC-JAR-3X8"DC-X.OVER-1*5"HWDP. DEPTH OUT: 152 M.	15/12/88
3	224.1	28.0	17 1/2"BIT-9.5"MWD-26"H.O.-BITSUB-8"NMDC-9X8"DC-JAR-3X8"DC-9X5"HWDP DEPTH OUT: 611 M.	16/12/88
4	242.7	22.0	17.5"BIT-9.5"MWD-9.5"DC-17.5"STAB-9.5"DC-XOVER-8"MONEL-9X8"DC-JAR-X.OVER-9X5"HWDP DEPTH OUT: 1020 M.	17/12/88
5	270.3	21.0	17.5"BIT-17.5"STAB-26"H.O.-15X8"DC-JAR-4X8"DC-X.OVER-9X5"HWDP. DEPTH OUT: 1020 M.	18/12/88
6	311.2	13.0	17.5"BIT-BITSUB-2X9.5"DC-17.5"STAB-X.OVER-9.25"MWD-X.OVER-8"MONEL-15X8"DC-JAR-4X8"DC-X.OVER-9X5"HWDP-DARTSUB. DEPTH OUT: 2414 M.	21/12/88
7	319.0	39.0	12 1/4"BIT-BITSUB-8"SHORTDC-8"MONEL-12 1/4"STAB-20X8"DC-JAR-3X8"DC-X.OVER-9XHWDP-DARTSUB. DEPTH OUT: 2904 M.	02/01/89
8	320.0	19.5	12 1/4"BIT-JUNKSUB-BITSUB-SHOCKSUB-8"MONEL-12 1/4"STAB-20X8"DC-JAR-3X8"DC-X.OVER-9XHWDP-DARTSUB. DEPTH OUT: 3095 M.	03/01/89
9	319.0	24.0	12 1/4"BIT-BITSUB-SHOCKSUB-8"MONEL-12 1/4"STAB-20X8"DC-JAR-3X8"DC-X.OVER-9X5" HWDP-DARTSUB. DEPTH OUT:3329 M.	09/01/89
10	316.3	16.0	12 1/4"BIT-JUNKSUB-BITSUB-8"MONEL-12 1/4"STAB-20X8"DC-JAR-3X8"DC-X.OVER-9X5"HWDP-DARTSUB. DEPTH OUT:3331 M.	12/01/89
11	315.0	8.0	10 1/8"REV.CIRC.JUNK BASKET-8"MONEL-12 1/4"STAB-20X8"DC-JAR-3X8"DC-X.OVER-5XHWDP. DEPTH OUT:3331.5 M	13/01/89
12	315.0	18.0	12 1/4"BIT-JUNK SUB-BITSUB-8"MONEL-12 1/4"STAB-20X8"DC-JAR-3X8"DC-X.OVER-5XHWDP. DEPTH OUT:3342 M.	14/01/89

B.H.A Number	Length (meters)	Hours	Description	Date Run
13	317.0	1.0	12 1/4"BIT-2XJUNK SUB-BITSUB-8"MONEL-12 1/4"STAB-20X8"X.OVER-JAR-3X8"DC-X.OVER-5XHWDP. DEPTH OUT:3342 M.	15/01/89
14	338.3	15.0	8.3/8"BIT-BITSUB-6.5"MONEL-8.3/8"STAB-6 3/8"DC-8.3/8" STAB-21X6 3/8"DC-JAR-3X6 3/8"DC-X.OVER-5XHWDP. DEPTH OUT:3422 M.	19/01/89
15	337.7	22.0	8 3/8"COREHEAD-30MX6 3/4"COREBARREL-X.OVER-20X6 3/8"DC-JAR-3X6 3/8"DC-X.OVER-9X5"HWDP-DARTSUB. DEPTH OUT:3429 M.	22/01/89
16	338.9	24.0	8 3/8"BIT-BITSUB-6 3/8"MONEL-8 3/8"STAB-6 3/8"DC-8 3/8"STAB-21X6 3/8"DC-JAR-3X6 3/8"DC-X.OVER-9X5"HWDP-DARTSUB. DEPTH OUT: 3562 M.	24/01/89
17	337.0	4.0	8 3/8"COREHEAD-6 3/4"COREBARREL-X.OVER-20X6 3/8"DC-JAR-3X6 3/8"DC-X.OVER-9X5"HWDP-DARTSUB. DEPTH OUT:3573 M.	25/01/89
18	338.9	12.1	8 3/8"BIT-BITSUB-8 3/8"MONEL-8 3/8"DC-STAB-6 3/8"DC-8 3/8"STAB-21X6 3/8"DC-JAR-3X6 3/8"DC-X.OVER-9X5"HWDP-DARTSUB. DEPTH OUT: 3739 M.	26/01/89
19	338.0	28.0	8 3/8"BIT-BITSUB-6 3/8"MONEL-8 3/8"STAB-6 3/8"DC-8 3/8"STAB-21X6 3/82DC-JAR-3X6 3/8"DC-X.OVER-9X5"HWDP-DARTSUB. DEPTH OUT: 3739 M.	28/01/89

APPENDIX 8.

DIRECTIONAL DATA SUMMARY

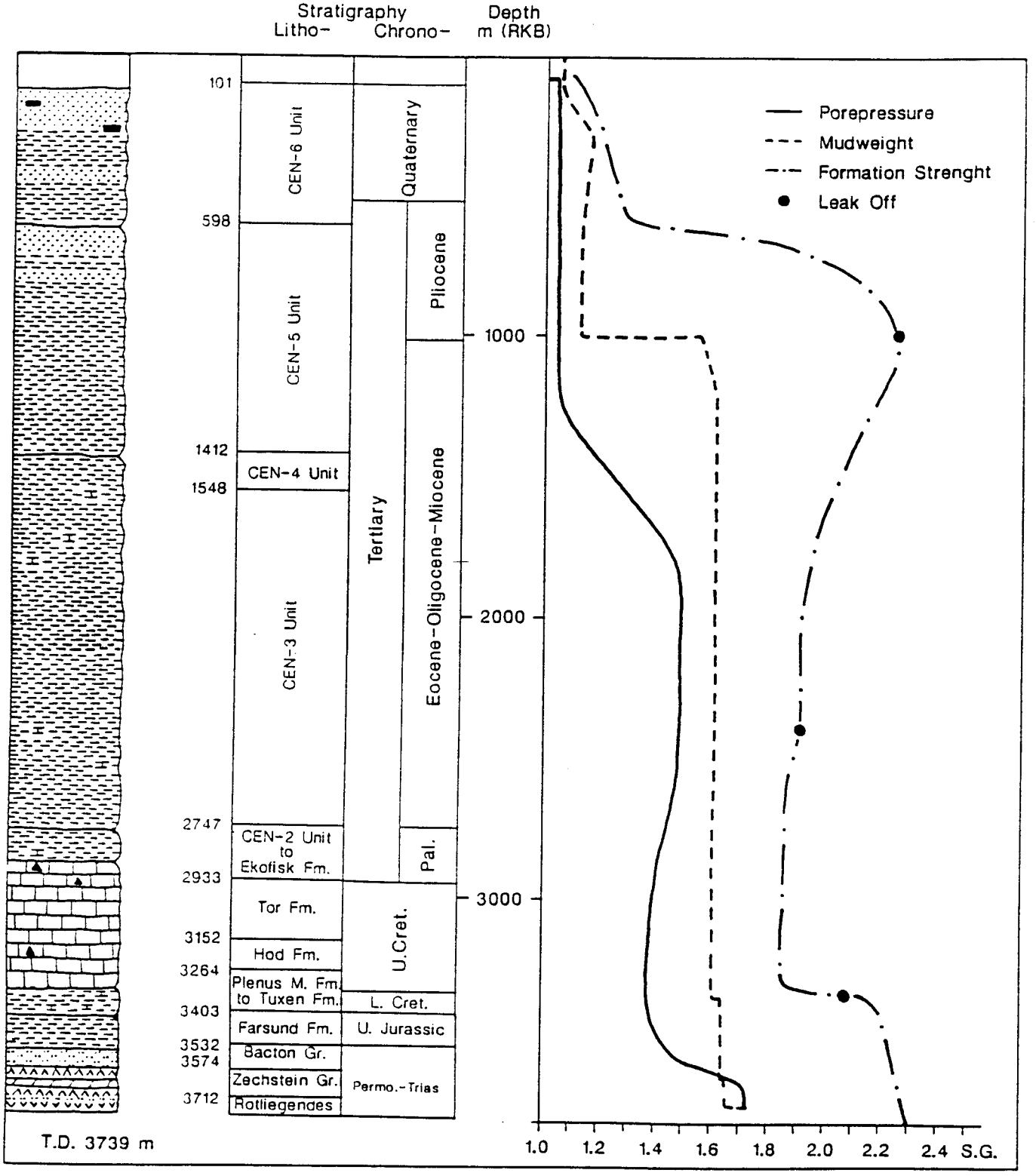
M.D. R/T (m)	V.D. R/T (m)	Inc (deg)	Azm (deg)	E/W (m)	N/S (m)	Clos (m)	Clos Azm (deg)	DLS (deg/ 30 m)
36.0	36.0	0.00	0.00	0.00	0.00	0.00	0.00	0.000
101.0	101.0	0.00	0.00	0.00	0.00	0.00	0.00	0.000
122.0	122.0	0.50	0.00	0.00	0.09	0.09	0.00	7.688
144.0	144.0	0.40	0.00	0.00	0.26	0.26	0.00	1.468
251.0	251.0	0.40	265.00	-0.49	0.71	0.87	325.52	0.014
348.0	348.0	1.10	261.00	-1.75	0.56	1.84	287.70	2.330
393.0	393.0	1.80	264.00	-2.88	0.41	2.91	278.10	5.023
444.0	443.9	2.20	263.00	-4.65	0.21	4.65	272.57	2.533
479.0	478.9	2.50	262.00	-6.07	0.02	6.07	270.20	2.768
548.0	547.8	2.90	270.00	-9.31	-0.21	9.31	268.73	1.874
597.0	596.7	3.90	269.00	-12.22	-0.23	12.22	268.92	6.590
672.0	671.6	2.20	269.00	-16.21	-0.30	16.21	268.94	7.319
720.0	719.6	1.60	270.00	-17.80	-0.31	17.80	268.99	4.036
768.0	767.6	1.30	276.00	-19.01	-0.25	19.01	269.24	2.018
815.0	814.6	1.10	284.00	-19.98	-0.08	19.98	269.77	1.374
922.0	921.6	0.60	312.00	-21.37	0.66	21.38	271.76	1.509
1015.0	1014.6	0.40	349.00	-21.76	1.35	21.80	273.55	0.694
1061.0	1060.6	0.20	319.00	-21.86	1.57	21.92	274.10	1.404
1107.0	1106.6	0.30	342.00	-21.96	1.74	22.03	274.53	0.702
1215.0	1214.6	0.30	338.00	-22.15	2.27	22.27	275.85	0.000
1261.0	1260.6	0.10	332.00	-22.22	2.42	22.35	276.20	1.404
1369.0	1368.6	0.10	328.00	-22.32	2.58	22.46	276.59	0.000
1465.0	1464.6	0.10	352.00	-22.37	2.74	22.54	276.97	0.000
1560.0	1559.6	0.40	273.00	-22.65	2.99	22.85	277.53	1.020
1660.0	1659.6	0.50	232.00	-23.39	2.76	23.55	276.74	0.323
1771.0	1770.6	0.70	168.00	-23.77	1.73	23.83	274.15	0.582
1867.0	1866.5	0.90	233.00	-24.21	0.54	24.22	271.27	0.675
1970.0	1969.5	0.50	164.00	-24.59	-0.59	24.59	268.64	1.254
2076.0	2075.5	1.10	207.00	-24.72	-2.02	24.81	265.32	1.828
2173.0	2172.5	1.70	193.00	-25.53	-4.25	25.88	260.56	1.998
2378.0	2377.4	1.50	215.00	-27.85	-9.44	29.40	251.27	0.316
2517.0	2516.4	2.00	200.00	-29.80	-13.20	32.59	246.11	1.162
2623.0	2622.3	3.00	320.00	-33.57	-13.86	36.32	247.56	3.207
2716.0	2715.2	1.00	210.00	-36.32	-14.10	38.97	248.78	6.945
2805.0	2804.2	1.25	220.00	-37.33	-15.53	40.43	247.41	0.907
2904.0	2903.1	2.00	235.00	-39.39	-17.42	43.07	246.14	2.447
3095.0	3094.0	2.25	160.00	-41.37	-23.71	47.68	240.19	0.466
3213.0	3211.9	1.75	323.00	-43.89	-25.07	50.54	240.26	1.430
3306.0	3304.9	0.75	190.00	-45.45	-25.45	52.08	240.75	3.473
3404.0	3402.9	1.75	300.00	-47.10	-26.22	53.90	240.90	3.312

M.D. R/T (m)	V.D. R/T (m)	Inc (deg)	Azm (deg)	E/W (m)	N/S (m)	Clos (m)	Clos Azm (deg)	DLS (deg/ 30 m)
3422.0	3420.9	1.50	290.00	-47.56	-26.00	54.20	241.33	4.487
3562.0	3560.8	2.00	220.00	-51.44	-27.04	58.11	242.27	1.170
3739.0	3737.6	3.00	215.00	-56.14	-33.16	65.20	239.43	1.825

APPENDIX 9.

LEAK OFF AND PORE PRESSURE DATA SUMMARY

PORE PRESSURE, MUDWEIGHT AND FORMATION STRENGHT CURVES



APPENDIX 10.

CASING AND WELLHEAD SUMMARY

10. CASING AND WELLHEAD SUMMARY

10.1 Wellhead Summary

A Vetco wellhead system was used consisting of the following:;

A Section: 20" wellhead welded on the 20" casing with a 31" base plate on the 30" conductor. The wellhead section was equipped with a 2V 210 Grayloc hub up, 207 bar working pressure.

B Section: 13 3/8" wellhead with 5V 137 Grayloc hub up, 690 bar working pressure.

C Section: 9 5/8" wellhead with U 112 Grayloc hub up, 1035 bar working pressure.

10.2 Casing Summary

Casing size (in)	Setting depth (m)	Weight LB/FT	Grade	Threads	Length used (m)
30"	151 m	1.5" MT	X-52	Drill-Quip H-60-D	161 m
20"	1002 m	106	K-55	Big Omega	1003
13 3/8"	1344	72	P-110	Butt	1346
	2397	72	N-80	Butt	1053
9 5/8"	3342	53.5	P-110	BDS	3007
	2111-2337	53.5	MW-155	BDS	226

APPENDIX 11.

COST SUMMARY

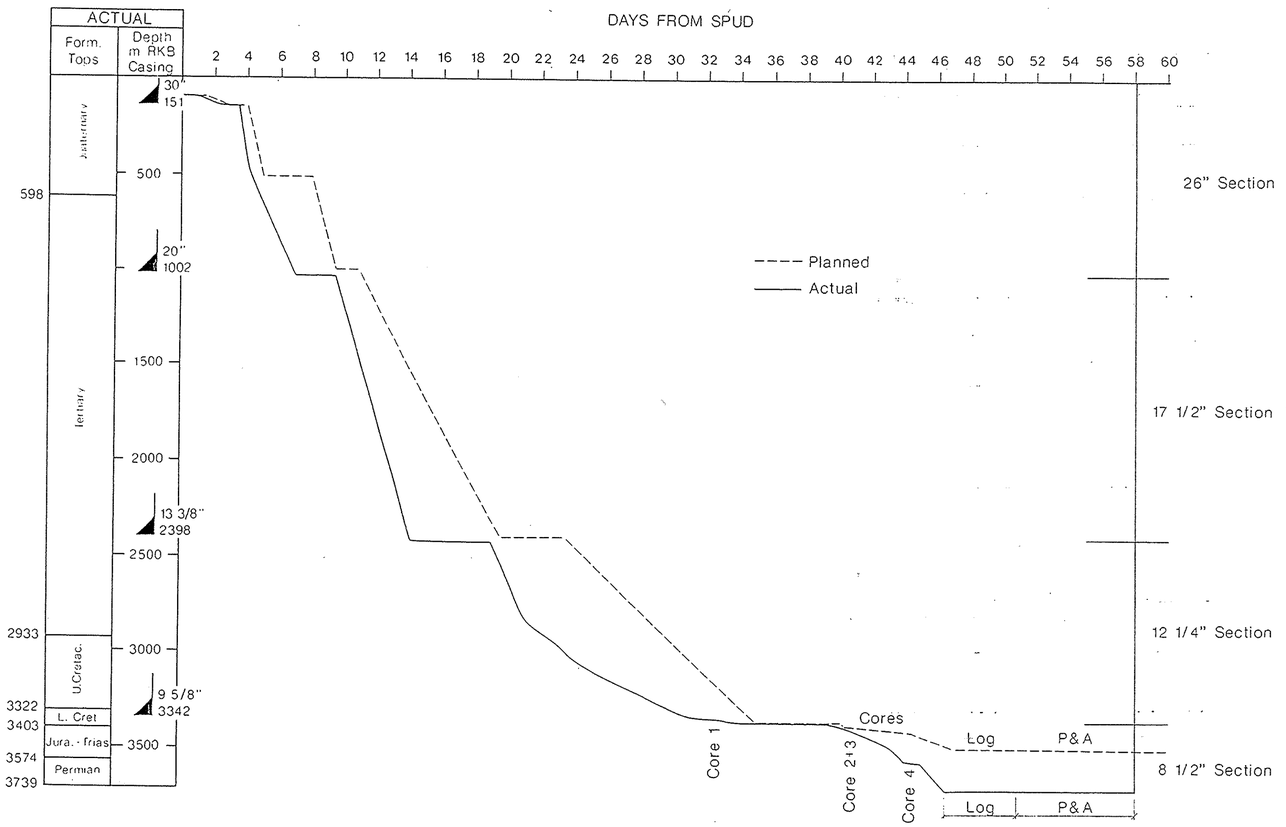
COST SUMMARY

	AFE (1000 DKK)		Final Cost (1000 DKK)		Deviation %
Consumables:		20,500		14,900	÷27%
- Wellhead/MLS	1,250		1,150		
- Conductor and Casing	8,800		8,300		
- Mud, Cement and Chemicals	4,400		2,750		
- Bits and Coreheads	1,650		1,350		
- Fuel and Lubes	3,900		1,150		
- Misc. Consumables	500		200		
Services:		33,300		32,900	÷1%
- Rig Contract	15,000		15,650		
- Mud,Mudlogging,Cementing and MWD Services	2,350		1,950		
- Electric Logging	3,200		3,400		
- Base Services	1,400		1,100		
- Supply/st.by Vessels	3,400		3,750		
- Helicopters	2,500		2,150		
- Site Survey	1,200		1,000		
- Geotechnicial Survey	1,500		1,000		
- Other Services	2,750		2,900		
Manpower:		4,400		4,600	+5%
- Operator's Manpower	2,800		2,900		
- Consultant's Manpower	1,600		1,700		
Other Costs:		850		1,250	+47%
- Travel	200		200		
- Communication	300		750		
- Reports and Analysis	350		300		
N.S.O.C.-D./Oil Spill Contingency		500	350	350	÷30%
Total		59,550		54,000	÷9%

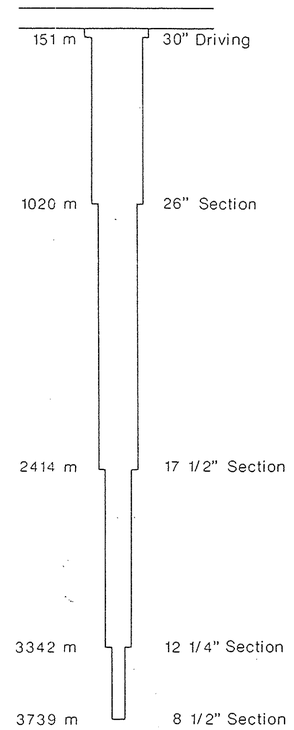
Final cost estimated as of May 1989.

OPERATION DATA

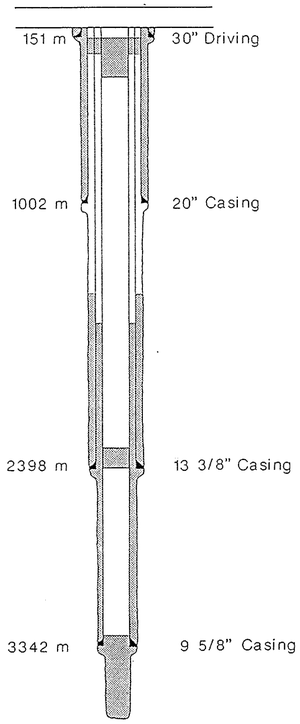
DRILLING CURVE



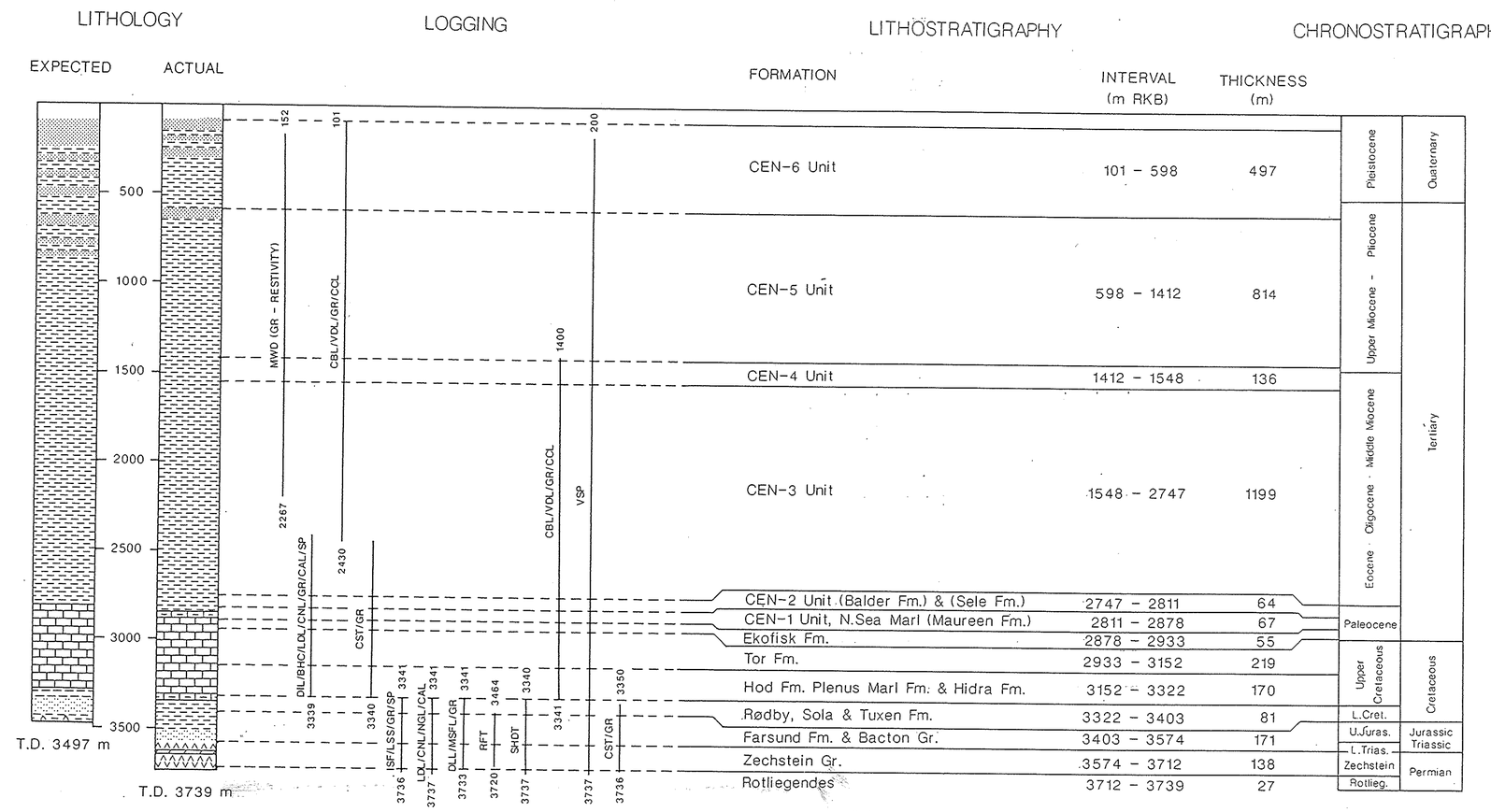
HOLE SIZE



CASING SCHEMATIC



EXPLORATION DATA



3357

GEUS

Report file no 3357

Enclosure (3/6)

LOG SYMBOLS

- CLAY, CLAYSTONE, SHALE
- SILT, SILTSTONE
- SAND, SANDSTONE
- CONGLOMERATE
- LIMESTONE
- DOLOMITE
- MARL, MARLSTONE
- ROCK SALT
- ANHYDRITE
- METAMORPHIC ROCK
- PLUTONIC ROCK
- VOLCANIC ROCK
- VOLCANIC ASH
- CALCAREOUS
- DOLOMITIC
- PYRITE

Dansk Operatørselskab i-s

danop

WELL DATA SHEET

TORDENSKJOLD-1 5503/3-2

Enclosure 1

DES	DRA	DATE	PROJ	NO	REV
KAB	SL	1989-05-16	4/86	M00472	

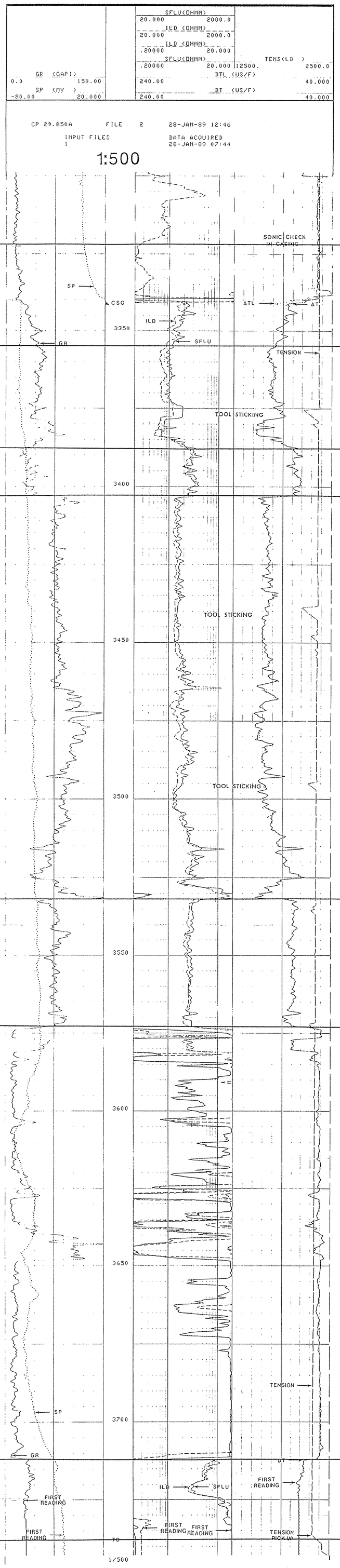
3357

GEUS

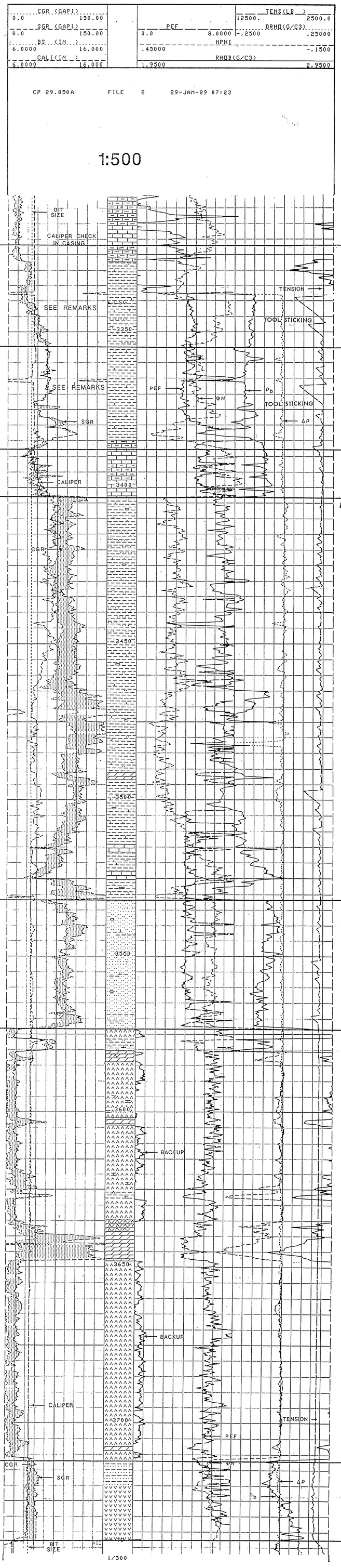
Report file no
3357

Enclosure (4/6)

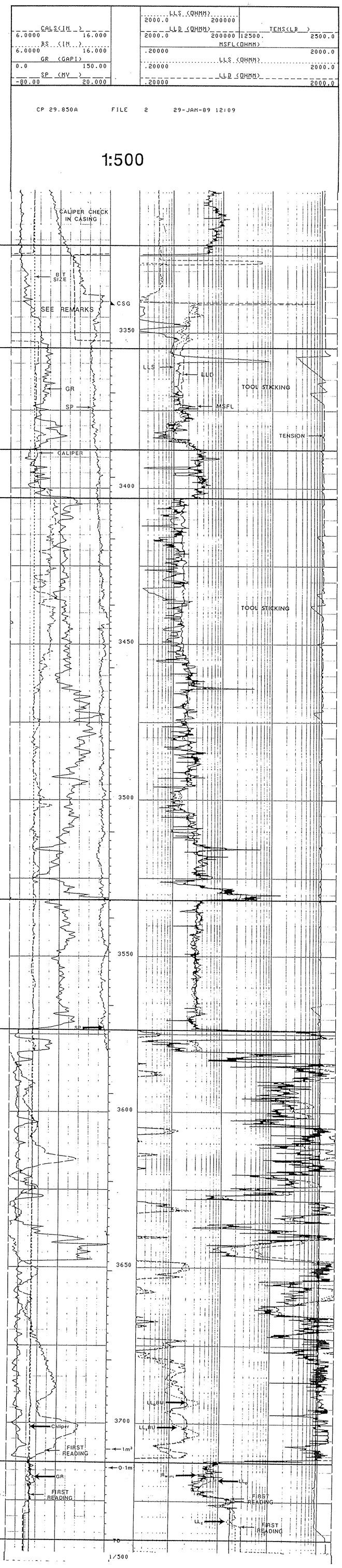
ISF / LSS / GR / SP

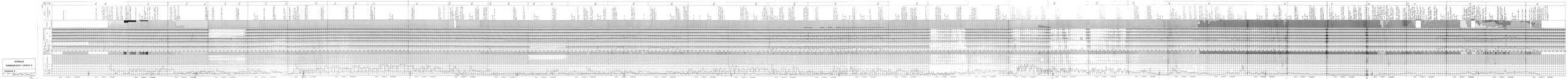


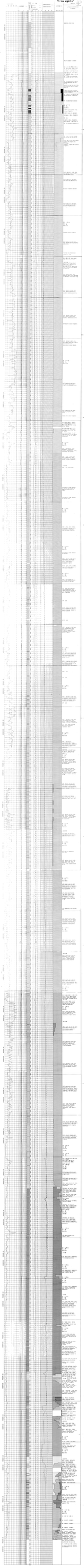
LDL / CNL / NGL



DLL / MSFL / GR




INTERLOS
"LWSDWLOLS = SSSS= 0
Distance =
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000



Dansk Operatørselskab i-s

danop

GRAPHIC WELL LOG

TORDENSKJOLD-1 5503/3-2

Enclosure 5

DES	DRA	DATE	PROJ	NO	REV
		1989-05-16	4/86		

GRAPHIC WELL LOG

Well TORD JSKJOLD-1 (5503/3-2) Sheet 1

dancp

ROP m/hr. (60 50 40 30 20 10 5 0)	Samp.qua.	Cuttings (20 40 60 80)	Depth m 1:500 (RKB)	Lithology	Porosity	Description (Lithology, Shows)	Fluor.	Cut	Ditch Gas (Total %, TG – CG)	Remarks (Casing, Cores, Logs, Tests; SWC, Deviation, Bit Data, Mud Data)	Age
			140			CLAY, med. GREY, SOFT, STICKY, TRACE BIVALVE SHELL FRAGMENTS				sample from 26" hole opener	
			150			SAND, CLAYEY, FINE-GRAINED, WELL SORTED, SUBROUNDED, TRACE GLAUCONITE AND SHELL FRAGMENTS A/A				sample from 17½" bit 151 30"	
			160			SAND, QUARTZ, LOOSE GRAINS, CLEAR, COLOURLESS, FINE-GRAINED, WELL SORTED, ROUNDED; BIVALVE SHELLS, TRACE.				FIRST SAMPLE FROM SHALE SHAKER (15/12-83, 16.00)	
			170			SAND, QUARTZ, LOOSE GRAINS, CLEAR, COLOURLESS, FINE-TO COARSE-GRAINED, POORLY SORTED, SUBROUNDED TO ROUNDED; SHELLS, MODERATE; LIGNITE, CLAYEY, RARE; RARE FINELY INTERLAMINATED CLAYSTONE, BROWNISH BLACK, AND SILTSTONE, LIGHT GREY, SLIGHTLY CALCAREOUS				17½" BIT 26" HOLE OPENER	
			180								
			190			SAND, QUARTZ, LOOSE GRAINS, CLEAR, COLOURLESS, FINE-TO COARSE-GRAINED, SUBANGULAR TO ROUNDED; WOOD FRAGMENTS, LIGHT BROWN, MODERATE TO ABUNDANT; LIGNITE, RARE; AT 200 M MINOR ROCK FRAGMENTS(?), BLACK, HARD, FINE TO VERY COARSE					QUATERNARY
			200			NO SHOWS					

SEE NEXT SHEET

GRAPHIC WELL LOG

Well TORO SKJOLD-1 (5505/5-2)

Sheet 2

danop

ROP m/hr.	Samp.qua.	Cuttings	Depth m 1:500	Lithology	Porosity	Description Lithology Shows	Fluor.	Cut	Ditch Gas Total % TG - CG	Remarks Casing / SWC Cores / Deviation Logs / Bit Data Tests / Mud Data	Age
60 50 40 30 20 10 5 0		20 40 60 80	220 230 240 250 260 270 280	G		GRAINS; CLAYSTONE/SILTSTONE A/A, RARE DISAPPEARING AT 238 M; MICA, RARE; SHELL FRAGMENTS, RARE TO MODERATE; PYRITE, TRACE; GLAUCONITE, TRACE; FORAMINIFERA, TRACE. SAND, QUARTZ, LOOSE GRAINS, CLEAR, COLOURLESS, FINE- TO COARSE-GRAINED BECOMING FINER DOWNWARD, POORLY SORTED, SUBANGULAR TO ROUNDED; WOOD A/A, RARE TO MINOR; LIGNITE, RARE TO MINOR, SHELL FRAGMENTS, TRACE TO RARE; MICA, TRACE TO RARE; GLAUCONITE, TRACE TO RARE; PYRITE, TRACE. NO SHOWS					QUATERNARY

GRAPHIC WELL LOG

Well TOR. NSKJOLD-1 (5503/3-2) Sheet 3

danop

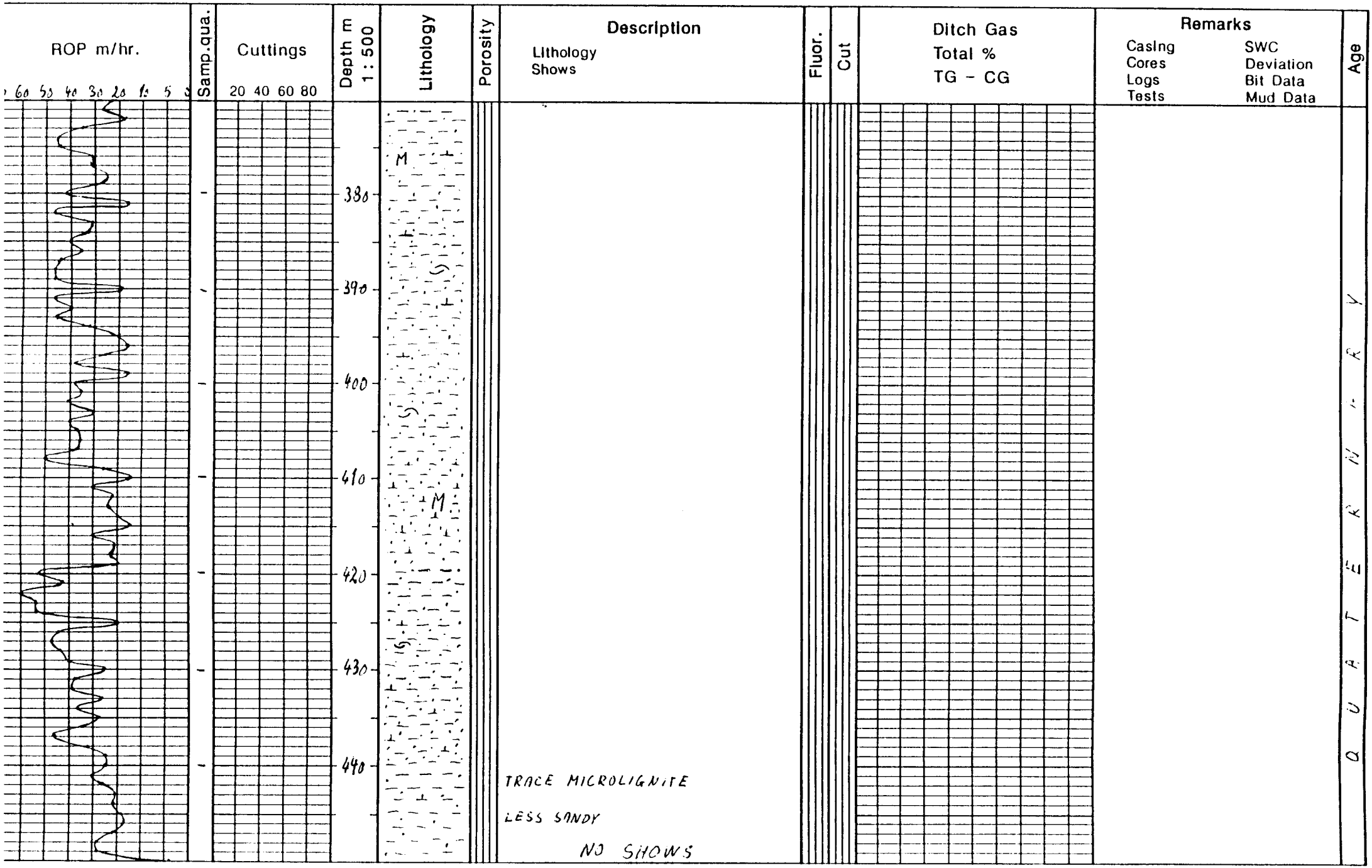
ROP m/hr. (60 50 40 30 20 10 5 0)	Samp.qua.	Cuttings (20 40 60 80)	Depth m 1:500	Lithology	Porosity	Description: Lithology, Shows	Fluor.	Cut	Ditch Gas: Total %, TG – CG	Remarks: Casing, Cores, Logs, Tests, SWC, Deviation, Bit Data, Mud Data	Age
			300	M, G		CLAY, MEDIUM DARK GREY, SOFT, SANDY (VERY FINE TO FINE GRAINS) GRADING TO SILTY DOWNWARD, VARYINGLY CALCAREOUS; MICA, TRACE TO RARE; SHELL FRAGMENTS, TRACE TO RARE; MICROLIGNITE, TRACE FROM 440 M; GLAUCONITE, TRACE TO LOCALLY RARE; PYRITE, TRACE.				15/12 16/12	QUATERNARY
			310								
			320	M							
			330								
			340								
			350	M							
			360			NO SHOWS					

GRAPHIC WELL LOG

Well TORD SKJOLD-1 (5503/3-2)

Sheet 4

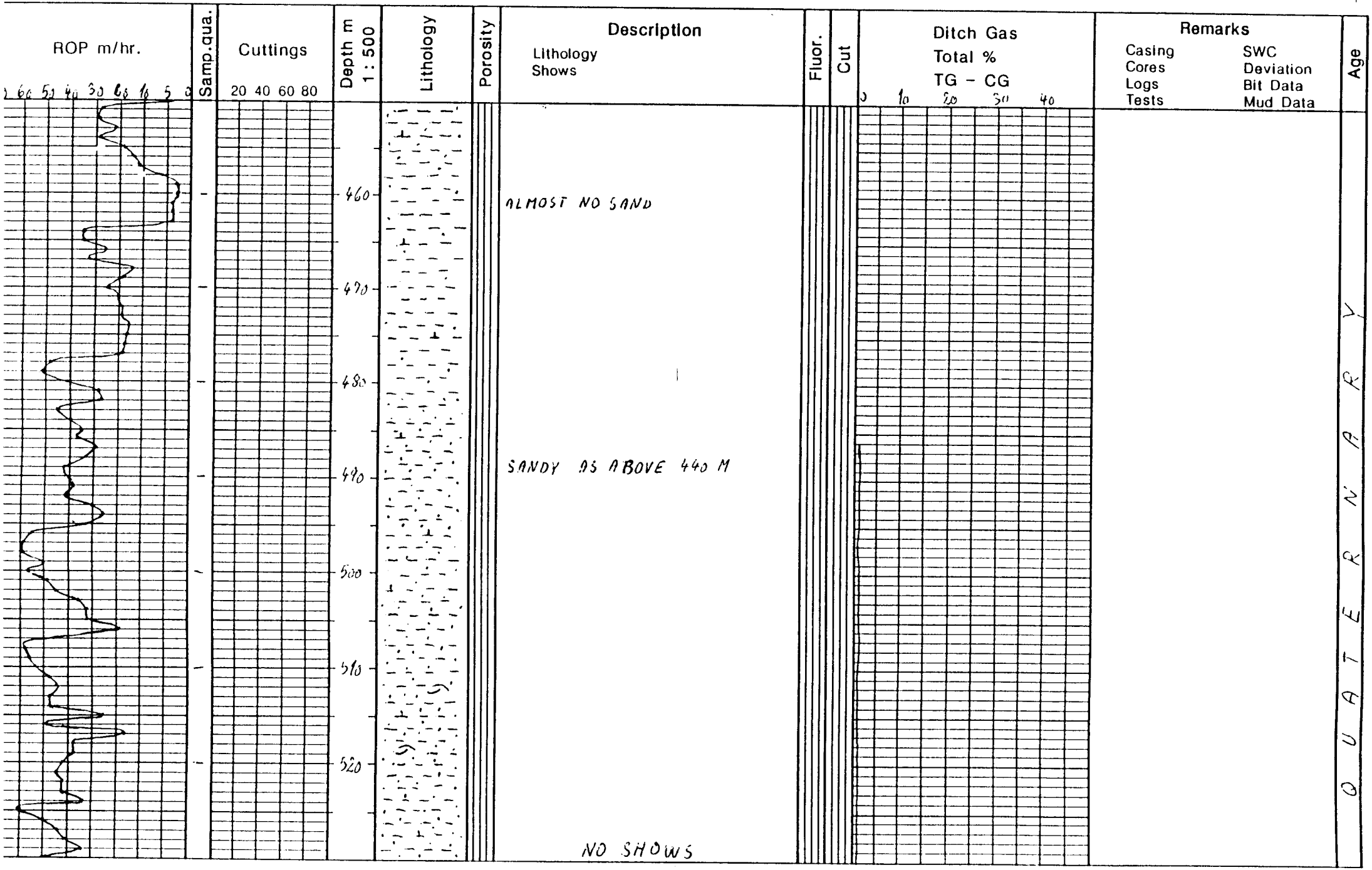
danop



GRAPHIC WELL LOG

Well TORD SKJOLD-1 (3503/3-2) **Sheet** 5

danop

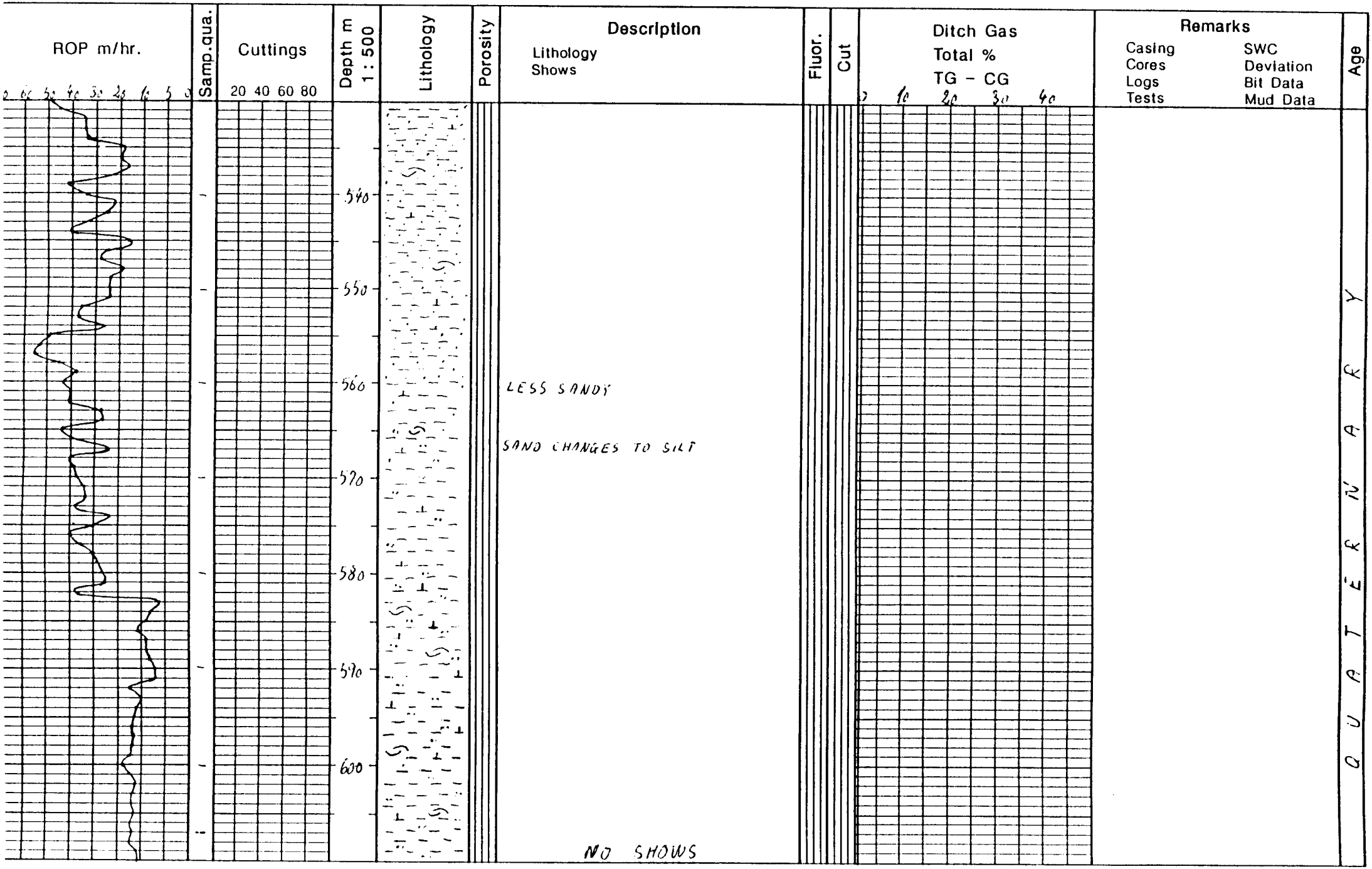


GRAPHIC WELL LOG

Well TORD. SKJOLD 1 (5503/3-2)

Sheet 6

danop

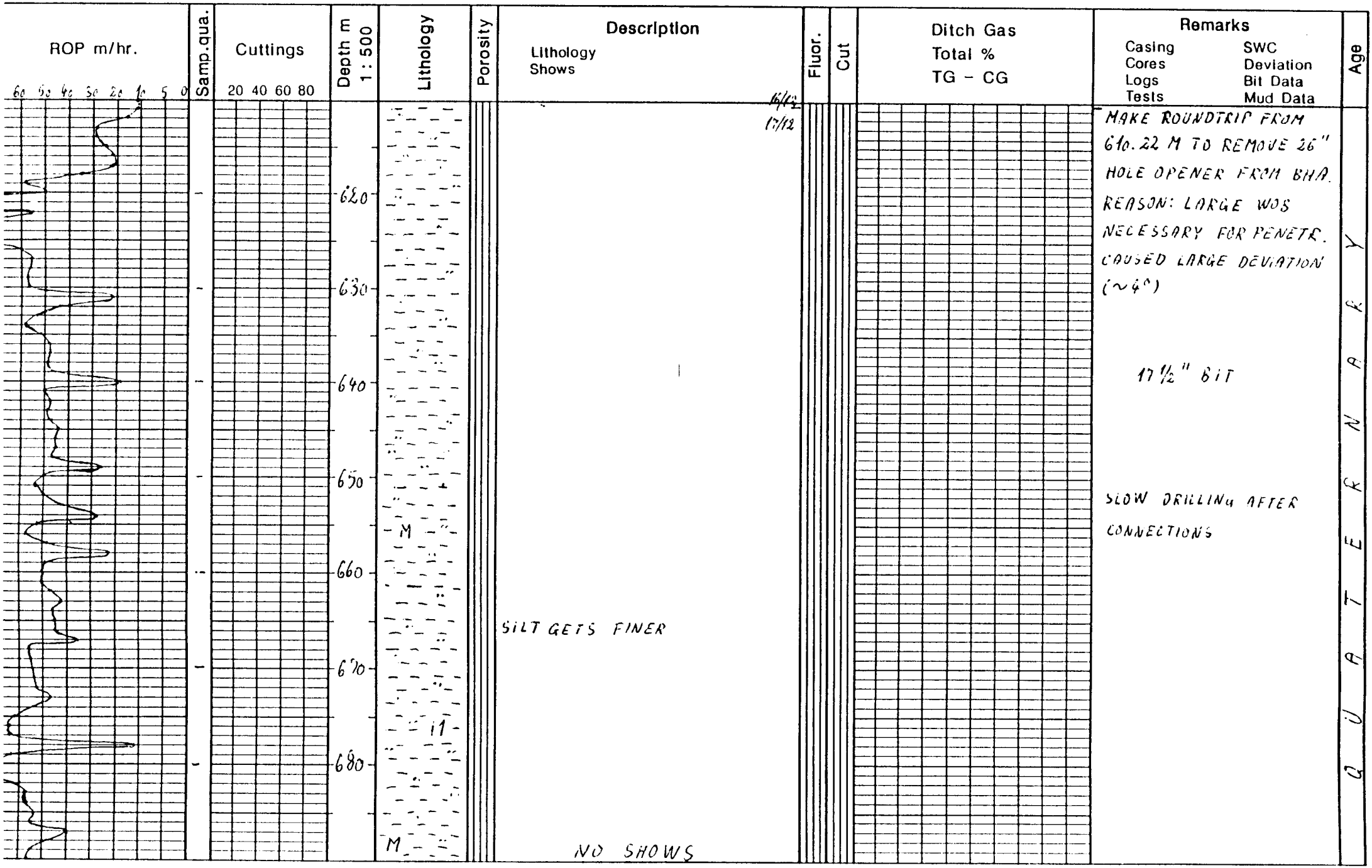


GRAPHIC WELL LOG

Well TORO SKJOLD-1 (6503/3-2)

Sheet 7

danop



GRAPHIC WELL LOG

Well TORD... JSKJOLD - 1 (7503/3-2)

Sheet 8

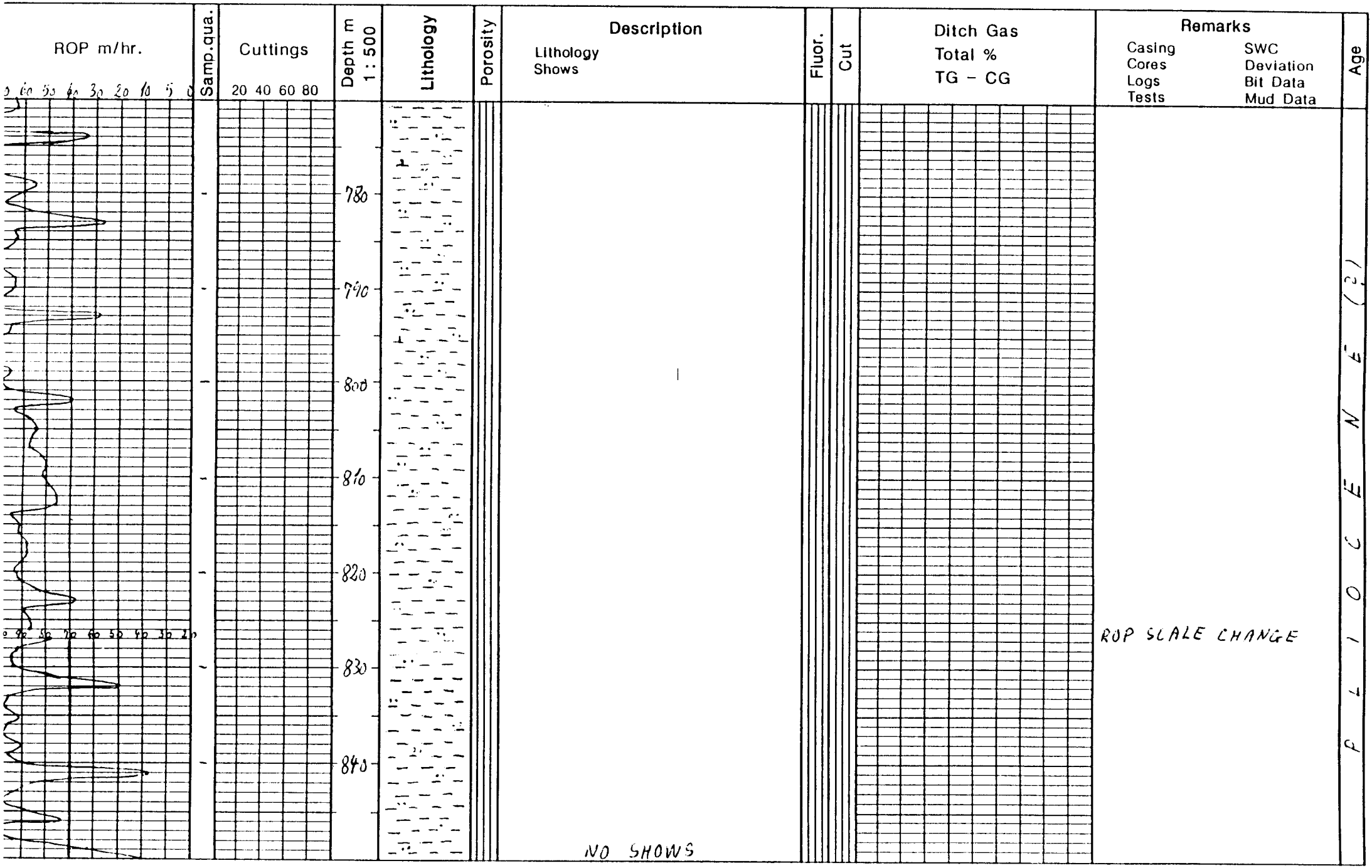
danop

ROP m/hr. 60 50 40 30 20 10 5 0	Samp.qua.	Cuttings 20 40 60 80	Depth m 1:500	Lithology	Porosity	Description: Lithology, Shows	Fluor.	Cut	Ditch Gas: Total %, TG - CG	Remarks: Casing, Cores, Logs, Tests, SWC, Deviation, Bit Data, Mud Data	Age
			700	M							PLIOCENE (?)
			710			SILT GETS STILL FINER					
			720								
			730								
			740								
			750			CLAY GETS SLIGHTLY BLOCKY					
			760			NO SHOWS					

GRAPHIC WELL LOG

Well TORUNSKJOLD-1(5503/3-2) Sheet 9

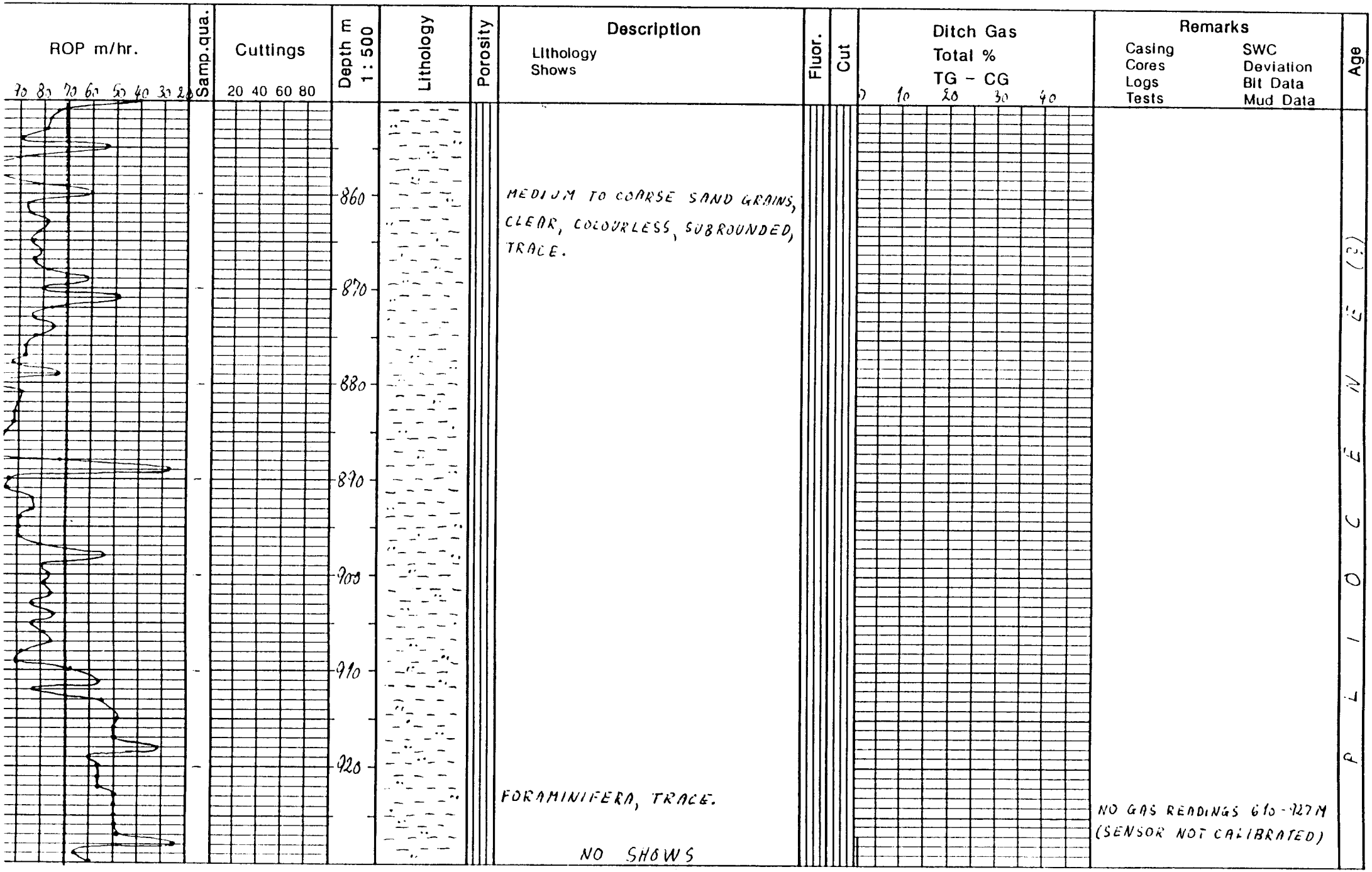
danop



GRAPHIC WELL LOG

Well TORD VSKJOLD - 1 (5503/3-2) Sheet 10

danop

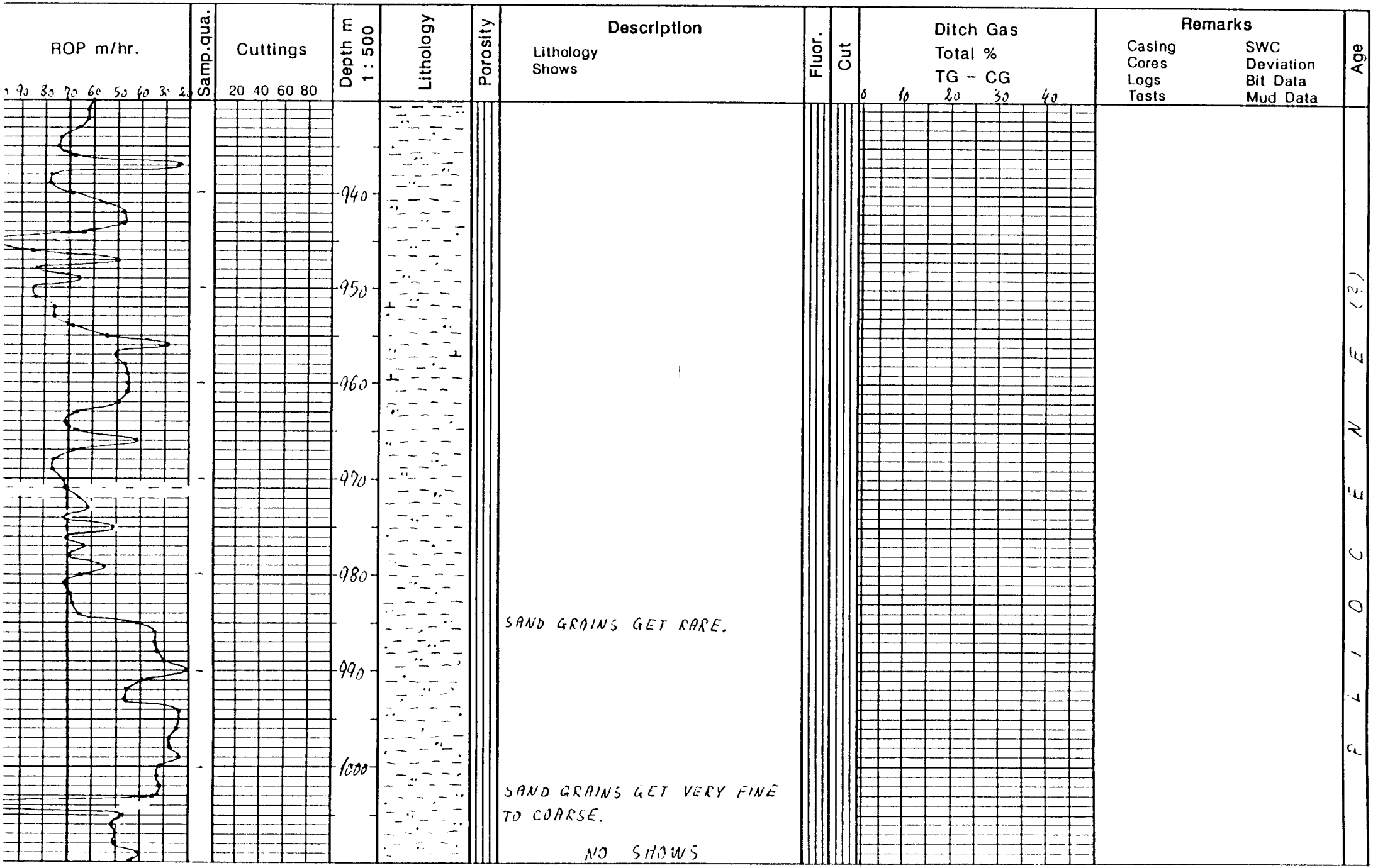


GRAPHIC WELL LOG

Well TORU ISKJOLD-1 (5503/3-2)

Sheet 11

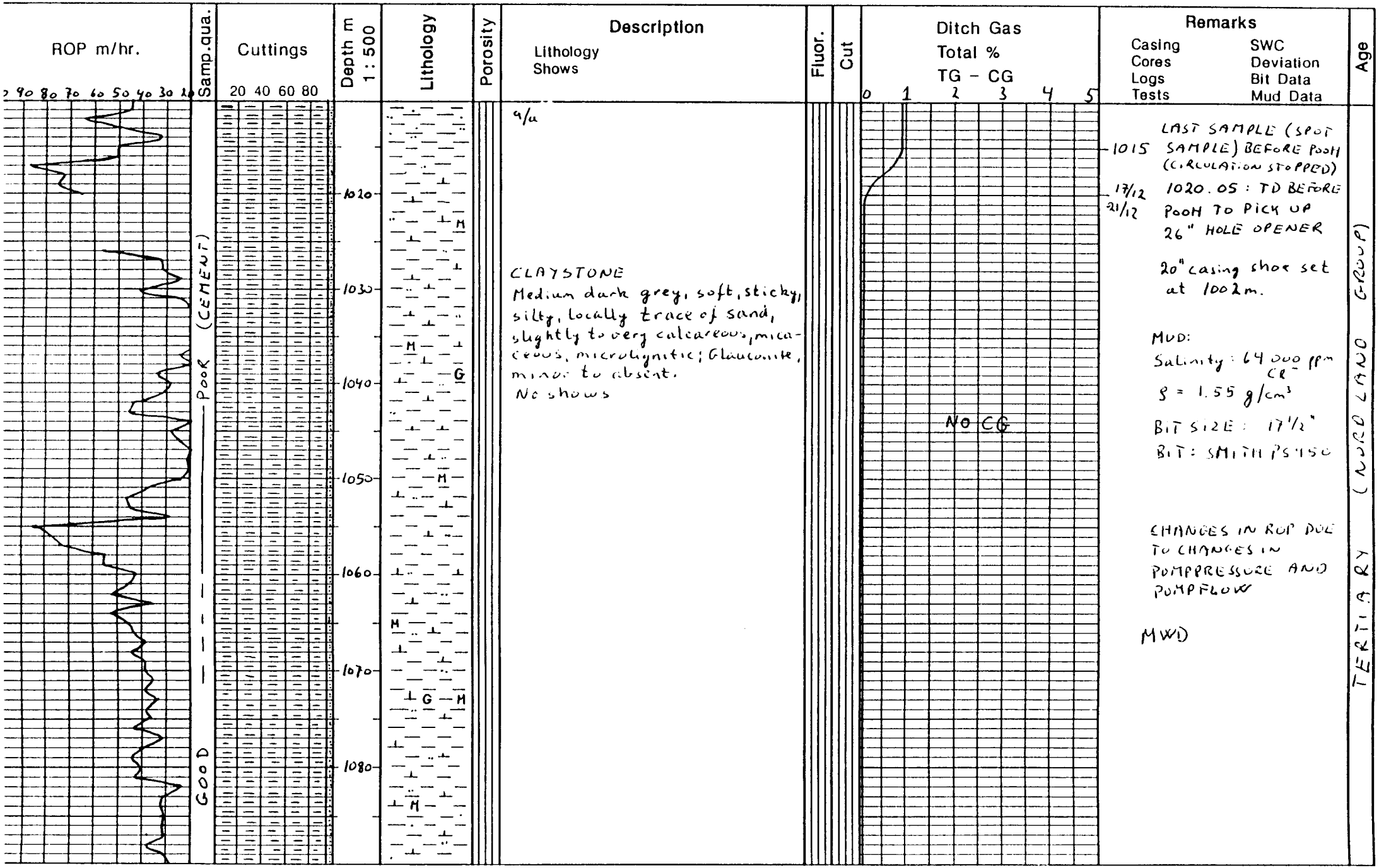
danop



GRAPHIC WELL LOG

Well TORU_NSK30LO-1(5503/3-2) Sheet 12

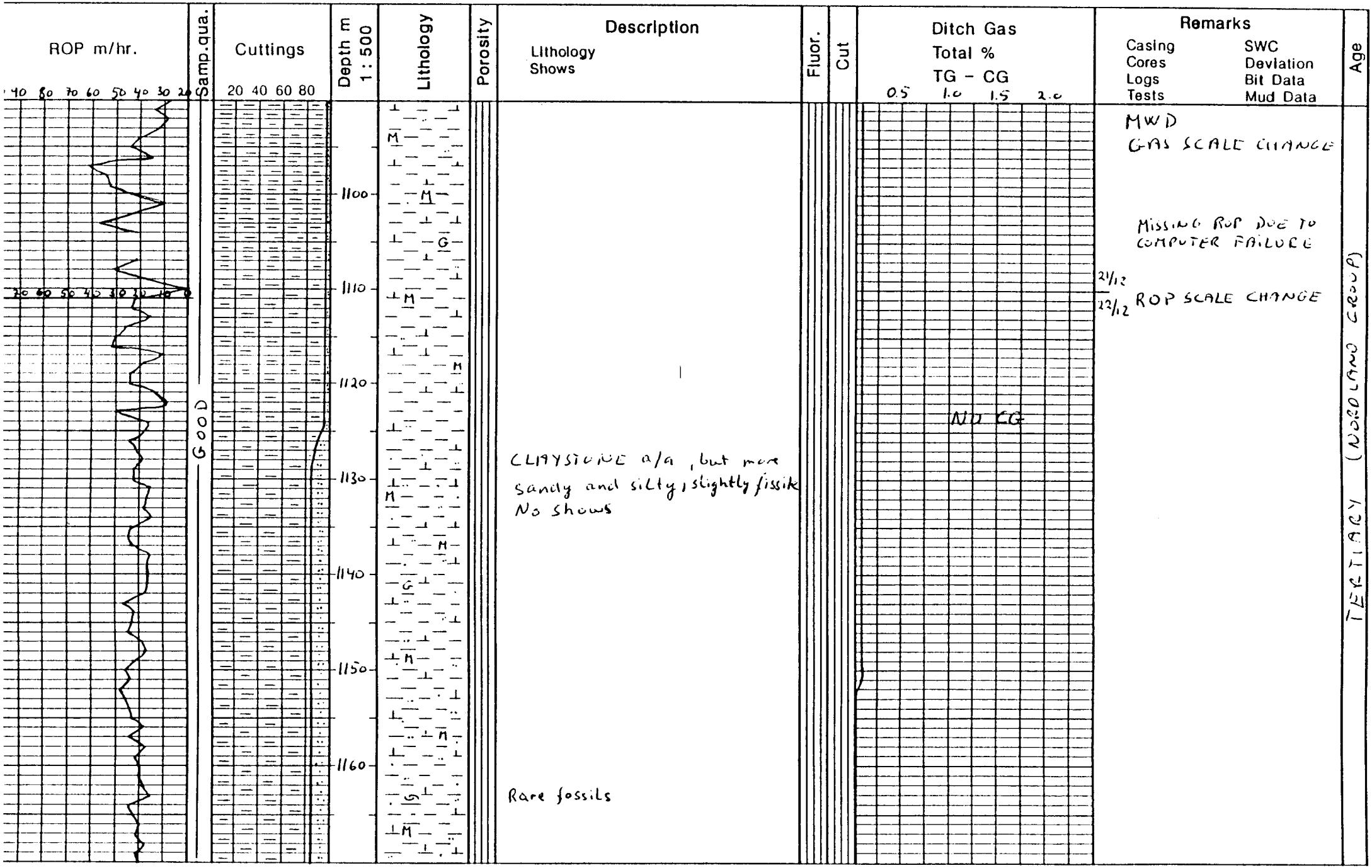
danop



GRAPHIC WELL LOG

Well TONE 1SK30LD-1 (5503/3-2) Sheet

danop

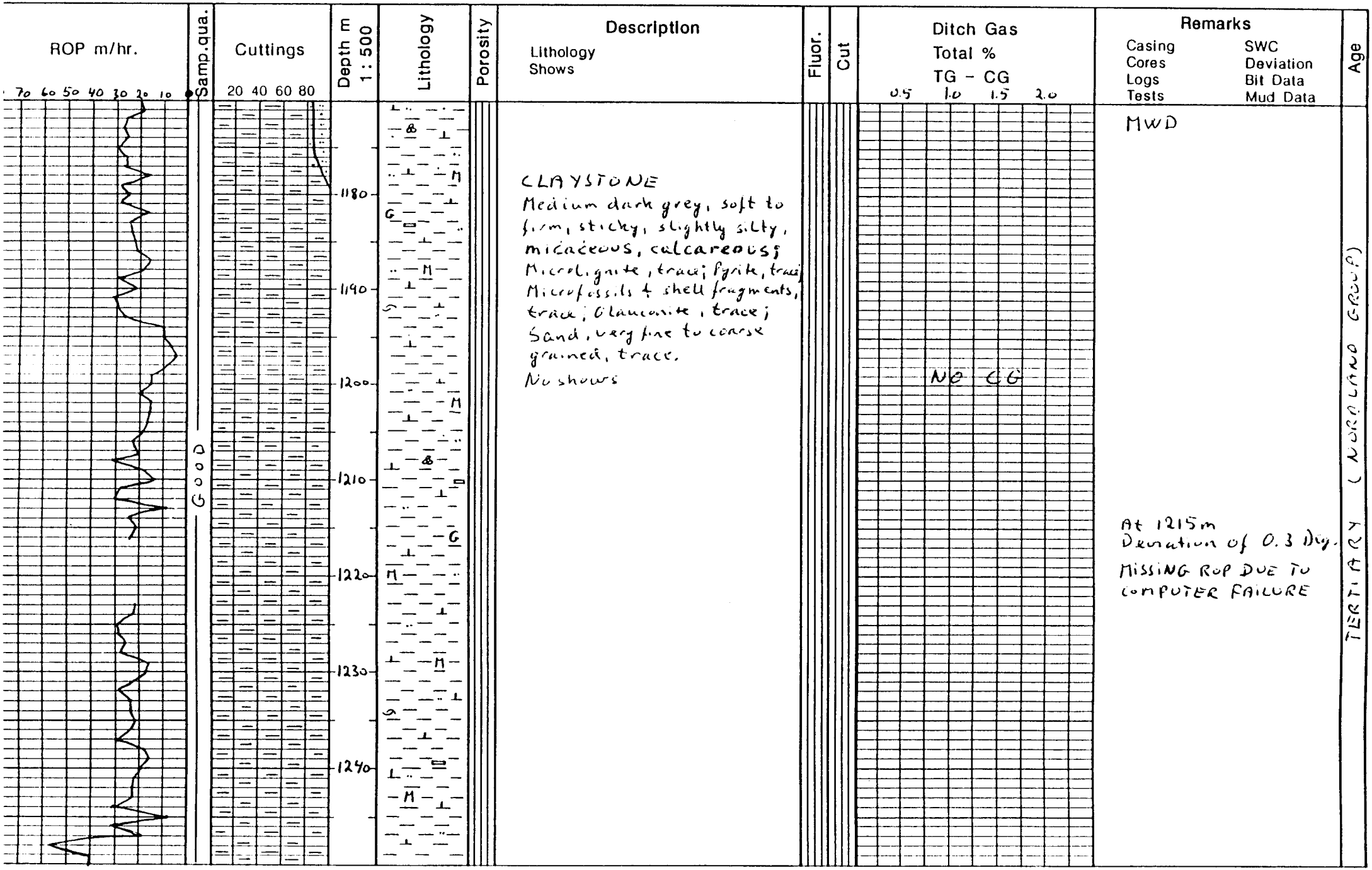


GRAPHIC WELL LOG

Well TUK USKJOLD-1 (5503/3-2)

Sheet 1

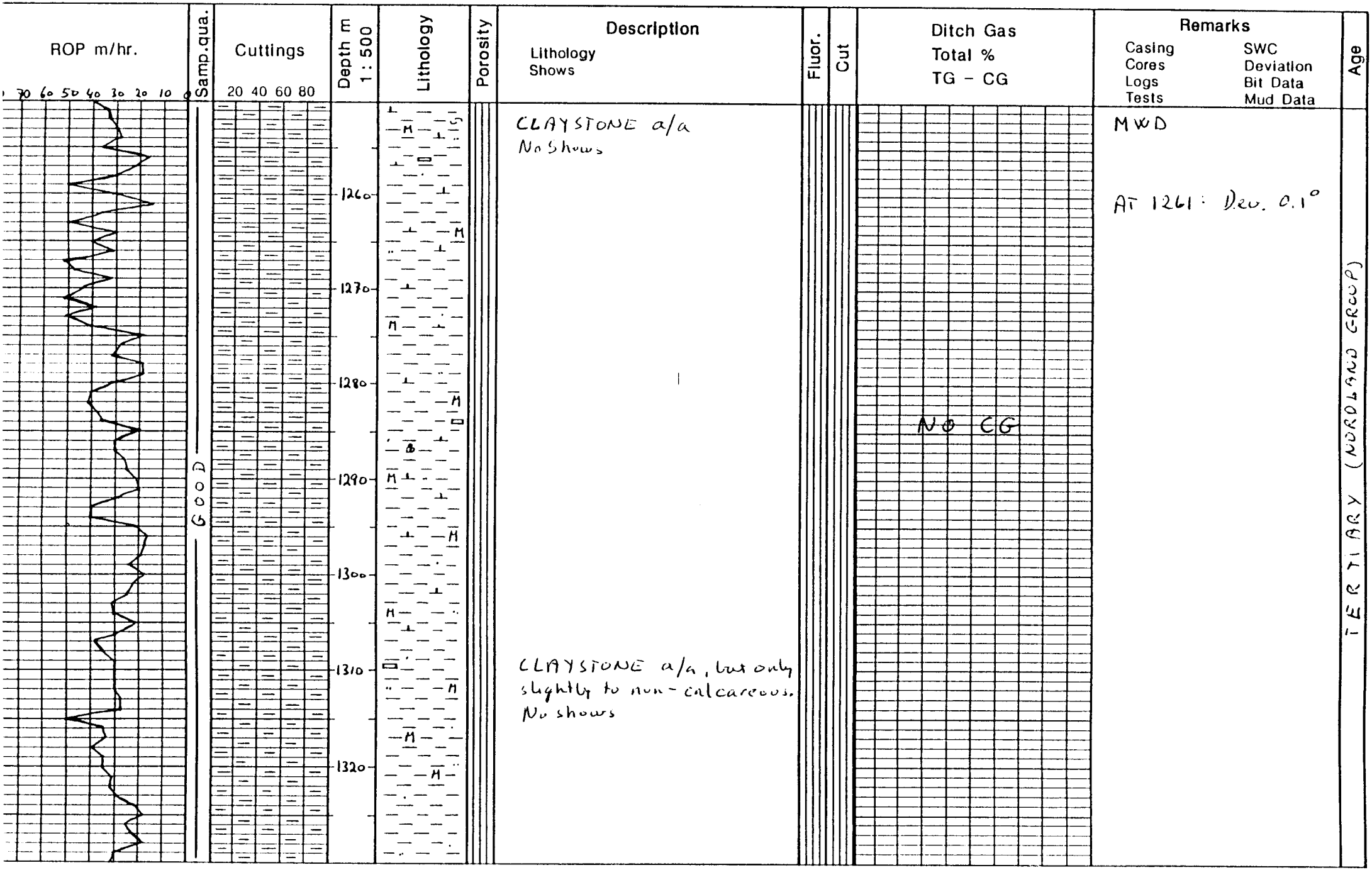
danop



GRAPHIC WELL LOG

Well TOR. NSK3OLD-1 (5503/3-2) Sheet ..

danop



GRAPHIC WELL LOG

Well TORD 35K30LD-1 (5503/3-2)

Sheet 16

danop

ROP m/hr. 70 60 50 40 30 20 10 0

Samp.qua. GOOD

Cuttings 20 40 60 80

Depth m 1:500 — 1340, 1350, 1360, 1370, 1380, 1390, 1400

Lithology

Porosity

Description

Lithology
Shows

CLAYSTONE a/a
No shows

Stringers of
CLAYSTONE
light olive grey, hard, sl. silty, very calcareous, micaceous.

Fluor.

Cut

Ditch Gas
Total %
TG – CG
0.5 1.0 1.5 2.0

NO CG

Remarks

Casing
Cores
Logs
Tests
SWC
Deviation
Bit Data
Mud Data

CHANGE OF GASSCALE

MWD

AT 1360m: Gasline test; Detector OK
At 1369m: Deviation of 0.1 Deg.

22/12
23/12

Age

TERTIARY (NORDLAND GROUP)

GRAPHIC WELL LOG

Well TORDENSKJOLD - 1 (5503/3-2) Sheet 1

danop

ROP m/hr. 80 70 60 50 40 30 20 10

Samp.qua. GOOD

Cuttings 20 40 60 80

Depth m 1:500

Lithology

Porosity

Description
Lithology
Shows

Fluor.

Cut

Ditch Gas
Total %
TG - CG
0.5 1.0 1.5 2.0

Remarks
Casing SWC
Cores Deviation
Logs Bit Data
Tests Mud Data

Age

1420
1430
1440
1450
1460
1470
1480

CLAYSTONE
Medium dark grey, firm, slightly silty, calcareous, slightly fissile, micaceous, locally laminated; Pyrite, trace.
No shows

stringers of CLAYSTONE, as at 1408m

CLAYSTONE
Dusky brown to dusky yellowish brown, firm to moderately hard, slightly to non-calcareous, micaceous; Microlignite, rare to trace; Pyrite, minor to trace; shell fragments, trace to absent
No shows

MWD NORDLAND G.R.

No CG

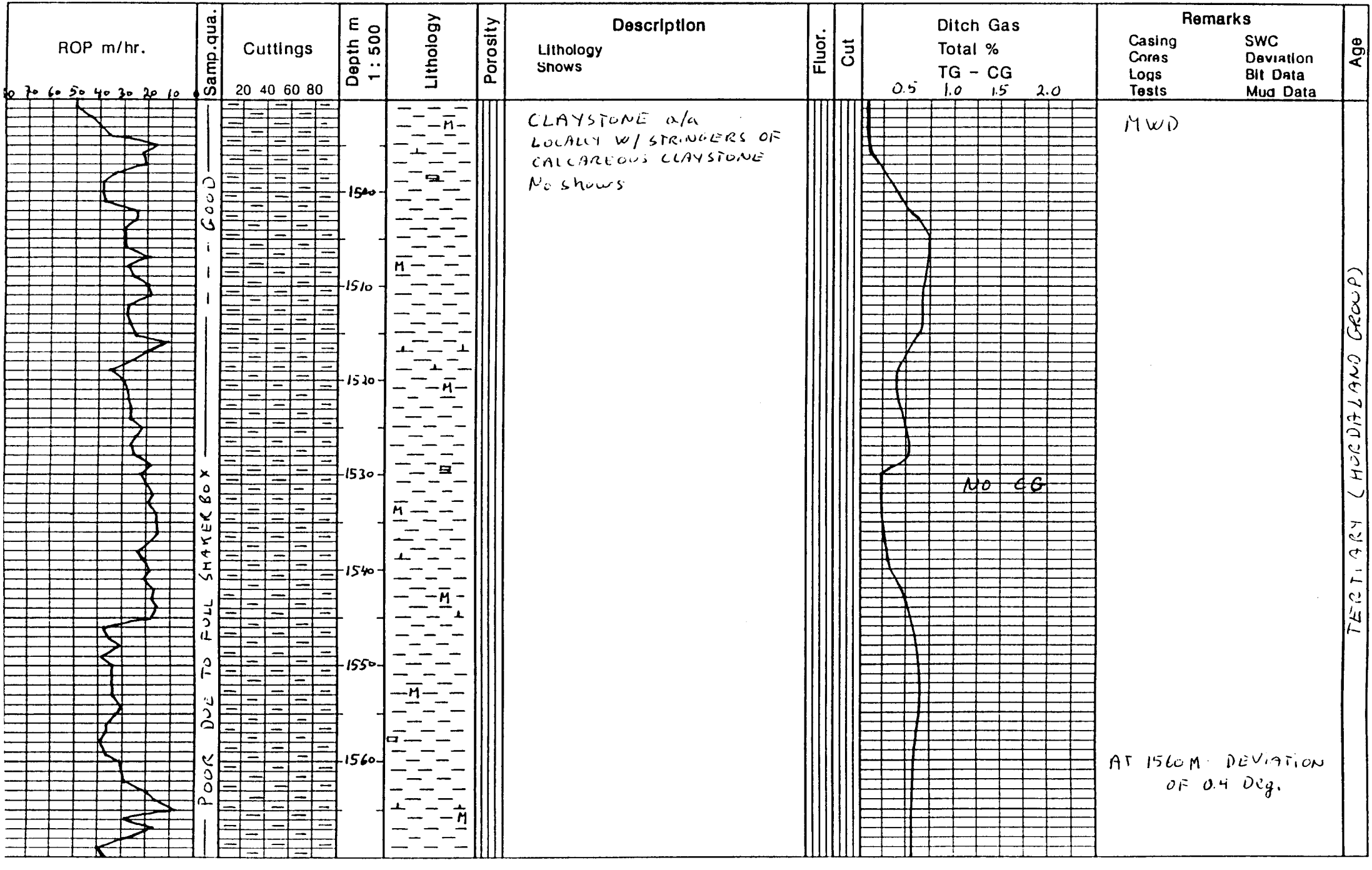
At 1465m Dev. of 0.1 Deg.

TERTIARY (HORDALAND GROUP)

GRAPHIC WELL LOG

Well TORDENSKJOLD - 1 (5503/3-2) Sheet 18

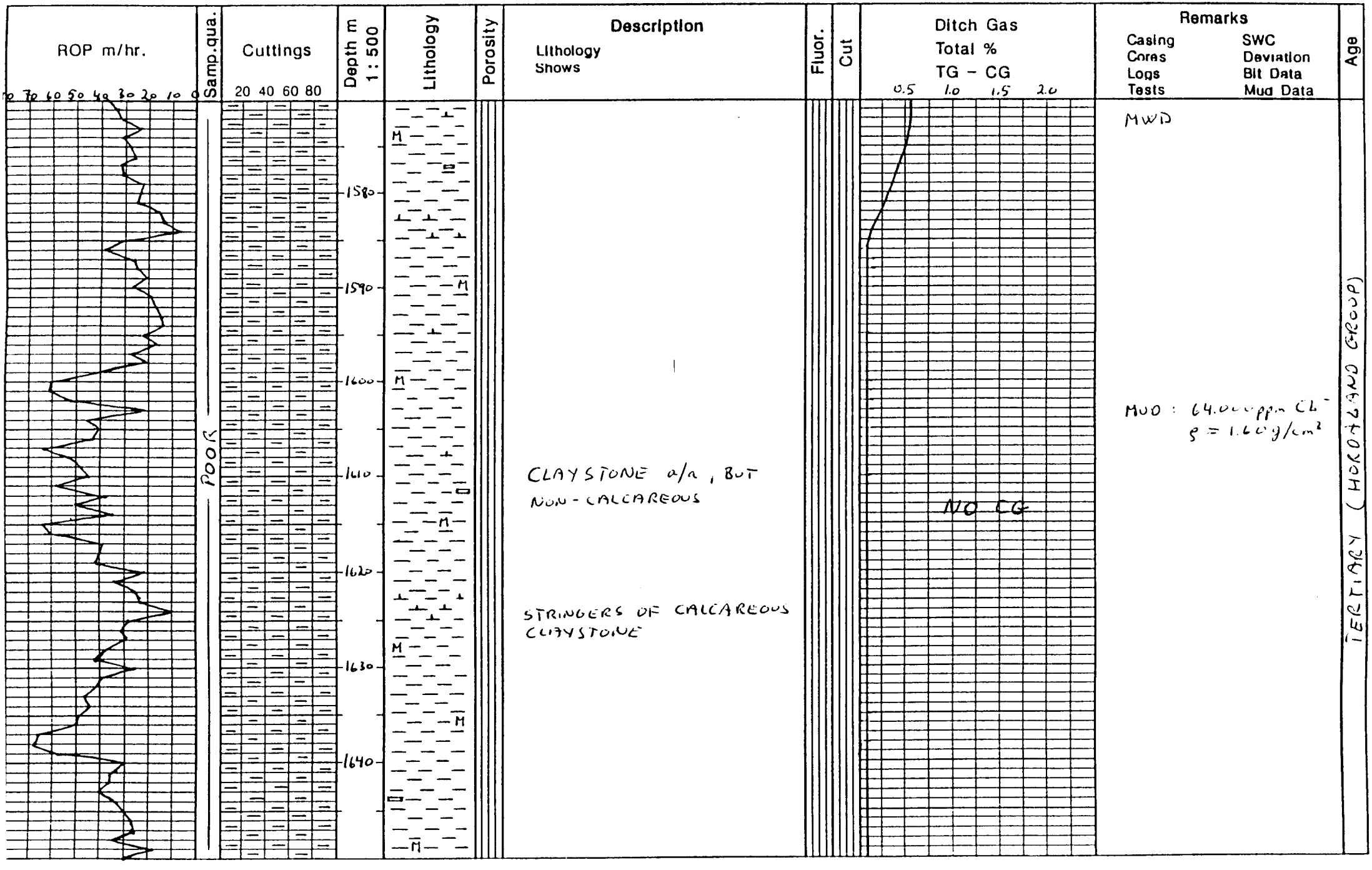
danop



GRAPHIC WELL LOG

Well TO. DENSKJOLD - 1 (5503/3-2) Sheet 19

danop



GRAPHIC WELL LOG

Well TO.. DENSKJOLD - 1 (5503/3-2) Sheet 20

danop

ROP m/hr. 80 70 60 50 40 30 20 10 0

Samp.qua. — Poor — Good

Cuttings 20 40 60 80

Depth m 1:500

Lithology

Porosity

Description
Lithology
Shows

Fluor.

Cut

Ditch Gas
Total %
TG – CG
0.5 1.0 1.5 2.0

Remarks
Casing SWC
Cores Deviation
Logs Bit Data
Tests Mud Data

Age

1660

1670

1680

1690

1700

1710

1720

CLAYSTONE a/a
No shows

STRINGERS OF
MARLSTONE
LIGHT OLIVE GREY, HARD, MICACEOUS

STRINGERS RICH IN
PHOSPHERITE (?) (Ph)
DUSKY GREEN, HARD TO VERY HARD,
AGGREGATES & SINGLE GRAINS.

STRINGERS OF
LIMESTONE
LIGHT OLIVE GREY, HARD,
SLIGHTLY TO VERY ARGILLACEOUS

NO CG

MWD

At 1666 Dev. of
0.5 Deg.

TERTIARY (HORDALAND GROUP)

GRAPHIC WELL LOG

Well TO.. DENSKJOLD - 1 (5503/3-2) Sheet 21

danop

ROP m/hr. 80 70 60 50 40 30 20 10 0

Samp.qua. GOOD

Cuttings 20 40 60 80

Depth m 1:500

Lithology

Porosity

Description
Lithology
Shows

CLAYSTONE a/a
No shows

PHOSPHORITIC STRINGER?

1750

CLAYSTONE a/a,
GETTING CALCAREOUS DOWN-
WARDS AND WITH INCREASING
AMOUNT OF DESSIMINATED
PYRITE

1760

1770

PHOSPHORITIC STRINGER?

1780

1790

LIMESTONE STRINGER a/a

1800

Fluor.

Cut

Ditch Gas
Total %
TG - CG
0.5 1.0 1.5 2.0

NO CG

Remarks
Casing SWC
Cores Deviation
Logs Bit Data
Tests Mud Data

MWD

At 1771 m Dev.
of 0.7 Deg.

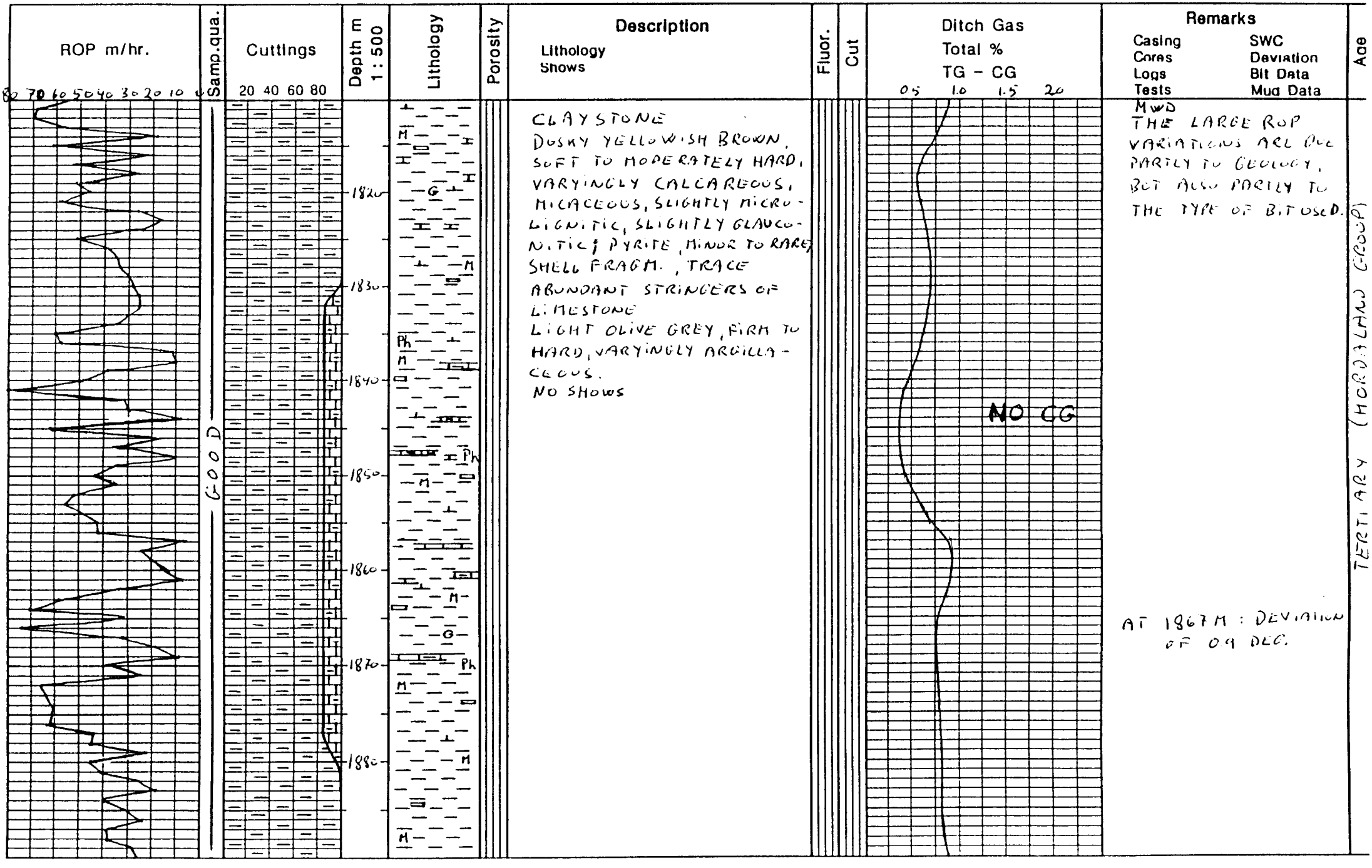
23/12 — 24/12 AT 1801 TRIP TO THE SHOE AT 1062M (12 hr.)

Age

TERTIARY (HORDALAND GROUP)

GRAPHIC WELL LOG

Well TO. DENSKJOLD - 1 (5503/3-2) Sheet 2



GRAPHIC WELL LOG

Well TORDENSKJOLD - 1 (5503/3-2) Sheet 23

danop

ROP m/hr. 80 70 60 50 40 30 20 10 0

Samp.qua.

GOOD

Cuttings 20 40 60 80

Depth m 1:500

1900
1910
1920
1930
1940
1950
1960

Lithology

Porosity

Description

Lithology
Shows

CLAYSTONE a/a
W/STRINGERS OF
LIMESTONE a/a AND
DOLOMITE
BROWNISH GREY TO DARK
YELLOWISH BROWN, HARD,
VARYINGLY ARGILLACEOUS.
NO SHOWS

Fluor.

Cut

Ditch Gas
Total %
TG - CG
0.5 1.0 1.5 2.0

NO CG

Remarks

Casing
Cores
Logs
Tests
SWC
Deviation
Bit Data
Mud Data

MWD

Age

TERTIARY (HORDALAND GROUP)

GRAPHIC WELL LOG

Well TC. DENSKJOLD - 1 (5503/3-2) Sheet 24

danop

ROP m/hr. 80 70 60 50 40 30 20 10 0

Samp.qua. — GOOD

Cuttings 20 40 60 80

Depth m 1:500

1980
1990
2000
2010
2020
2030
2040

Lithology

Porosity

Description

Lithology
Shows

CLAYSTONE
DUSKY YELLOWISH BROWN TO DARK GREY, FIRM TO MODERATE HARD, NON TO VARYINGLY CALCAREOUS, MICACEOUS, SLIGHTLY MICROLIGNITIC; PYRITE, DISPERSED, LAMINAE, AGGREGATES, RARE TO TRACE; PHOSPHORITE (?), TRACE; GLAUCONITE, TRACE

STRINGERS OF LIMESTONE & DOLOMITE A/A.
NO SHOWS

Fluor.

Cut

Ditch Gas
Total %
TG – CG
0.5 1.0 1.5 2.0

NO CG

Remarks

Casing — SWC
Cores — Deviation
Logs — Bit Data
Tests — Mud Data

MWD
AT 1970 M : DEVIATION OF 0.5 DEG.

24/12
25/12

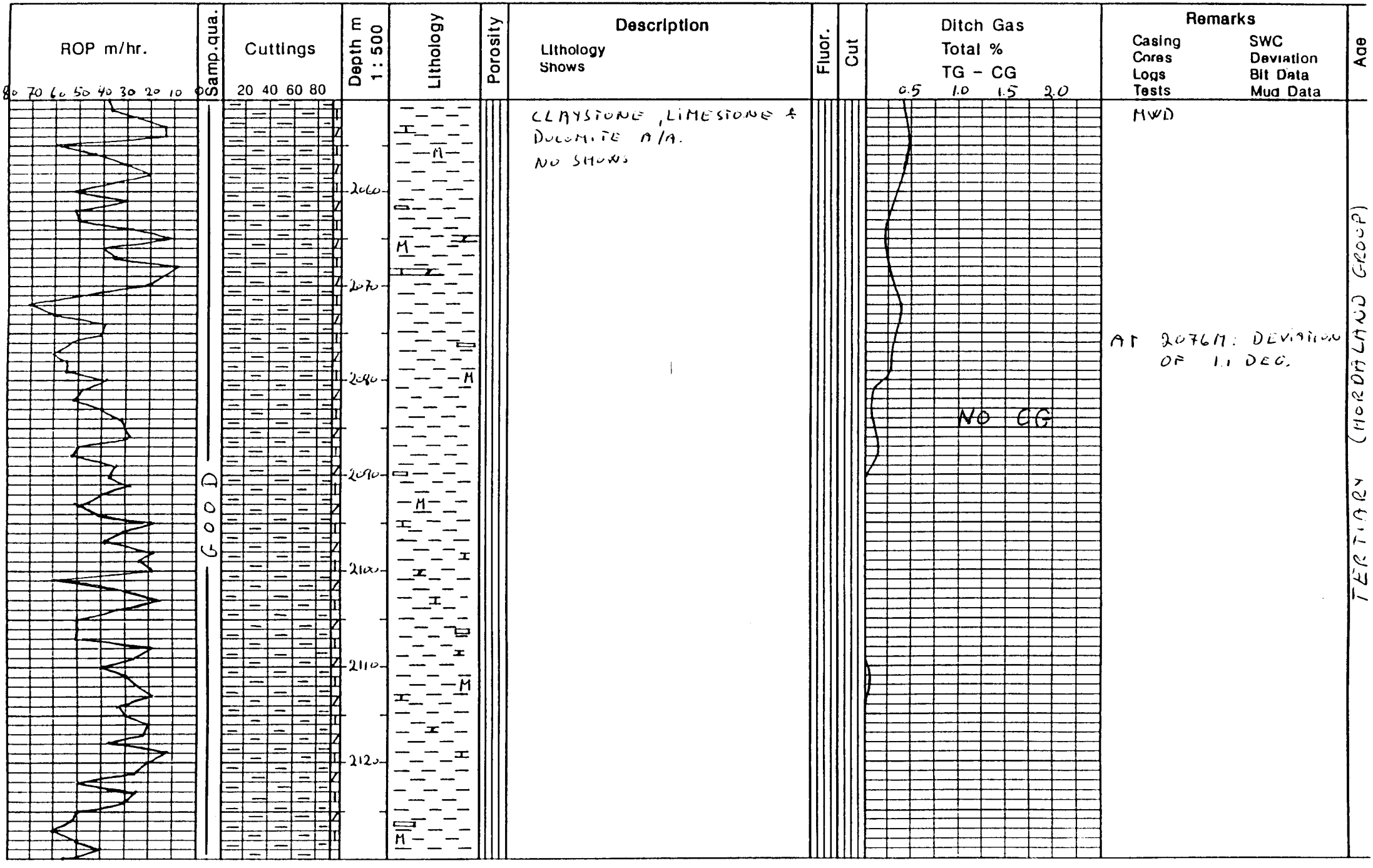
Age

TERTIARY (NORDLAND GROUP)

GRAPHIC WELL LOG

Well TC. DENSKJOLD - 1 (5503/3-2) Sheet 2

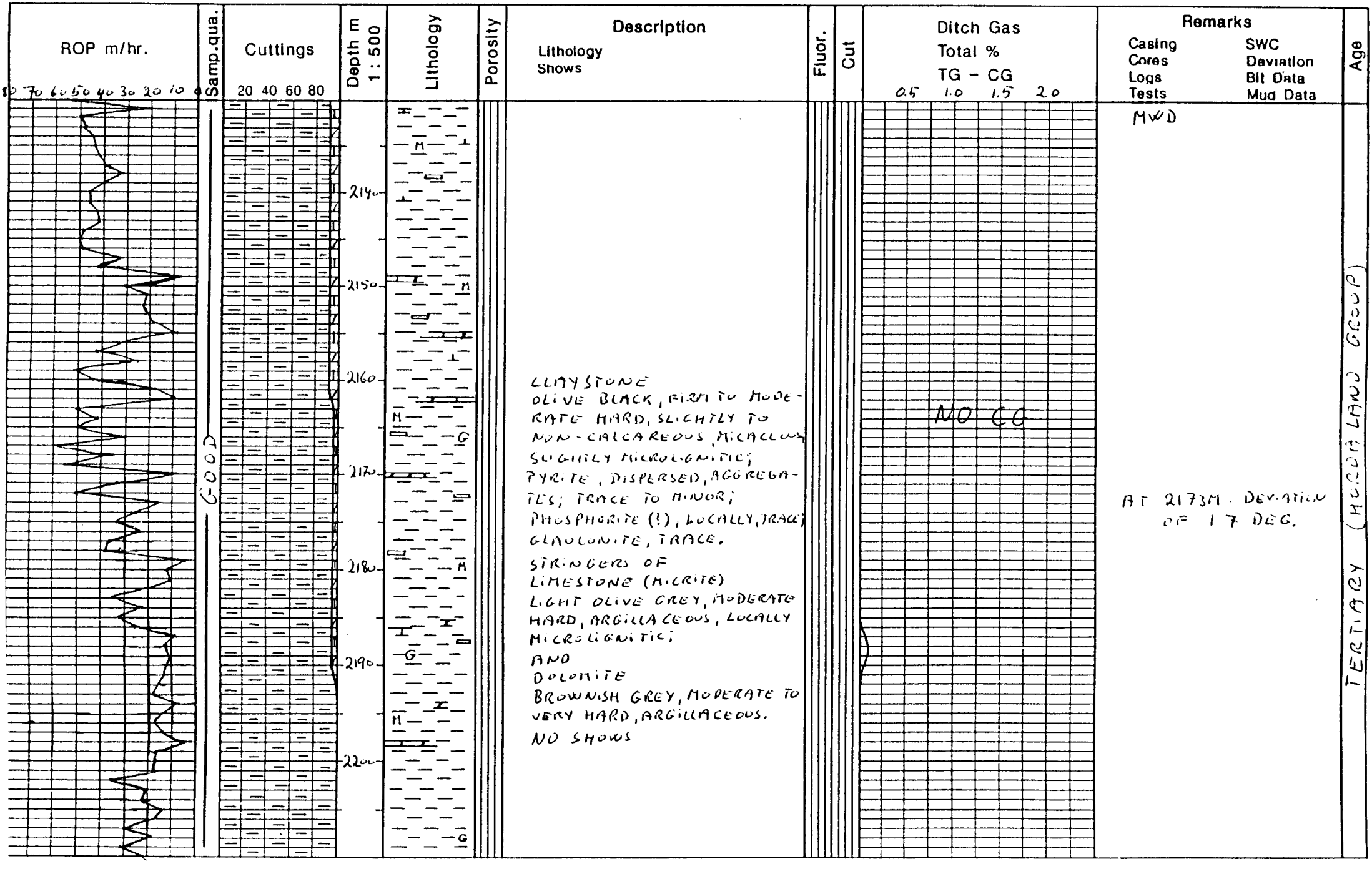
danop



GRAPH.) WELL LOG

Well To DENSKJOLD-1(5503/3-2) Sheet 26

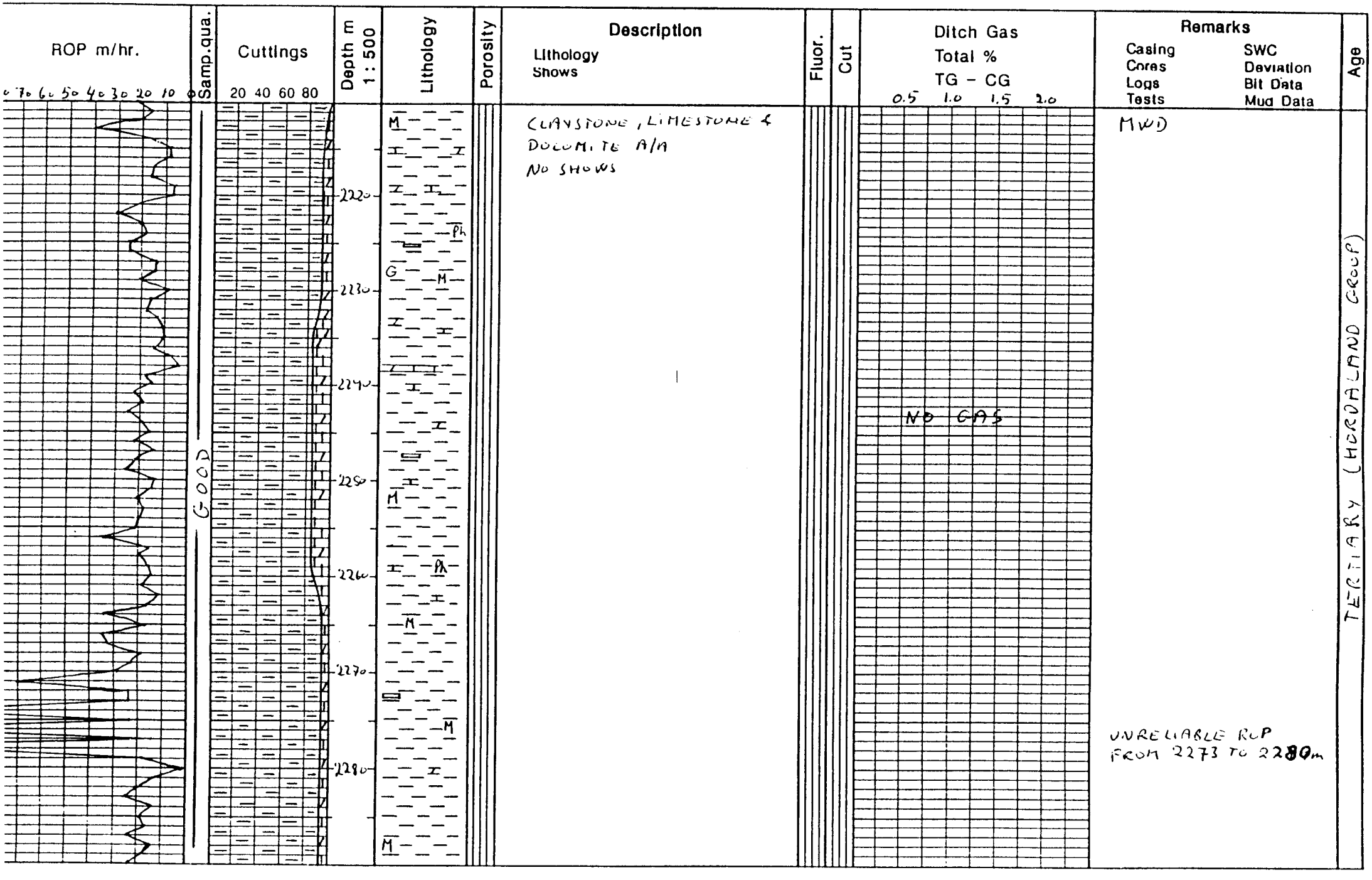
danop



GRAPH WELL LOG

Well TO DENSKJOLD - 1 (5503/3-2) Sheet 27

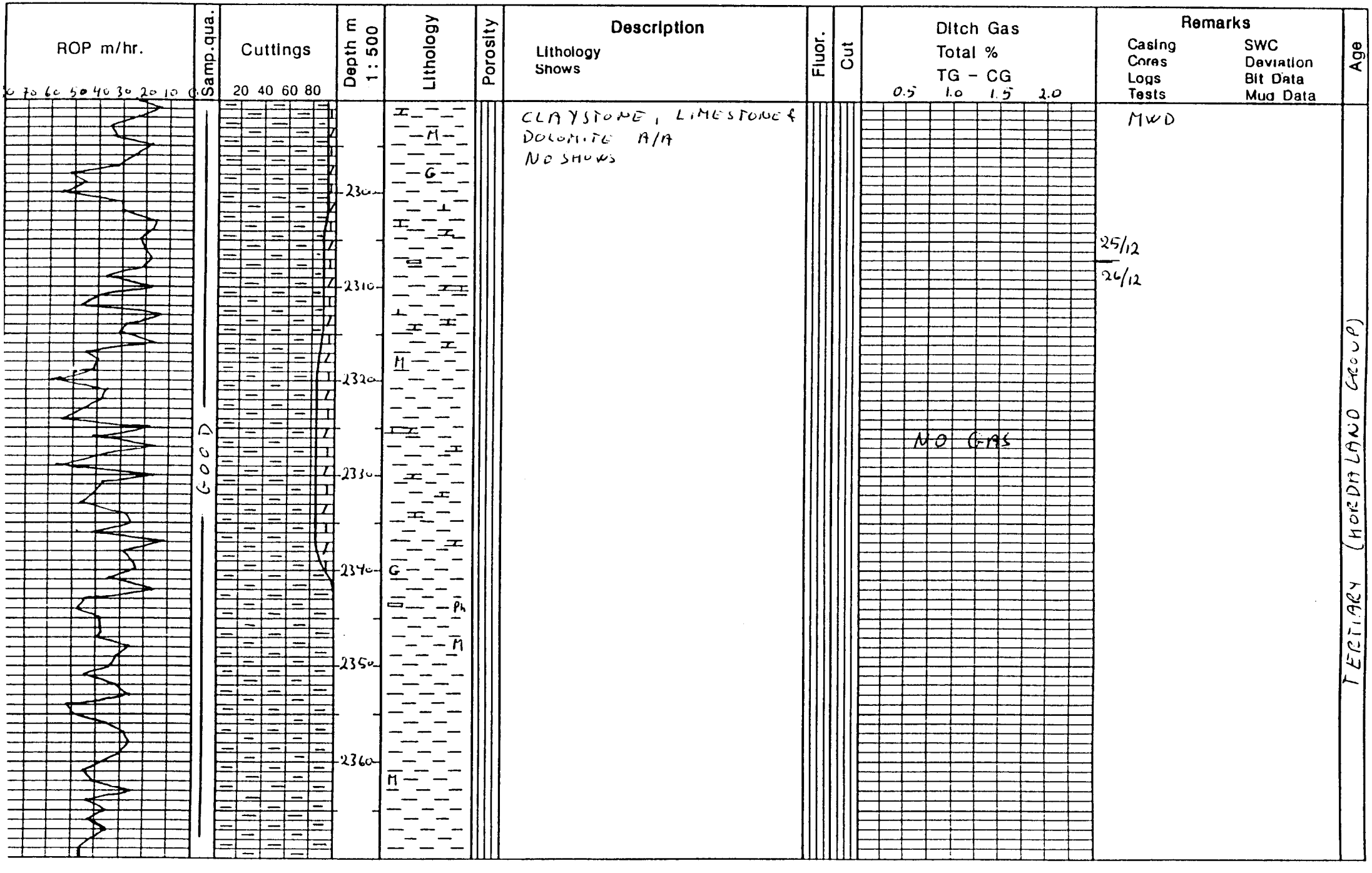
danop



GRAPH WELL LOG

Well TC DENSKJOLD - 1 (5503/3-2) Sheet 28

danop



GRAPHI WELL LOG Well Tor ENSKJOLD - 1 (5503/3-2) Sheet 29

danop

ROP m/hr. 70 60 50 40 30 20 10 0

Samp.qua.

Cuttings 20 40 60 80

Depth m 1:500

Lithology

Porosity

Description
Lithology
Shows

CLAYSTONE, LIMESTONE & DOLOMITE M/A
NO SHOWS

Fluor.

Cut

Ditch Gas
Total %
TG - CG
0.5 1.0 1.5 2.0

NO GAS

Remarks
Casing Cores Logs Tests
SWC Deviation Bit Data Mud Data

MWD

AT 2378 m : DEVIATION OF 1.5 DEG.

26/12 POOH BEFORE CASING.

Age

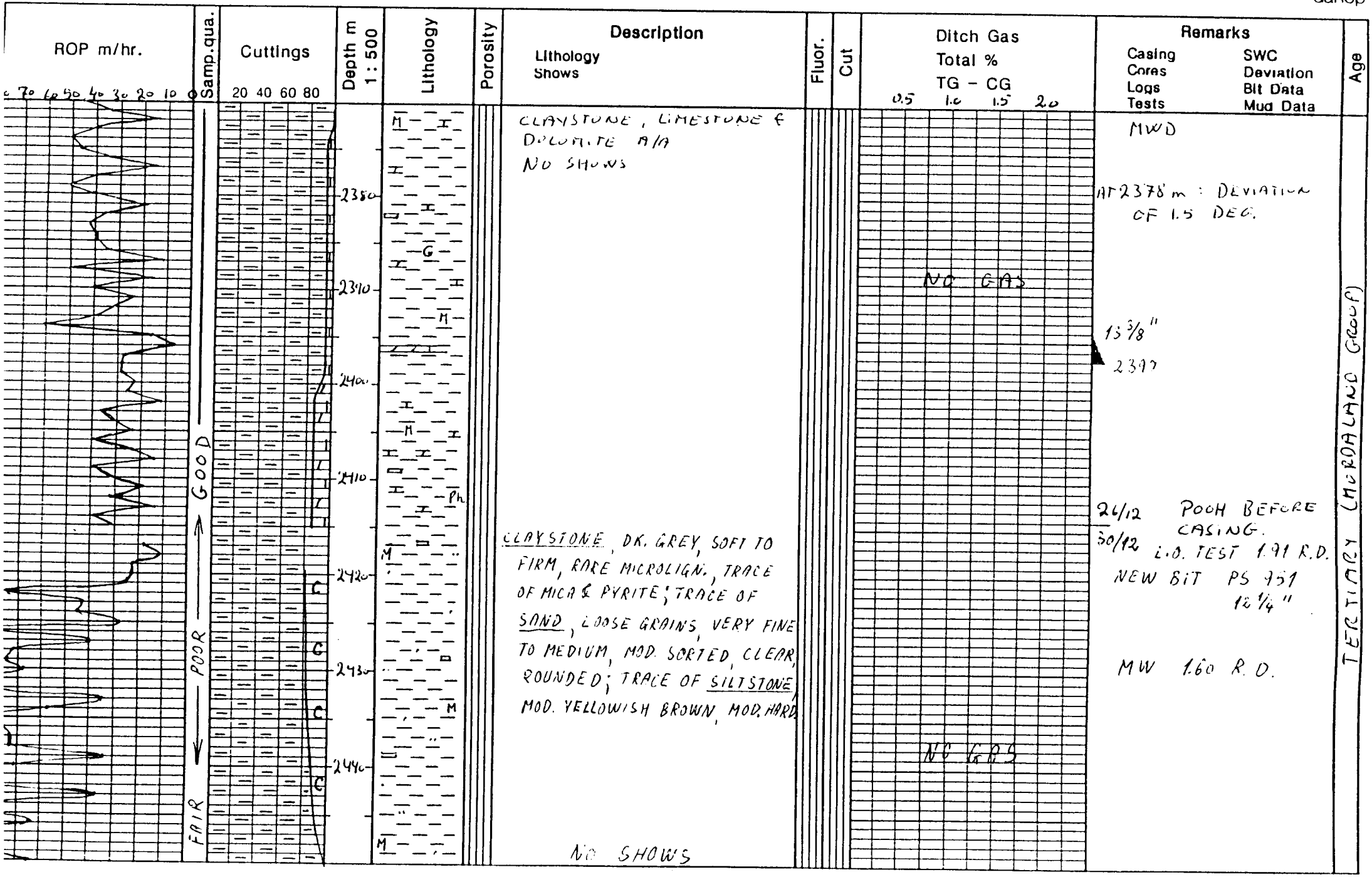
TERTIARY (HORDALAND GROUP)

GOOD

2380
2390
2400
2410
2420
2430
2440

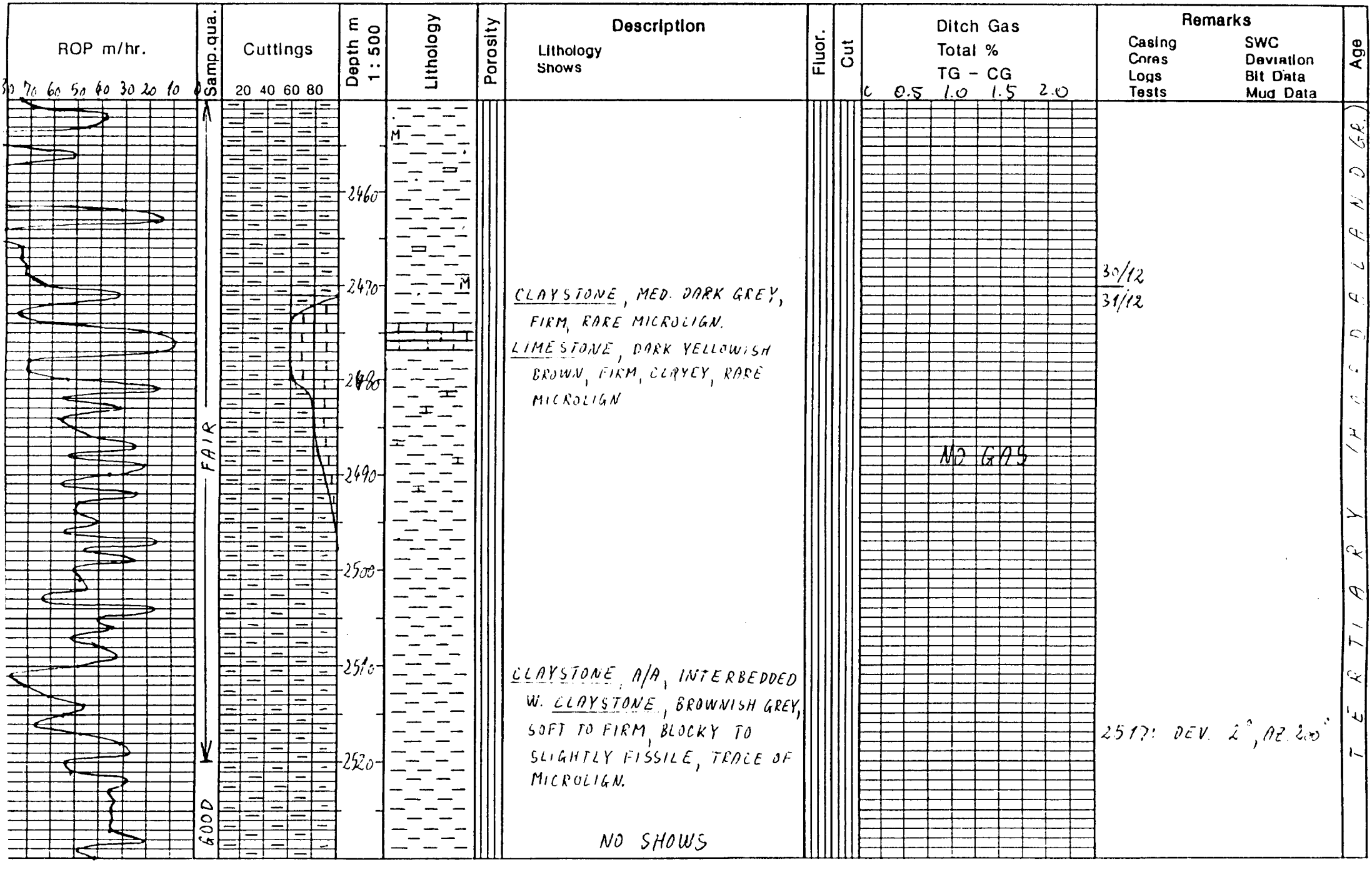
GRAPH WELL LOG Well TO DENSKJOLD-1(5503/3-2) Sheet 29

danop



GRAPH ; WELL LOG Well TO DENSKJOLD-1(5503/3-2) Sheet 3

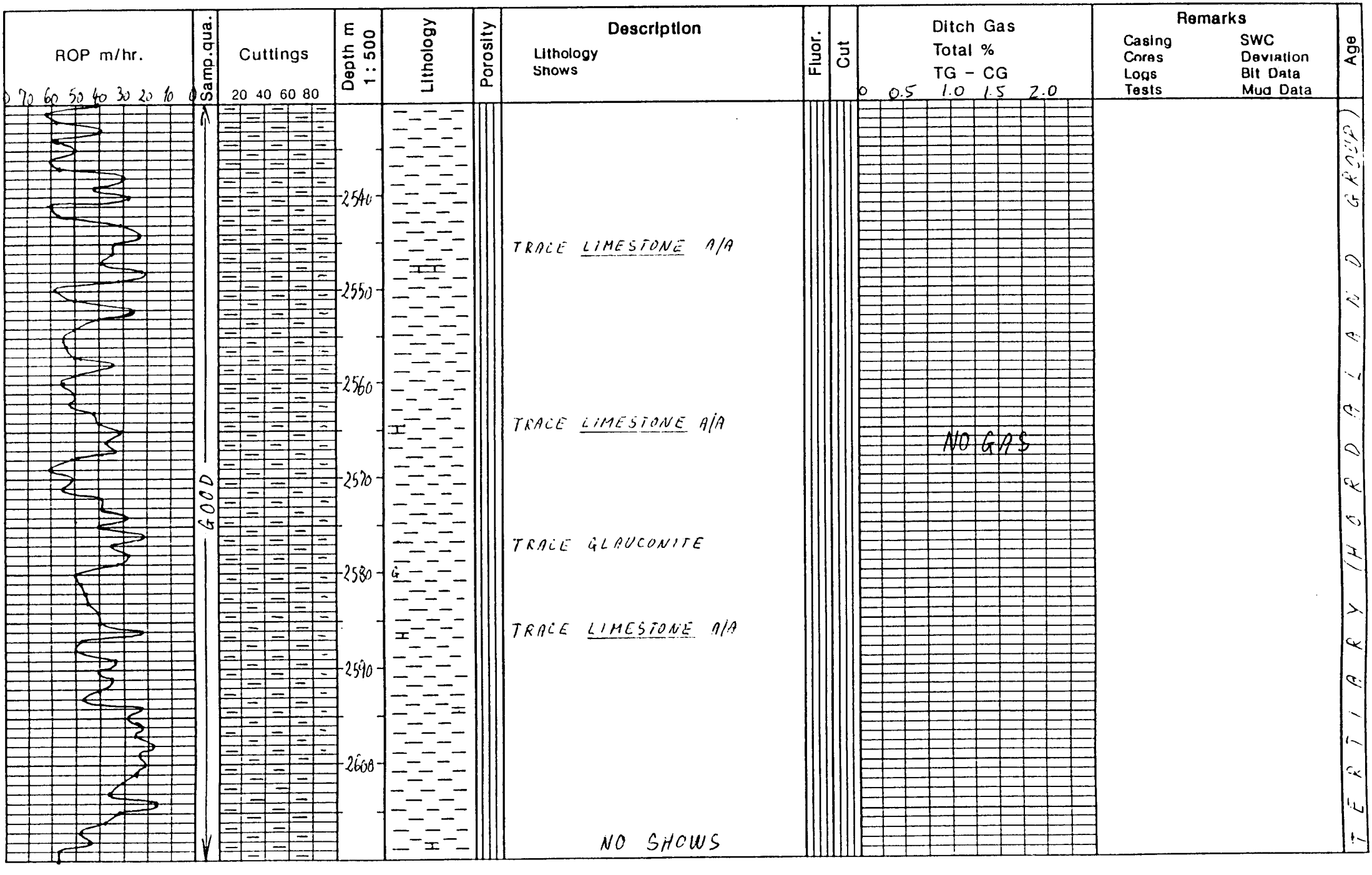
danop



GRAPHIC WELL LOG

Well TO. DENSKJOLD - 1 (5503/3-2) Sheet 3

danop



GRAPHIC WELL LOG

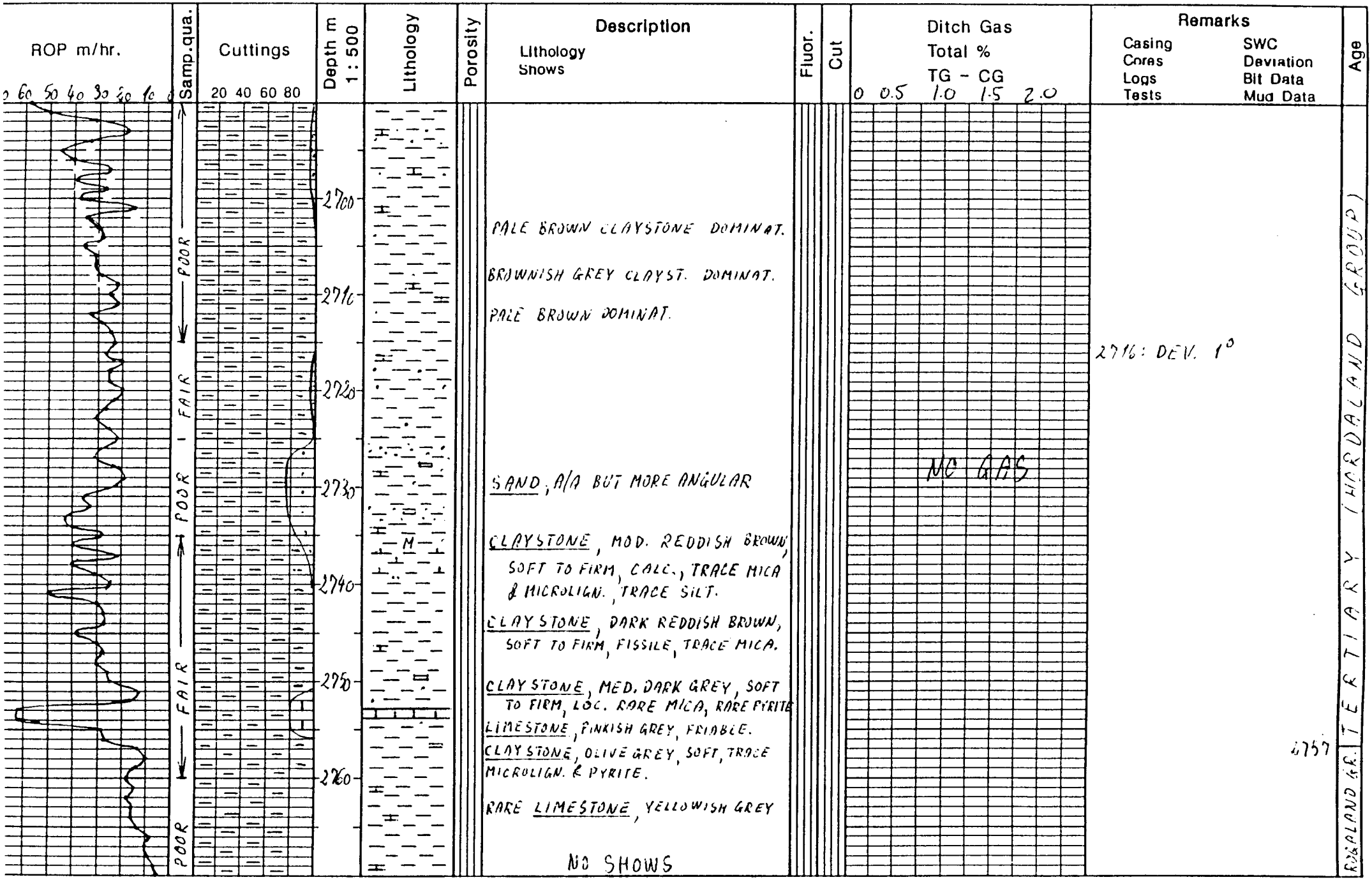
Well TO. DENSKJOLD - 1 (5503/3-2) Sheet 3

danop

ROP m/hr.	Samp.qua.	Cuttings	Depth m 1:500	Lithology	Porosity	Description (Lithology, Shows)	Fluor.	Cut	Ditch Gas Total % TG – CG	Remarks (Casing, Cores, Logs, Tests; SWC, Deviation, Bit Data, Mud Data)	Age
70 60 50 40 30 20 10 0	GOOD / POOR	20 40 60 80	2620 2630 2640 2650 2660 2670 2680			TRACE LIMESTONE A/A CLAYSTONE, DARK GREENISH GREY, FIRM TO MOD HARD, TRACE OF MICROLIGN. & GLAUCONITE, INTERBEDDED W. CLAYSTONE, BROWNISH GREY, A/A. SAND, LOOSE GRAINS, VERY FINE TO MED., MOD SORTED, CLEAR, SUBROUNDED TO ROUNDED, ALSO BLACK GRAINS (MICROLIGN.?) LIMESTONE, PALE YELLOWISH BROWN, FRIABLE. TRACE OF CLAYSTONE, PALE BROWN, SOFT. TRACE OF PYRITE. TRACE OF LIMESTONE A/A NO SHOWS			0 0.5 1.0 1.5 2.0 NO GAS	2623: DEV. 3°, AZ. 326°	TERTIARY (MORDALF)

RAPHIC WELL LOG Well TORDENSKJOLD - 1 (5503/3-2) Sheet 22

danop



GRAPHIC WELL LOG

Well TORDENSKJOLD - 1 (5503/3-2) Sheet 34

danop

ROP m/hr. 70 60 50 40 30 20 10 0

Samp.qua. — POOR — GOOD — FAIR — GOOD

Cuttings 20 40 60 80

Depth m 1:500 — 2780, 2790, 2800, 2810, 2820, 2830, 2840

Lithology

Porosity

Description
Lithology
Shows

CLAYSTONE, MED. DARK GREY, SOFT TO FIRM, TRACE MICROLIGN.

TRACE MED. BLUISH GREY CLAYST.

TRACE GREYISH ORANGE PINK LIMEST.

TRACE CALCAR SILTST.

CLAYSTONE, LIGHT OLIVE GREY, MOD. HARD, RARE MICA, TRACE MICROLIGN. & PYRITE & GLAUCONITE.
RARE LIMEST. A/A

LIMESTONE, GREYISH ORANGE PINK, HARD.

CLAYSTONE, GREENISH GREY, FIRM, SLIGHTLY FISSILE.

CLAYSTONE, MOD. REDDISH BROWN, FIRM.

CLAYSTONE, OLIVE GREY, MOD HD.

GREENISH GREY A/A

Fluor.

Cut

Ditch Gas
Total %
TG – CG
0 0.5 1.0 1.5 2.0

NO GAS

Remarks
Casing — SWC
Cores — Deviation
Logs — Bit Data
Tests — Mud Data

MW 1.60 R.D.

31/12
1/1

2805 : NEW BIT
M x DG 12 1/4"
DEV. 1 1/4°

1/1
1/1

Age

TERTIARY (PLIOCENE) GROUP

GRAPHIC WELL LOG

Well TORD. SKJOLD-1 (5503/3-2)

Sheet 3.

danop

ROP m/hr. 70 60 50 40 30 20 10 0

Samp.qua. GOOD — FAIR — GOOD — GOOD IE

Cuttings 20 40 60 80

Depth m 1:500 2860 2870 2880 2890 2900 2910 2920

Lithology

Porosity

Description
Lithology
Shows

CLAYSTONE, OLIVE GREY TO OLIVE BLACK, SOFT TO HARD, SLIGHTLY CALC.

CLAYSTONE, GREENISH GREY TO MED GREY, FIRM TO MOD HARD, VERY CALC., TRACE PYRITE

CLAYST. GRADING TO MARLSTONE

ABUND. MARLST., VERY LIGHT GREY, FIRM

CHALK, WHITE, FIRM, RARE PYRITE.

VERY SLOW STREAM. DULL WHITE CUT

CLAYSTONE, OLIVE GREY.

SAND, LOOSE GRAINS, CLEAR, VERY FINE.

CHERT, LIGHT BROWN, YELLOWISH GREY, SEMITRANSLUCENT.

CAVINGS

BEDS OF GREENISH GREY CLAYST.

TRACE LOOSE SAND GRAINS & CALC.

SILTSTONE

LAMINAE OF GREYISH ORANGE PINK CHALK

BEDS OF HARD LIMESTONE (INDURATED CHALK)

NO SHOWS

Fluor.

Cut

Ditch Gas
Total %
TG – CG
0 0.5 1.0 1.5 2.0

NO GAS

Remarks
Casing — SWC
Cores — Deviation
Logs — Bit Data
Tests — Mud Data

MW 1.60 K.D.

SPOT SAMPLES

ROP SCALE CHANGE

INCR. WOB, RPM

2/1
3/1 2964: NEW BIT SMITH F27D 12¼"

DEV.: 2°, AZ. 235°

SPOT SAMPLES

Age

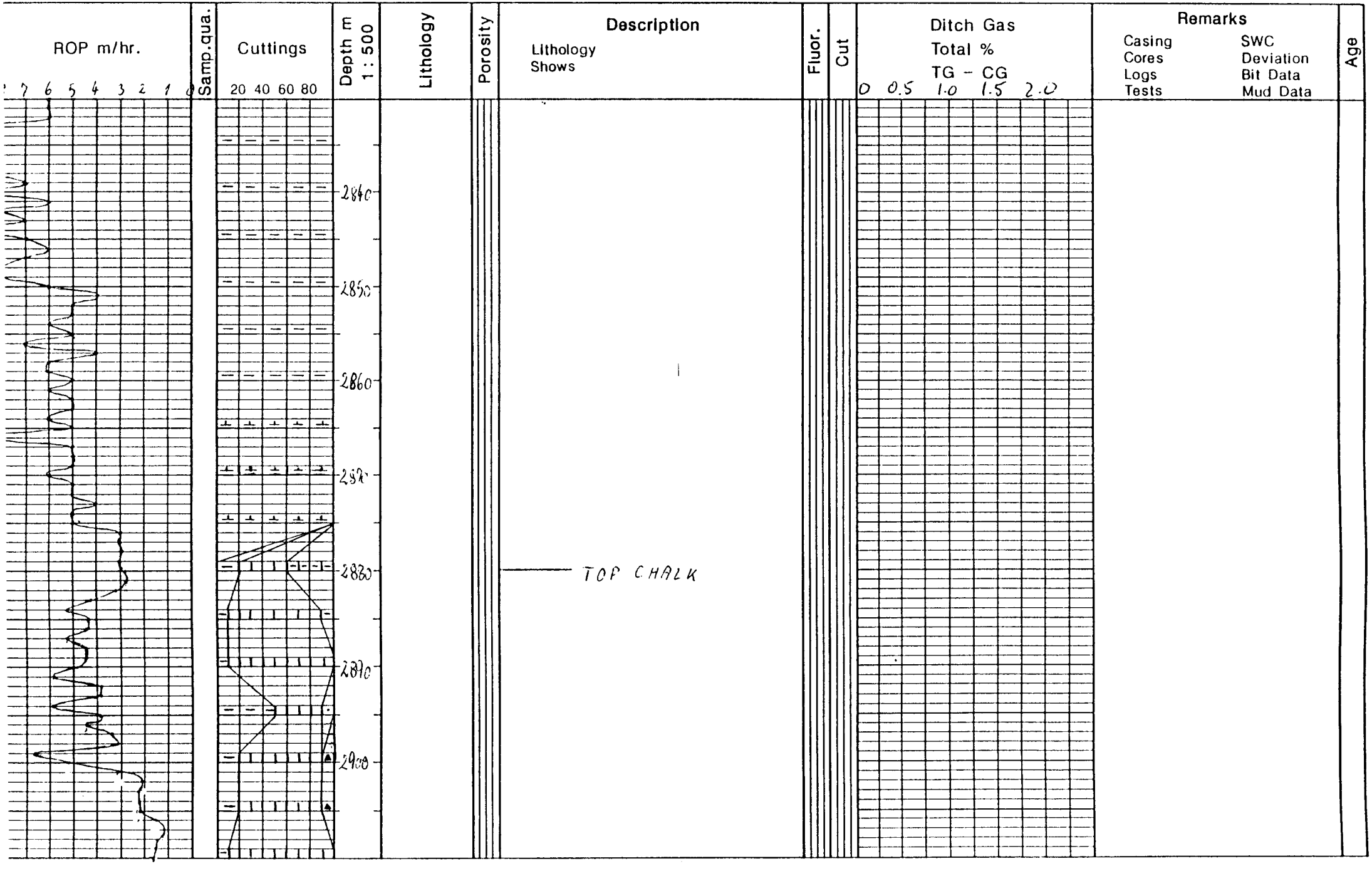
TERTIARY (ROGALAND GROUP)

DANIAN (?) (CHALK GROUP)

GRAPHIC WELL LOG

Well TORSKJOLD-1 (5503/3-2) Sheet 35. (ROP NEW SCALE)

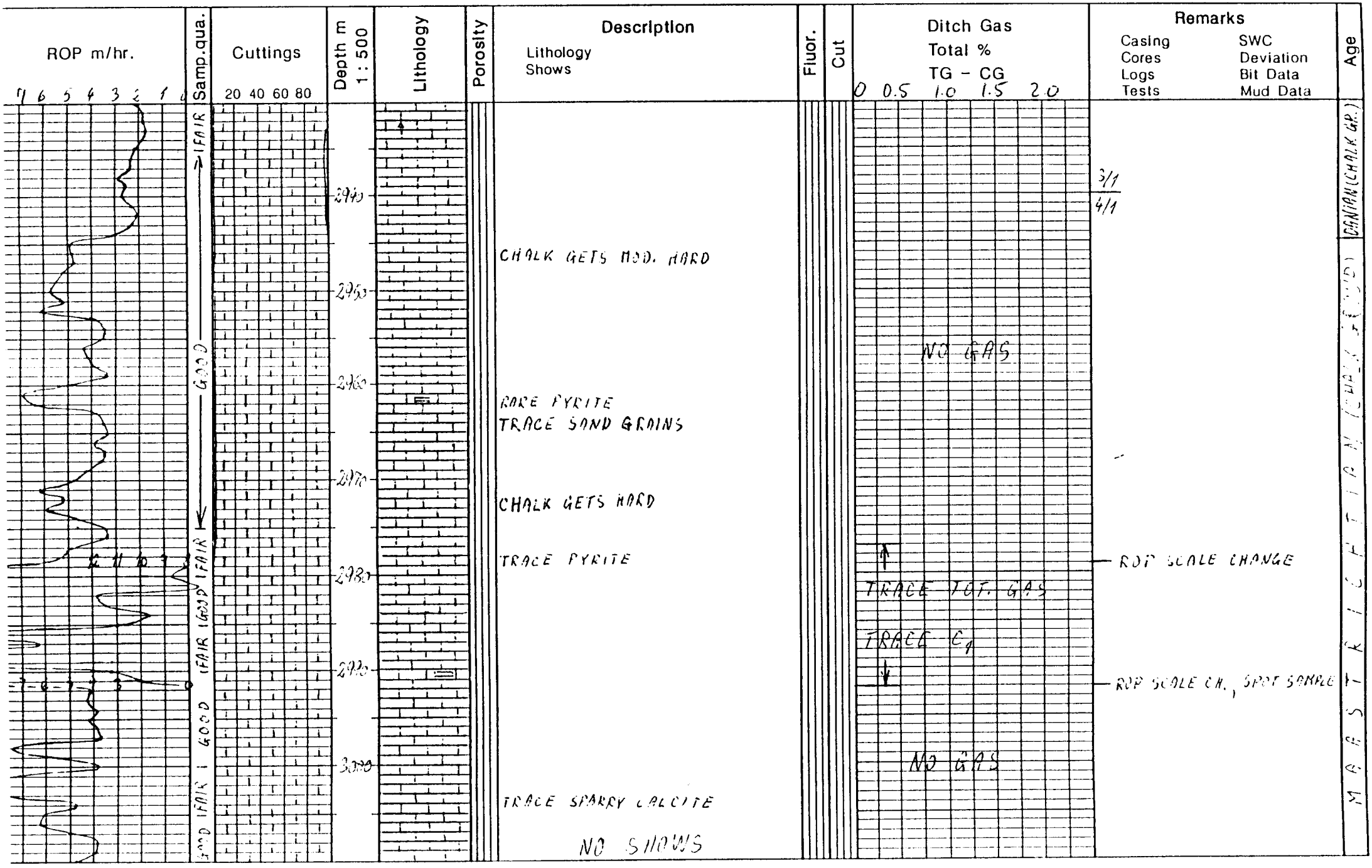
danop



GRAPHIC WELL LOG

Well TORDA SKJOLD-1 (5503/3-2) Sheet 3c

danop



GRAPHIC WELL LOG

Well TORD SKJOLD-1 (5503/3-2) Sheet 32

danop

ROP m/hr. 7 6 5 4 3 2 1 0	Samp.qua.	Cuttings 20 40 60 80	Depth m 1:500	Lithology	Porosity	Description: Lithology, Shows	Fluor.	Cut	Ditch Gas: Total %, TG – CG 0 0.5 1.0 1.5 2.0 2.5	Remarks: Casing, Cores, Logs, Tests; SWC, Deviation, Bit Data, Mud Data	Age
	GOOD		3020							MW: 1.60 S.G.	UPPER CRETACEOUS (CHALK GROUP)
			3030			RARE CLAYSTONE, GREENISH GREY & REDDISH BROWN.				4/1 5/1	
			3040			TRACE PYRITE.			NO GAS		
			3050								
			3060						C1		
			3070								
			3080			NO SHOWS			NO GAS	SPOT SAMPLE	

GRAPHIC WELL LOG

Well TORI ISKJOLD-1 (3503/3-2) Sheet 38

danon

ROP m/hr. 7 6 5 4 3 2 1 0

Samp.qua.

Cuttings 20 40 60 80

Depth m 1:500

Lithology

Porosity

Description
Lithology
Shows

Fluor.

Cut

Ditch Gas
Total %
TG - CG
0.5 1.0 1.5 2.0

Remarks
Casing SWC
Cores Deviation
Logs Bit Data
Tests Mud Data

Age

FAIR CTGS SL. CHURNED-LAM.

10% CAVINGS 3156

FAIR

WHITE

RED

3100
3110
3120
3130
3140
3150
3160

CHALK, A/A ; CLAYSTONE, CAVINGS

INCR. AMOUNT OF IRON FILINGS AT 3120
NORMAL AMT. AT 3121

CHALK: WHITE-VERY LIGHT GREY, A/A FIRM-MOD. HARD, RARE CHERT, MICRO-CRYPTOXLN, BLOCKY, NO DISPOR LOCALLY TR MICROFOSSILS, RARE SILT GRAINS. NO SHOWS

CHALK: GREY RED - PALE RED, SOFT - FIRM, TR ARGILLACEOUS, RARE SILT GRAINS.

MINOR CHALK: LIGHT GREY, SLIGHTLY ARGILLACEOUS

CHALK, REDDISH & WHITISH A/A, PARTLY INTERLAMINATED.

GAS LESS THAN 0.01%

0.2.4.6.8.1.0 2 3 4 5 6

GAS LESS THAN 0.01%

SURVEY 3095m 2¼°/160°
5/1 #6 TI B4 IG
3095 NEW BIT
6/1 NB #7 PDI 12¼"
DIAMOND BOART
Mw 1.60 sg
TIGHT HOLE TRIPPING OUT AND IN: 2934 - 3024, MAX. OVERPULL: 20 T
(FRACTURED ZONE?)
SPOT SAMPLE
6/1
7/1
GAS SCALE CHANGE
START 3M SAMPLING
TORQUE TREND CHANGE 3153M

UPPER CRETACEOUS (CAMPANIAN)

GRAPHIC WELL LOG

Well TOR. NSKNLD 1(SS83/32) Sheet 3

danop

ROP m/hr. 7 6 5 4 3 2 1 0

Samp.qua. FAIR

LESS 5% CAVINGS

Cuttings 20 40 60 80

GREY

WHITE

CAVINGS

Depth m 1:500

3180

3190

3200

3210

3220

3230

3240

Lithology

Porosity

Description

Lithology
Shows

CHALK, LIGHT GREY - MED LIGHT GREY, FIRM - HARD, [illegible], MOTTLED P/P.

WHITISH & GREY CHALK PARTLY INTER-LAMINATED

TRACE PYRITE & SILT GRAINS

NO SHOWS

CHALK: PRED. OFF WHITE - VERY LIGHT GREY, MINOR LIGHT - MED. LIGHT GREY, FIRM - MOD. HARD, BLOCKY, TR - SLIGHTLY ARGILLACEOUS, RARE SILT - VF SAND GRAINS, TRACE MICROFOSSILS

CHALK: PRED. PALE RED - PINK GREY FLAKY - BRITTLE (BIT ACTION?)

CHALK GRADING LIMESTONE:

MEDIUM LIGHT GREY - VERY LIGHT GREY MOD. HARD - HARD OCC VERY HARD, BRITTLE, LOCALLY SILICIFIED - CHERTY, TRACE - SL ARGILLACEOUS, RARE GLAUCONITE GRAINS, NO VIS. POROSITY

TRACE CHERT: LIGHT OLIVE GREY, VERY HARD, BRITTLE, LOCALLY TRANSLUCENT.

Fluor.

Cut

Ditch Gas
Total %
TG - CG
0 0.2 0.4 0.6 0.8 1 2 3 4 5 6

TRACE TOT. GAS

Remarks
Casing
Cores
Logs
Tests
SWC
Deviation
Bit Data
Mud Data

Age

Mw 1.60 RD

7/1
8/1 STATIC MUD LOSS 6 BBLS WHILE CUT & SLIP ON TRIP IN

BIT #7 100% worn 1 3/4° 323°

8/1 NBH 8 3213 m

7/1 SMITH F270 12 1/4"

GAS TRAP & EQUIPMENT CHECKED OK.

INCR. WOB AT 3248

GRAPHIC WELL LOG

Well TOR ENSKJOLD - 1 (5503/3-2) Sheet 4c

danop

ROP m/hr. 8 7 6 5 4 3 2 1 0

Samp.qua.

Cuttings 20 40 60 80

Depth m 1:500

Lithology

Porosity

Description

Lithology
Shows

Fluor.

Cut

Ditch Gas
Total %
TG – CG
0 0.2 0.4 0.6 0.8 1 2 3 4 5 6

Remarks

Casing | SWC
Cores | Deviation
Logs | Bit Data
Tests | Mud Data

Age

GOOD

GREY

RED

GREY

GOOD

GREEN GREY

RED

CAVINGS

CAVINGS

3260

3270

3280

3290

3300

3310

3320

LIMESTONE: YELLOW – PINK GREY – LIGHT OLIVE GREY, HARD – V. HARD, SILICIFIED – RECRYSTALLIZED GRADING CHERT, MINOR SOFT – CHALKY, NON ARGILLACEOUS, TRACE GREEN GLAUCONITE, DENSE, NO VIS. POROSITY

CHERT: a/a

SAND: PRED. CLEAR QTZ, LOOSE, VERY FINE, SUBANGULAR, MOD – W. SORTED, MINOR HEAVY MINERALS, HEMATITE STAIN, GARNET ?, LOCALLY ABUNDANT GLAUCONITE IN CALCITE CMT. SANDSTONE. No SHOWS

MARL GRADING CHALK: GREY RED – PALE RED BROWN, SOFT – FIRM, MINOR HARD – BRITTLE, TR SILT – VF SAND, ARG – VERY ARGILLACEOUS, TRACE GLAUCONITE
NO SHOWS

DOLOMITE: GREY RED, HARD – V. HARD, BRITTLE, ARGILLACEOUS, TR. GLAUCONITE

LIMESTONE: YELLOW GREY – LIGHT OLIVE GREY, MINOR LT – DARK GREEN GREY, TRACE MICA, ELSE A/A.

CLAYSTONE: DARK GREY – GREY BLACK, FIRM – MOD. HARD, CALC GRADING MARL, WHITE CHALKY LAMINAE, TRACE CARBONACEOUS

CHERT: YELLOW ORANGE – YELLOW BROWN VERY HARD, SPLINTERY

LIMESTONE: VERY LIGHT GREY – OFF WHITE GRADING CHALK.

MARL: GREY RED – PALE RED BROWN, SOFT.

TRACE SAND: LOOSE, VERY FINE, TR GLAUC.

LIMESTONE / MARL: MOD – PALE – DARK RED BROWN, SOFT – FIRM, STICKY, TRACE SILTY.

MARL: GENERALLY A/A BECOMING MORE ARGILLACEOUS

0.02%

Mw 1.60 sg

9/1

10/1

← FALSE DRILL BREAK ?
JAR COMPRESSED.

10/1

11/1

3/4°
190°
3306m

CIRC B.U. 3329 m
NO TRIP GAS.
1 LOST CONE
PULLED BACK TO SHOE FOR SURVEY @ 3327 m.
11/1 HOLE PROBLEMS AT 3329 m

GRAPHIC WELL LOG

Well TORLUSKJOLD-1 (5503/3-2) Sheet 4.

danop

ROP m/hr. 8 7 6 5 4 3 2 1 0 | Samp.qua. | Cuttings 20 40 60 80 | Depth m 1:500 | Lithology | Porosity | Description (Lithology, Shows) | Fluor. | Cut | Ditch Gas Total % TG – CG | Remarks (Casing, Cores, Logs, Tests, SWC, Deviation, Bit Data, Mud Data) | Age

Samp.qua.: GOOD … GOOD

Cuttings: RED … GREY … BLACK … Cement … GY … RED … GY … RED … BLK … WH + RED + GY

Depth m 1:500: 3340 … 3350 … 3360 … 3370 … 3380 … 3390 … 3400

MARL/CLAYSTONE: DARK RED BROWN – MOD. BROWN, SOFT – MOD. HARD, CALCAREOUS, TRACE SHELL FRAGMENTS, LOCAL RIPPLE LAMINATION

CLAYSTONE: MED. DARK GREY – BLACK, FIRM – HARD, CALCAREOUS, MICROLIGNITE?, SL. MICROMICACEOUS, "SILTY TEXTURE" DUE TO RECRYSTALLIZED CALCITE.

SAND: PRED CLR QTZ, LOOSE, VERY FINE – FINE, SUBANGULAR – SUBROUNDED, WELL SORTED, TRACE MICA. NO SHOWS

INTERBEDDED / LAMINATED:

CLAYSTONE / MARL: GY RED – MOD BROWN, GRY GRN AND

CLAYSTONE / MARL: LIGHT GREY – MED GREY, OCC MED DARK GREY, SOFT – FIRM, OCC MOD. HARD, CALCAREOUS

NO SHOWS.

LIMESTONE, WHITE TO VERY LIGHT GREY, FIRM, OCC. MODERATELY HARD, SLIGHTLY ARGILLACEOUS, CRYPTOCRYST., SLIGHTLY DOLOMITIC. NO SHOWS

CLAYSTONE, OLIVE BLACK, SOFT TO FIRM, CALC.; VERY MICROLIGNITIC.

LIMESTONE GRAD. MARL, WHITE – VERY LIGHT GREY, GREYISH RED – MOD. BROWN, LIGHT GREY – MED. LIGHT GREY, SLIGHTLY TO V. ARGILL., SOFT TO FIRM. NO SHOWS

CLAYSTONE, OLIVE BLACK, SOFT TO FIRM, MOD CALC., V MICROLGN., PYRITIC (?).

Ditch Gas: Total % / TG – CG

Remarks:

- CORE #1 CUT W/ JUNKBASKET.
- REC: 3331.0 – 3331.32 M
- NB#11 REED HA216J 15/1 #11 T48216
- CIRC. B.O 3342M. RAN LOGS
- 3342m
- NB # 13 CHRISTENSEN 5225 JUNK ON BOTTOM. 20/1
- 20/1
- 21/1
- 3340 – 3405m calcimetry 66/68 to 55/56
- INCREASED RPM + WOB
- CLEAR CHANGE IN TORQUE ALL OTHER PARAM CONST.

Age: EARLY CRETACEOUS – JURASSIC

GRAPH.. WELL LOG

Well TOR ENSKJOLD-1 (5503/3-2) Sheet 4

danop

ROP m/hr. 8 7 6 5 4 3 2 1 0

Samp.qua.

Cuttings 20 40 60 80

Depth m 1:500

Lithology

Porosity

Description
Lithology
Shows

CLAYSTONE, OLIVE BLACK N/A

SAND, LOOSE, CLEAR QUARTZ, VERY FINE - FINE, WELL SORTED, SUBANG - SUBROUNDED, OCC. ROUNDED.

CLAYSTONE, OLIVE GREY, SOFT FIRM, CALCAREOUS, VERY CARBONACEOUS, MICROPYRITIC (3), ABUND. CALCITE CRYST.

SH: GREY BLACK - BLACK, FRD. HARD - HARD, FISSILE, BLKY - SLIGHTLY CALC - PYRITIC, MICROMICACEOUS

CLAYSTONE, BROWNISH BLACK, HARD - VERY HARD SPLINTERY, MASSIVE, CALCAREOUS, PYRITIC (10% AT 3433M), MICROLIGNITIC, SLIGHTLY MICACEOUS, SLICKENSIDES

SAND INTERBEDDED WITH CLAYSTONE;

SAND, LOOSE, FRED CLEAR QTZ, FINE-GR., WELL SORT., SUBANGUL., TRACE PYRITE.

CLAYSTONE, N/A, TURNING DARK GREY AND FIRM.

CLAYSTONE: GREY BLACK, FIRM - MOD HD, OCC HARD, NON - LOC. SLIGHTLY CALCAREOUS, MICRO - MICACEOUS, TR CARBONACEOUS, RARE TRACE PYRITE.

AT 3468m:

CLAYSTONE: BROWN BLACK, FIRM, SUBFIS, MOD CALCAREOUS (WH SPECS OF CALC), MICROMICACEOUS, TR CARBONACEOUS AND

CLAYSTONE (FROM 3474m): OLIVE GREY - DARK GREY, SOME BRN GREY, FIRM, SUBFIS, NON - LOC SL CALCAREOUS, MICROMIC, CARB, SLTY, TR MICROPYR, "EARTHY".

TR DOL: OLV GREY, VERY HARD, BLOCKY, CRYPTOXLN.

Fluor.

Cut

Ditch Gas
Total %
TG - CG
.2 .4 .6 .8 1

0 %

0 %

0%

Remarks
Casing
Cores
Logs
Tests
SWC
Deviation
Bit Data
Mud Data

3410M 4 HRS SAT DUE TO PROBLEMS RECOVERING SWC'S AT 4

AT 3417M RPM INCREASE 15% > 175 RPM

2 1/1

22/1 CBHT CT 3 3/4"
Core #2: Cut 3422 - 3440m
Rec 3422 - 3426.6m (46.6%)

2 1/1 CB #2 CT 4 3
Core 4 3 Cut 3433 - 3436m
Rec 3434 - 3435.4m (52%)

AB 4 M

CALCIMETRY 2-3
W.O.B. 11-12 T
RPM. 175.

- 3448: BOTTOMS UP 23/4 2 1/4

Age

wbm Cly washed out

GRAPHIC WELL LOG

Well TOR1 . N.SK.10.1.0 (5503/3-2) Sheet 4.

danop

ROP m/hr. 8 7 6 5 4 3 2 1 0

Samp.qua.

Cuttings 20 40 60 80

Depth m 1:500

Lithology

Porosity

Description
Lithology
Shows

Fluor.

Cut

Ditch Gas
Total %
TG – CG
.2 .4 .6 .8 1

Remarks
Casing
Cores
Logs
Tests
SWC
Deviation
Bit Data
Mud Data

Age

Abn Cly from mtx in Sst

3500
3510
3520
3530
3540
3550
3560

CLAYSTONE, DARK GREY TO GREYISH BLACK, FIRM-MOD. HARD, LOC. SLIGHTLY CALCAREOUS, MICACEOUS, SLIGHTLY MICROLIGNITIC, TRACE-MINOR PYRITE, TRACE-MINOR LOOSE SILT GRAINS
LIMESTONE, WHITE, SOFT-FIRM, W. DARK STREAKS

TR CALCITE: WHITE CLEAR, CRYSTALLINE, LOC W/ PYRITE ON CRYSTAL FACES.

SILTSTONE, LIGHT OLIVE GREY, FIRM, VERY CALC.

CLAYSTONE: BROWN GREY, SOFT-FIRM, NON CALC. "EARTHY"

SILTSTONE: LIGHT GREY – WHITE, SOME: HARD – VERY HARD, WELL CMT (SIL/DOL CMT), TR CARB. TR MICRO PYR, LOC VF SDY GRDG SST. SHOWS: BRI YEL FLU NO VIS CUT, V SLO STRMG BL WH FLU CUT, BRI YEL WH-BL WH FLU RESD. NO VIS RESD

SLTST LOC V ARG OR CALC/DOL CMT BCMG CLY AT SURFACE

SD: CLEAR QUARTZ, VERY FINE – FINE, LOOSE, WELL SORTED, TR PYRITE. NO SHOWS

SST INTLAM SLST: PALE RD-GY RD, HARD, WELL CMT, TR MICA, NO VIS POR VF GRD SLT. 50-60° BEDDING

CALCIMETRY: 3543, PARTLY DOL.

FLOW CHECK SPOT SPLS.
24/1
25/1

TRIP GAS: TR/2

RR1 CB#2 CT 403
CORE NO 4
CUT 3562 – 3572 m
REC 3562 – 3569 7m

GRAPHIC WELL LOG

Well TORL. ISKJOLD-1 (5503/3-2)

Sheet 4

danop

ROP m/hr. 8 7 6 5 4 3 2 1 0	Samp.qua.	Cuttings 20 40 60 80	Depth m 1:500	Lithology	Porosity	Description Lithology Shows	Fluor.	Cut	Ditch Gas Total % TG - CG 0.2 0.4 0.6 0.8 1.0	Remarks Casing / SWC Cores / Deviation Logs / Bit Data Tests / Mud Data	Age
	1" = 30 metres		3580 3590 3600 3610 3620 3630 3640 3650			SST/SLST: LT BRN GY, VERY DOLOMITIC GRDG DOL. ANHYDRITE: WHITE, SOFT, AMORPH AND CRYSTALLINE, CLEAR CBS XLS, BRIT. INTBD WITH SHALE/CLAYSTONE: GREY BLACK, DARK GREY, MOD HD - HD, BLKY - FIS, SL CALC, MICRONIC, ABN SLICKENSIDES. ANHYDRITE: A/P AND ALSO - WHITE - PINKISH GREY, MOD. HARD - HARD, MICROXLN - GRANULAR, LOC DOLOMITIC AND - LT BRN GY - PALE YEL BRN, MOD. HARD, BRITTLE, ARGILLACEOUS, LOC DOLOMITIC GRDG DOLOSTONE, MICROXLN AND MINOR BEDS OF: DOLOSTONE: LIGHT BROWN GREY - PALE YELLOW BROWN, MOD HARD - HARD, BRITTLE, LOC ARG, MICROCRYSTALLINE - GRANULAR NC SHOWS DOLOSTONE: LT BRN GY - V PALE YEL BRN, HD, BRIT, SL ARG, MICROXLN - GRANULAR IR: V DULL YEL FLU NON - V SLOW STRNG YEL WH - BR WH FLU CUT, NO VIS CUT, BRI YEL FLU RESD, LT STRAW VIS R			TRIP GAS 1.75%	X RR 1 NBHOW [illegible]	

GRAPHIC WELL LOG

Well TOR ENSKJELD-1 (5503/3-2) Sheet 4

danop

ROP m/hr. 8 7 6 5 4 3 2 1 0

Samp.qua.

Cuttings 20 40 60 80

Depth m 1:500

Lithology

Porosity

Description

Lithology
Shows

ANHYDRITE :
INTERLAMINATED / INTERBEDDED.
1) WHITE, SOFT - FIRM, AMORPH, LOC. ARGILLACEOUS LAMINAE.
2) CLEAR, BRITTLE, CRYSTALLINE (GYPSUM?)
3) PINKISH GREY - LIGHT BROWN GREY - PALE YELLOW BROWN, MOD HARD - HARD, BRITTLE, ARGILLACEOUS, LOC DOLOMITIC (?) MICROCRYSTALLINE GRANULAR

MINOR THIN BEDS / LAM. OF

SHALE DARK GREY - GREY BLACK - BROWN BLACK, MOD. HARD - HARD, BLOCKY - FISSILE, SLIGHTLY DOLOMITIC, MICROMIC, LOCALLY SLICKEN SIDES.
AND MINOR BEDS OF

DOLOSTONE : LIGHT BROWN GREY - PALE YELLOW BROWN, MOD HARD - HARD, BRITTLE, LOC ARG., MICRO CRYSTALLINE - GRANULAR.

SAND(STONE): LOOSE GRAINS, COL. - LESS OR REDDISH, V. FINE - MED. GR., MOD. SORT., SUBANG.
INTERBEDD W. CLAYSTONE: MOD. REDDISH BROWN - GREYISH RED, FIRM, BLOCKY.

"VOLCANIC ROCK": V. LIGHT GREY W. V. ABUND. IRREGUL. BLACK DOTS & LAM.

Fluor.

Cut

Ditch Gas
Total %
TG - CG
.2 .4 .6 .8 1.0 2 3 4 5

CG: 0.15/0.25

CG: 0.21/0.25

CG: 0.34/0.25

CG: 0.27/0.25

CG: 0.2-0.2

CG: 0.22/0.1

Remarks

Casing SWC
Cores Deviation
Logs Bit Data
Tests Mud Data

MW: 1.63

MW: 1.64

MW: 1.65

Age

GRAPHIC WELL LOG

Well TOR NSKJOLD-1 (5503/3-2)

Sheet 46

danop

ROP m/hr. 8 7 6 5 4 3 2 1 0	Samp.qua.	Cuttings 20 40 60 80	Depth m 1:500	Lithology	Porosity	Description Lithology Shows	Fluor.	Cut	Ditch Gas Total % TG – CG 0.2 0.4 0.6 0.8 1.0 2 3 4 5	Remarks Casing / SWC Cores / Deviation Logs / Bit Data Tests / Mud Data	Age
		BLOCKY	3740			"VOLCANICS" W GREEN & PINK-PURPLE GRAINS CLAYSTONE: BROWNISH BLOCK, HARD, BLOCKY, SLIGHTLY DOLOMITIC T.D. : 3739 m 27. jan. 1989 (18.30)			FG 1.9 / C1	NW 165 27½ 3739 TD 3½	